

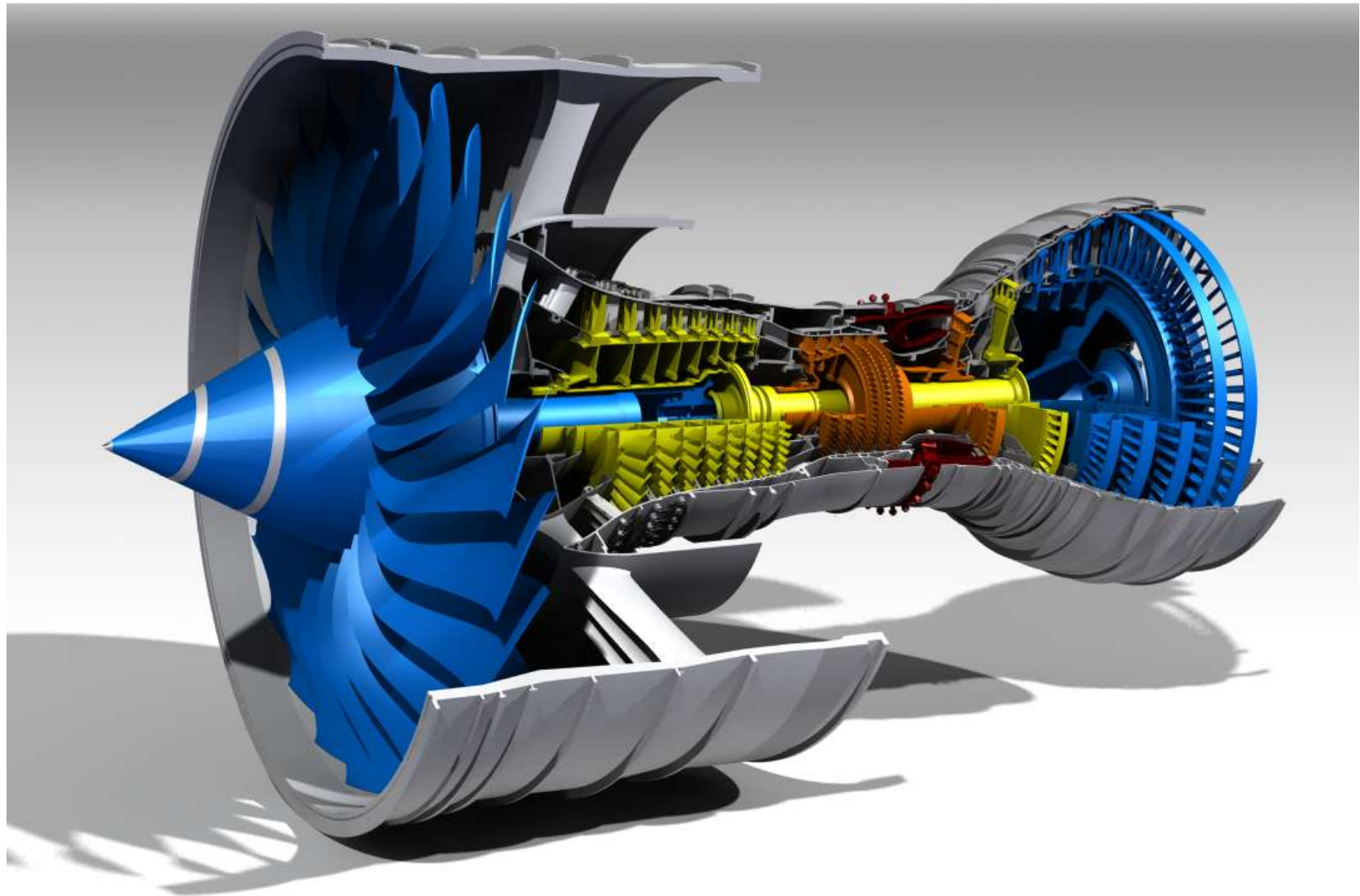
AE 4238

AE4-238 Aero Engine Technology

Combustion-I: Introduction

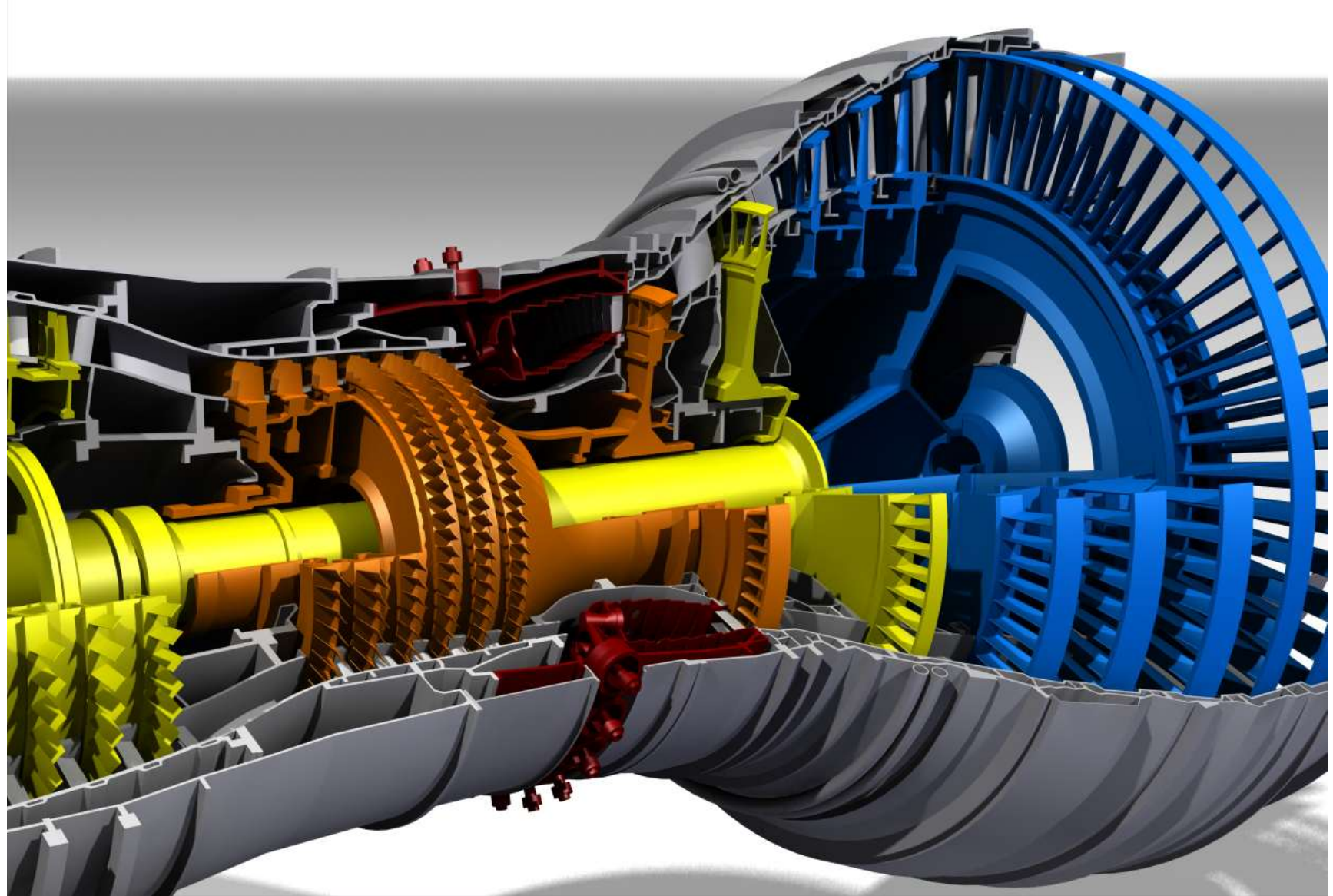
Prof. Dr. Arvind Gangoli Rao

A Turbofan Engine

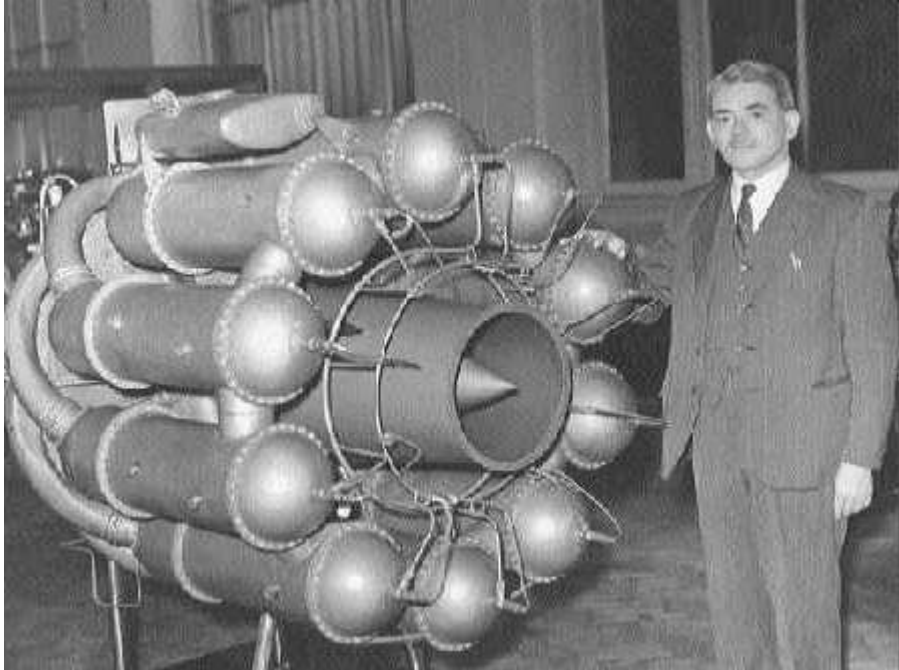


AE 4238

A turbofan engine

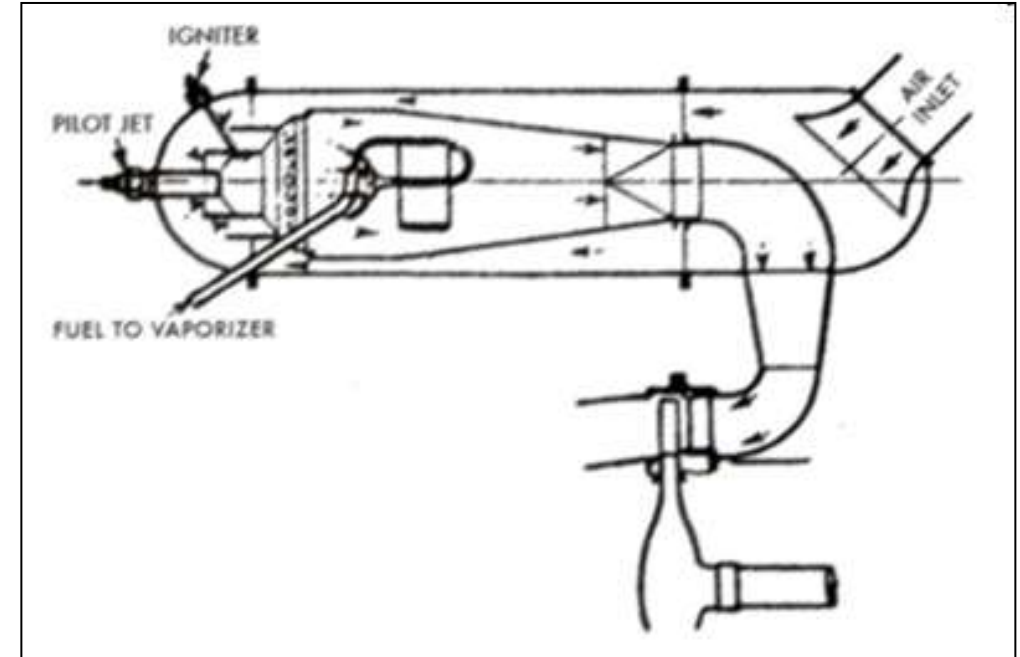
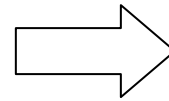


The first gas turbine

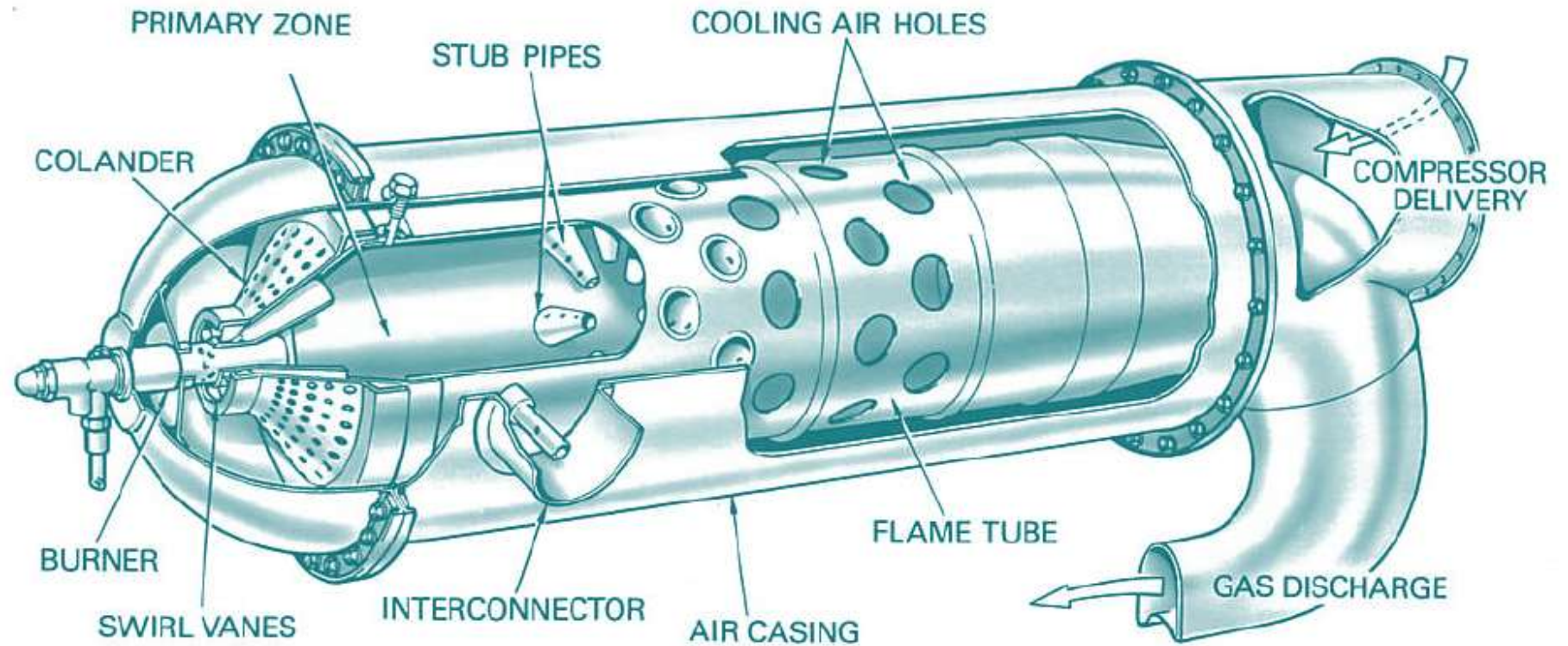


← This is where how it all started!!

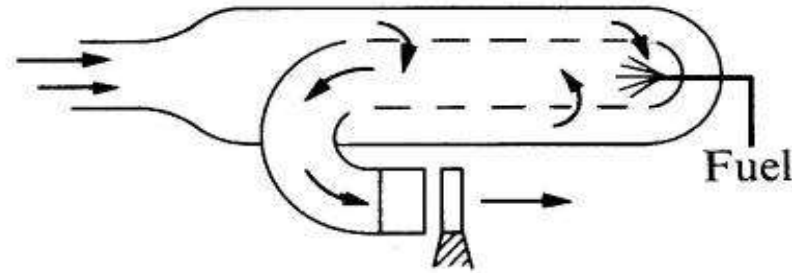
Combustion Chamber
combustion intensity more than 25x
greater than any other of that time



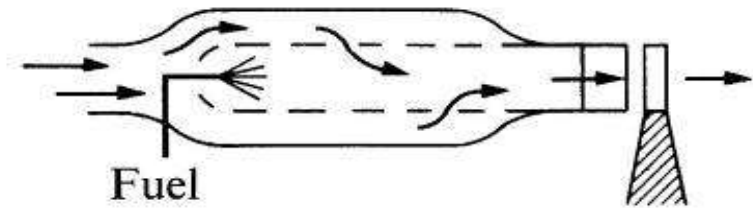
Whittle combustor



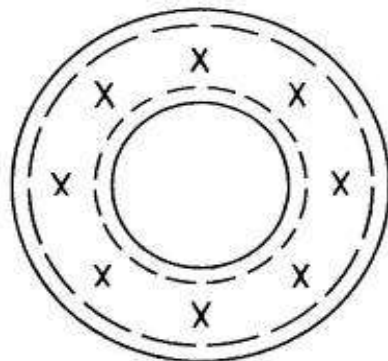
Different arrangements of combustors



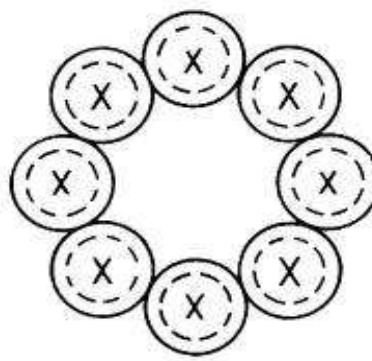
Reverse flow



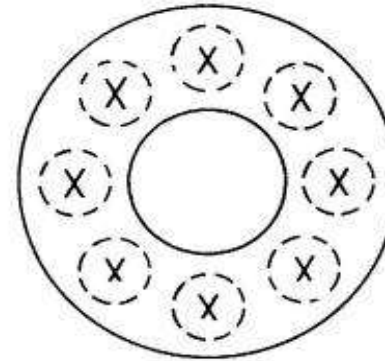
Straight-through flow



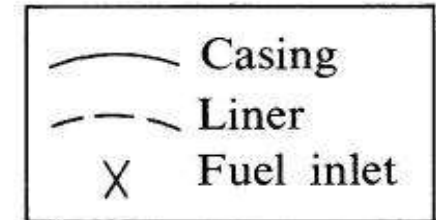
Annular combustor



Can or tubular combustor

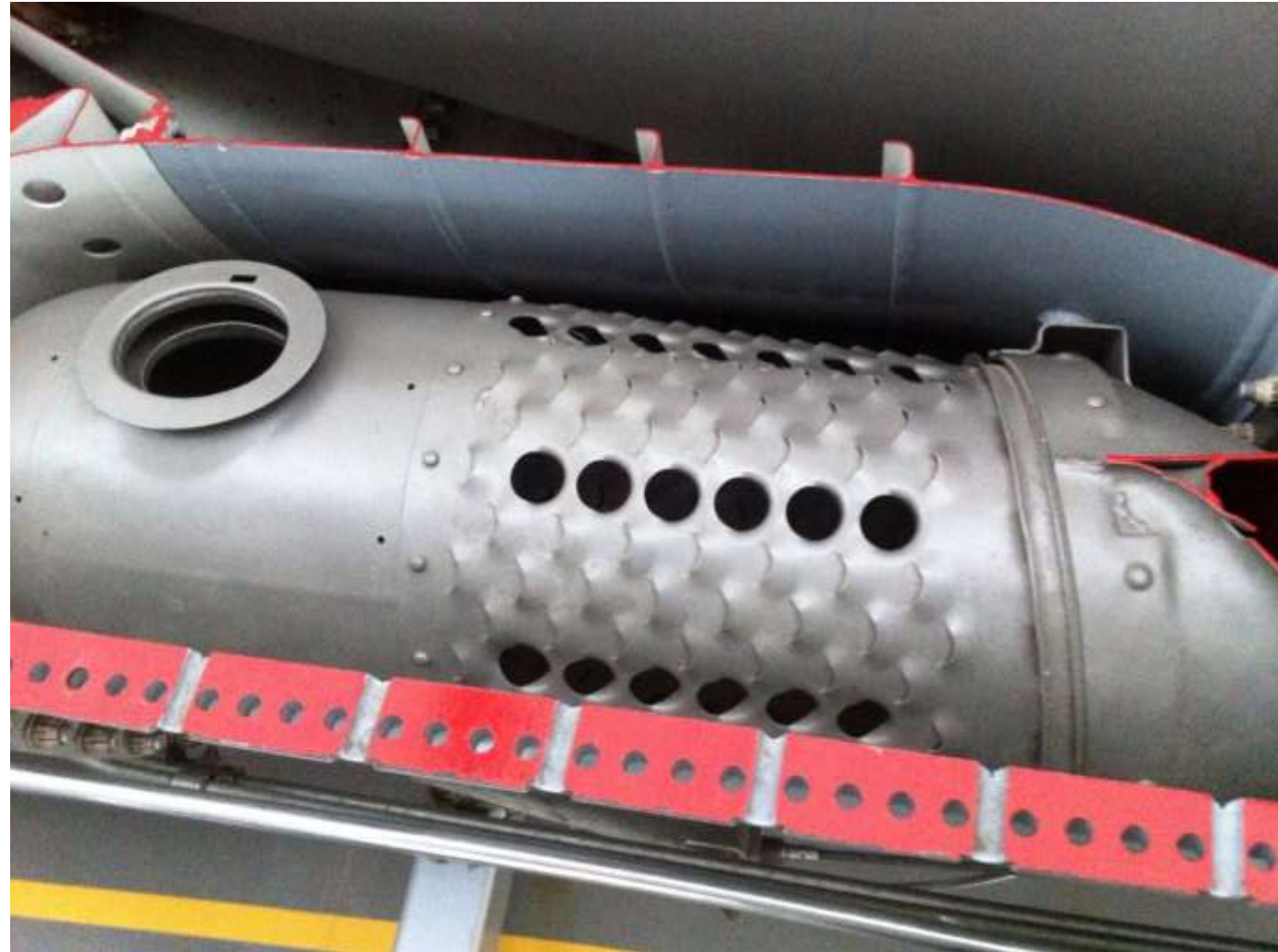


Can-annular combustor



Combustion chamber, longitudinal and cross sections.

Early can type combustor



Early can type combustor



Design in the 70's



AE 4238

AE4-238 Aero Engine Technology

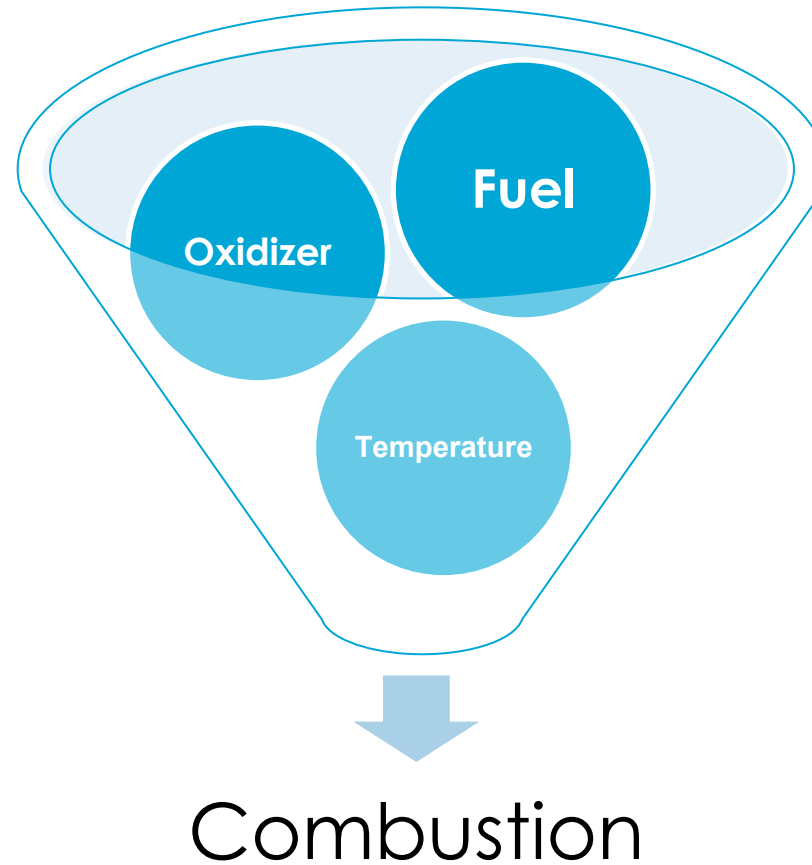
Combustion-II: Enthalpy

Prof. Dr. Arvind Gangoli Rao

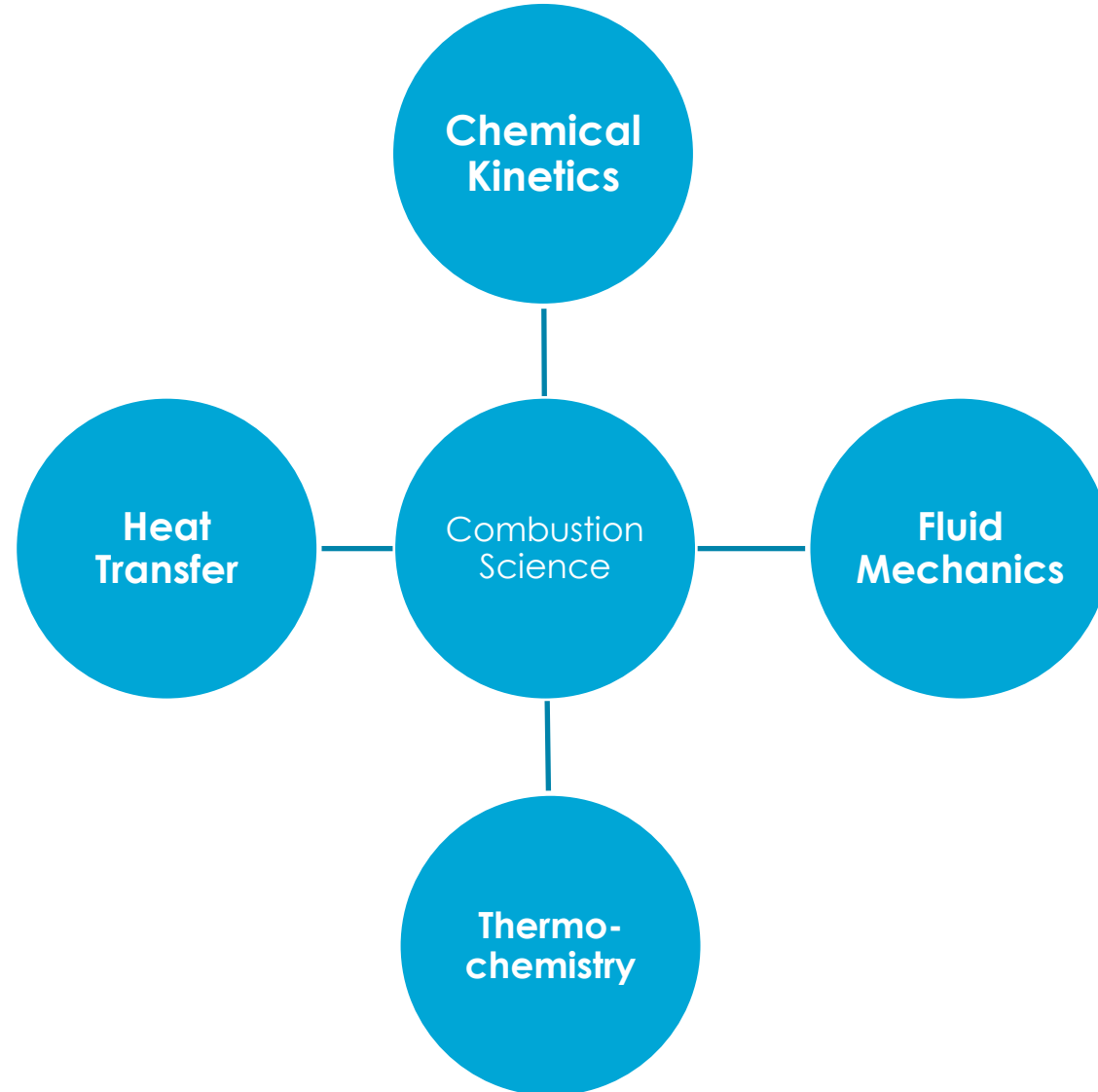
Contents of this lecture

- Elements of a combustion system
- Basics of enthalpy of combustion
- Thermodynamics of combustion

Essential elements for combustion



Combustion Science



Enthalpy

There are three types of Enthalpy

1. Sensible
2. Latent
3. Reaction

Sensible < Latent < Reaction

Enthalpy of formation

The enthalpy of formation h_f at a given temperature is defined as the heat released from producing 1 mole of a substance from its elements at that temperature.

By this definition, the enthalpy of formation is zero for the reference elements.

These elements are, for example, graphite (C), molecular hydrogen (H₂), molecular oxygen (O₂), molecular nitrogen (N₂), and molecular chlorine (Cl₂).

Enthalpy of formation and reaction

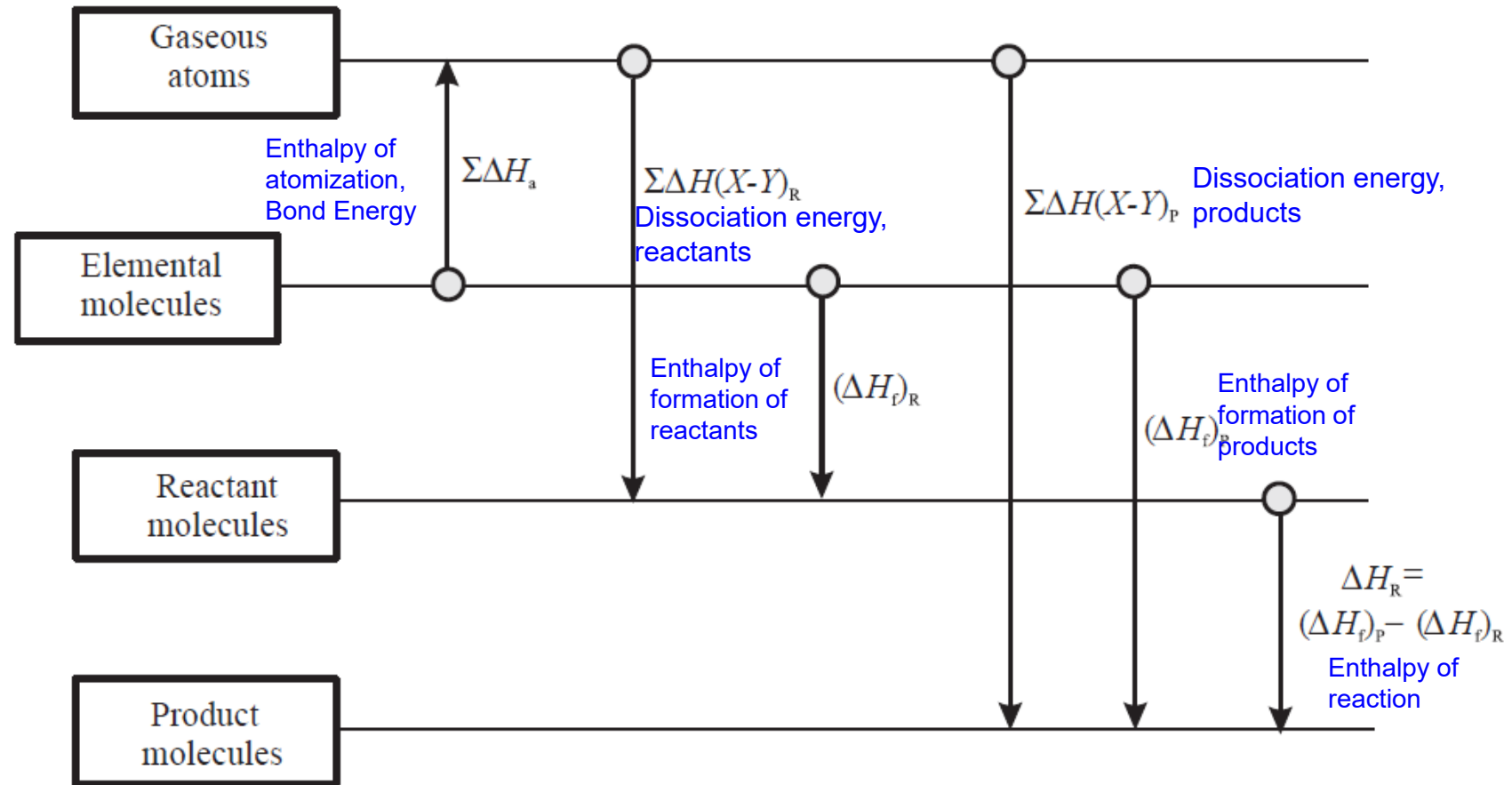
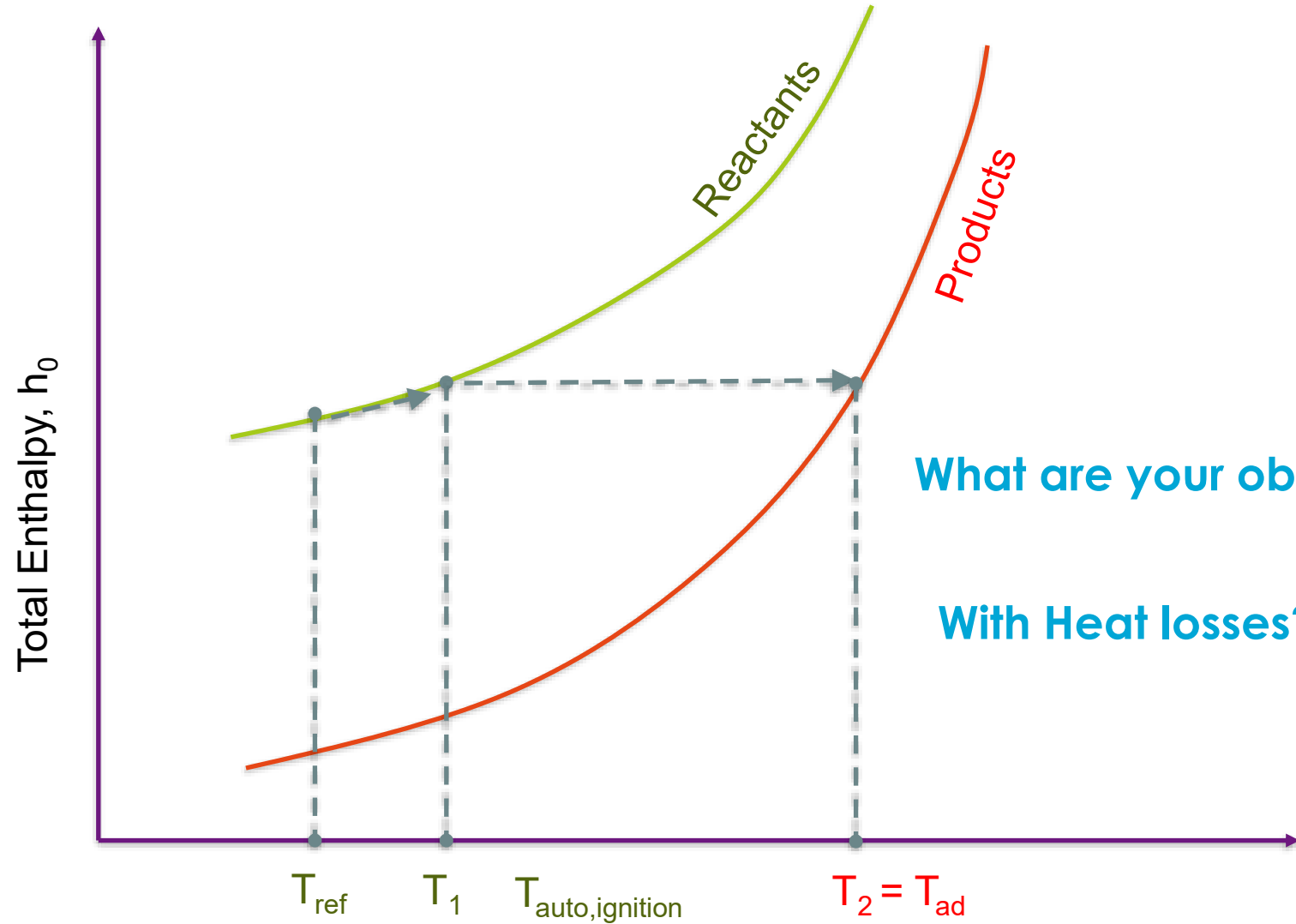


FIGURE 11.1

Relationship between enthalpies of formation and reaction.

Heat of reaction



What are your observations?

With Heat losses?

The heat of reaction

$$h_i(T) = h_{f,i}^0(T_{ref}) + \Delta h_{s,i}(T),$$

or

$$\Delta h_{s,i}(T) = h_i(T) - h_{f,i}^0(T_{ref}) = \int_{T_{ref}}^T c_{p,i} dT;$$

h_i - mass or molar specific enthalpy of species i .

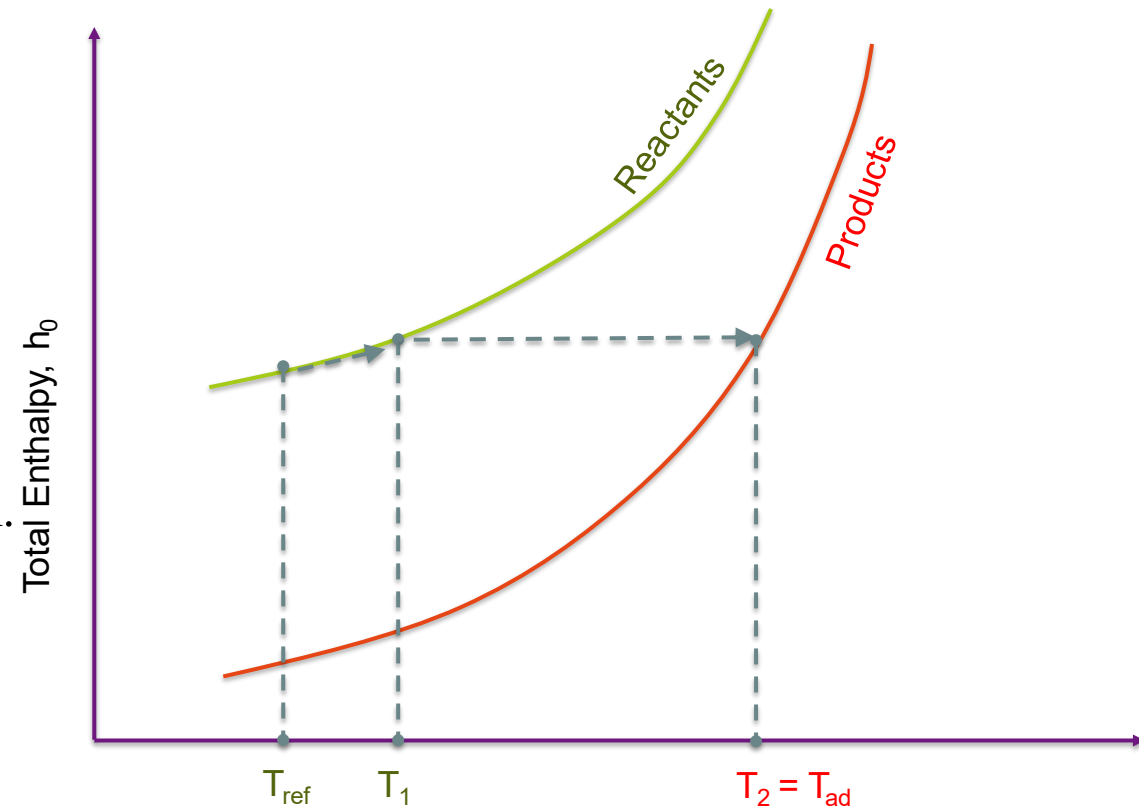
$h_{f,i}^0$ = enthalpy of formation of specie ' i '

$\Delta h_{s,i}$ = sensible enthalpy of formation of specie ' i '

h_i = enthalpy of specie ' i '

T_{ref} = reference temperature

$C_{p,i}$ = sp. heat @ const. pressure for specie ' i '



Enthalpy of formation

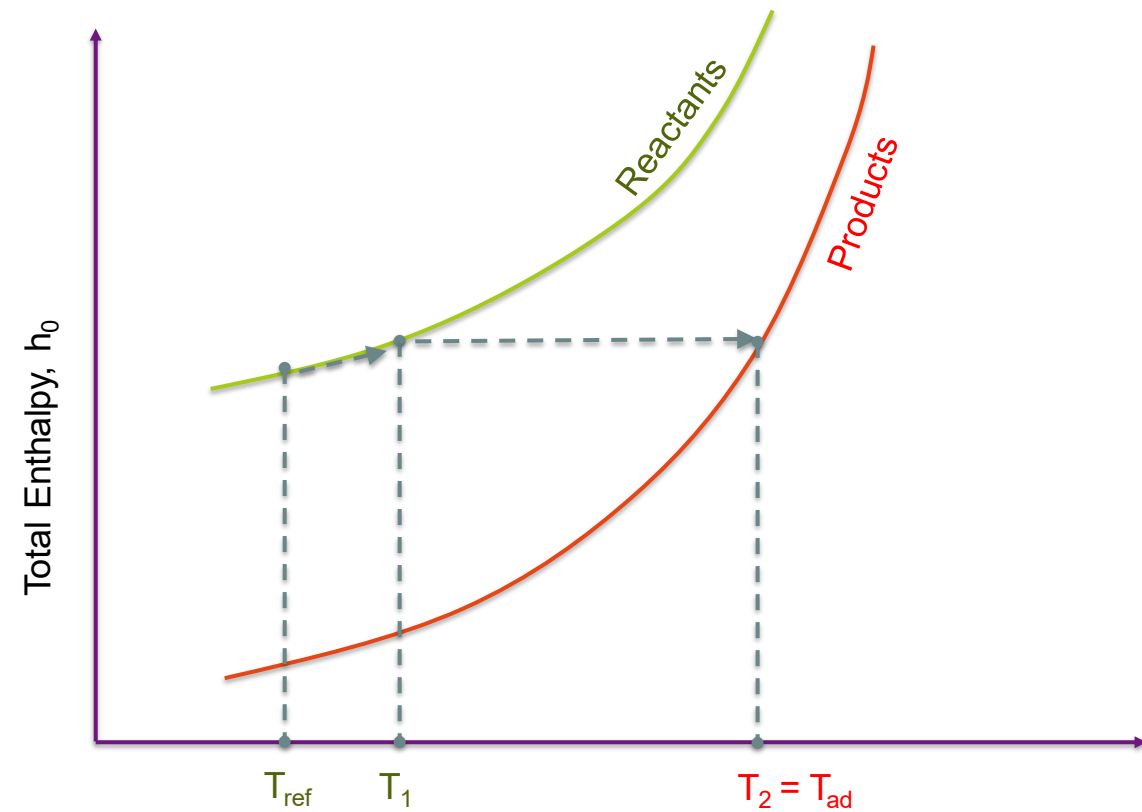
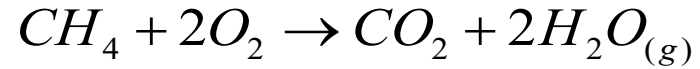
Table 1.2 Standard enthalpy of formation of key combustion species in the vapor state

Species	Name	$\bar{h}_{f,298}^0$ (kJ/mol)
H ₂ O	Water	-241.8
CO	Carbon monoxide	-110.5
CO ₂	Carbon dioxide	-393.5
CH ₄	Methane	-74.9
C ₂ H ₆	Ethane	-84.8
C ₃ H ₈	Propane	-104.7
C ₄ H ₁₀	Butane	-125.6
C ₈ H ₁₈	Octane	-208.4
C ₂ H ₄	Ethylene	52.5
C ₂ H ₂	Acetylene	226.7
CH ₃ OH	Methanol	-201.0
C ₆ H ₆	Benzene	82.9
H•	Hydrogen atom	218.0
O•	Oxygen atom	248.2
OH•	Hydroxyl radical	39.0

Calculate the HHV and LHV for Methane

Heat of reaction

$$(Q_P)_{25} = H_{\text{prod}} - H_{\text{react}}$$



$$(Q_P)_{\text{NTP}} = (\Delta H_f)_{\text{CO}_2} + 2(\Delta H_f)_{\text{H}_2\text{O}} - (\Delta H_f)_{\text{CH}_4} - 2(\Delta H_f)_{\text{O}_2}$$

$$(Q_P)_{\text{NTP}} = (-393522) + (-241827) - (-74897) - 2(0)$$

$$(Q_P)_{\text{NTP}} = -802279 \text{ kJ/k mol} \quad \text{Or } (Q_P)_{\text{NTP}} = -50142 \text{ kJ/kg}$$

Energy in various bonds

Table 11.1 Some Atomisation, Dissociation and Resonance Energies (Based on 25 °C)

Bond Atomisation (ΔH_a)	Energy MJ/kmol	Bond Dissociation ($\Delta H(X-Y)$)	Energy MJ/kmol	Resonance	Energy MJ/kmol
H-H	435.4	C-H	414.5	Benzene: C ₆ H ₆	150.4
C (graphite)	717.2	N-H	359.5	Naphthalene: C ₁₀ H ₈	255.4
(O = O) _{O₂}	498.2	O-H	428.7	Carbon dioxide: CO ₂	137.9
N≡N	946.2	H-OH	497.5	-COOH group	117.0
		C-O	351.7		
		C=O	698.1		
		C-C	347.5		
		C=C	615.5		
		C≡C	812.2		

Note: These values have been taken from different sources and may not be exactly compatible.

AE4-238 Aero Engine Technology

Combustion-III: Bond energy

Prof. Dr. Arvind Gangoli Rao

Contents of this lecture

- Energy in bonds
- Hess Law
- Properties of hydrocarbon combustion

Energy in various bonds

Table 11.1 Some Atomisation, Dissociation and Resonance Energies (Based on 25 °C)

Bond Atomisation (ΔH_a)	Energy MJ/kmol	Bond Dissociation ($\Delta H(X-Y)$)	Energy MJ/kmol	Resonance	Energy MJ/kmol
H–H	435.4	C–H	414.5	Benzene: C ₆ H ₆	150.4
C (graphite)	717.2	N–H	359.5	Naphthalene: C ₁₀ H ₈	255.4
(O = O) _{O₂}	498.2	O–H	428.7	Carbon dioxide: CO ₂	137.9
N≡N	946.2	H–OH	497.5	–COOH group	117.0
		C–O	351.7		
		C=O	698.1		
		C–C	347.5		
		C=C	615.5		
		C≡C	812.2		

Note: These values have been taken from different sources and may not be exactly compatible.

Bond energy

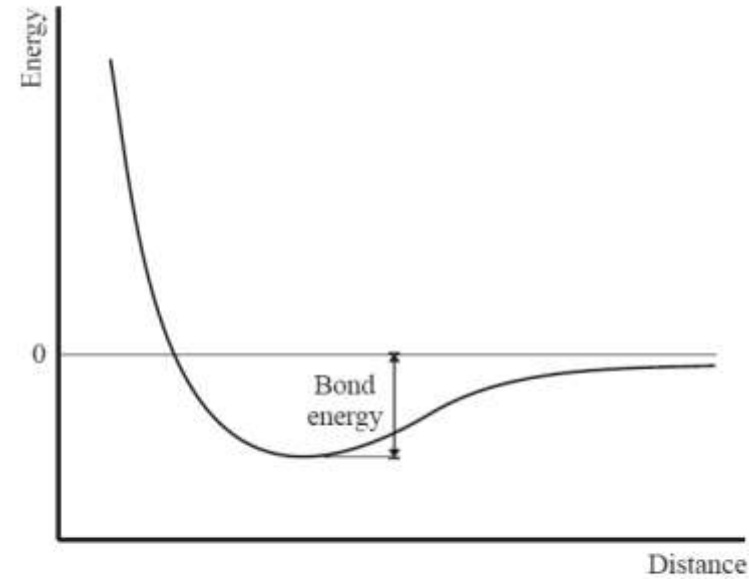


FIGURE 11.2

Variation of bond energy with distance between atoms.

- Energy required to separate two atoms.
- The bond energy is defined as the min. potential energy relative to when the atoms are separated at infinity
- The minimum point indicates equilibrium

Hess Law

If a reaction (isobaric or isochoric) is carried out in stages, the algebraic sum of the heat evolved in separate stages is equal to the total amount of heat released when the reaction occurs directly.

Which means that the heat liberated in a reaction is independent of the path of the reaction between the initial and the final states.

CH₄ reaction mechanism

The GRI 3.0 Reaction mechanism has around 53 species and 325 reactions

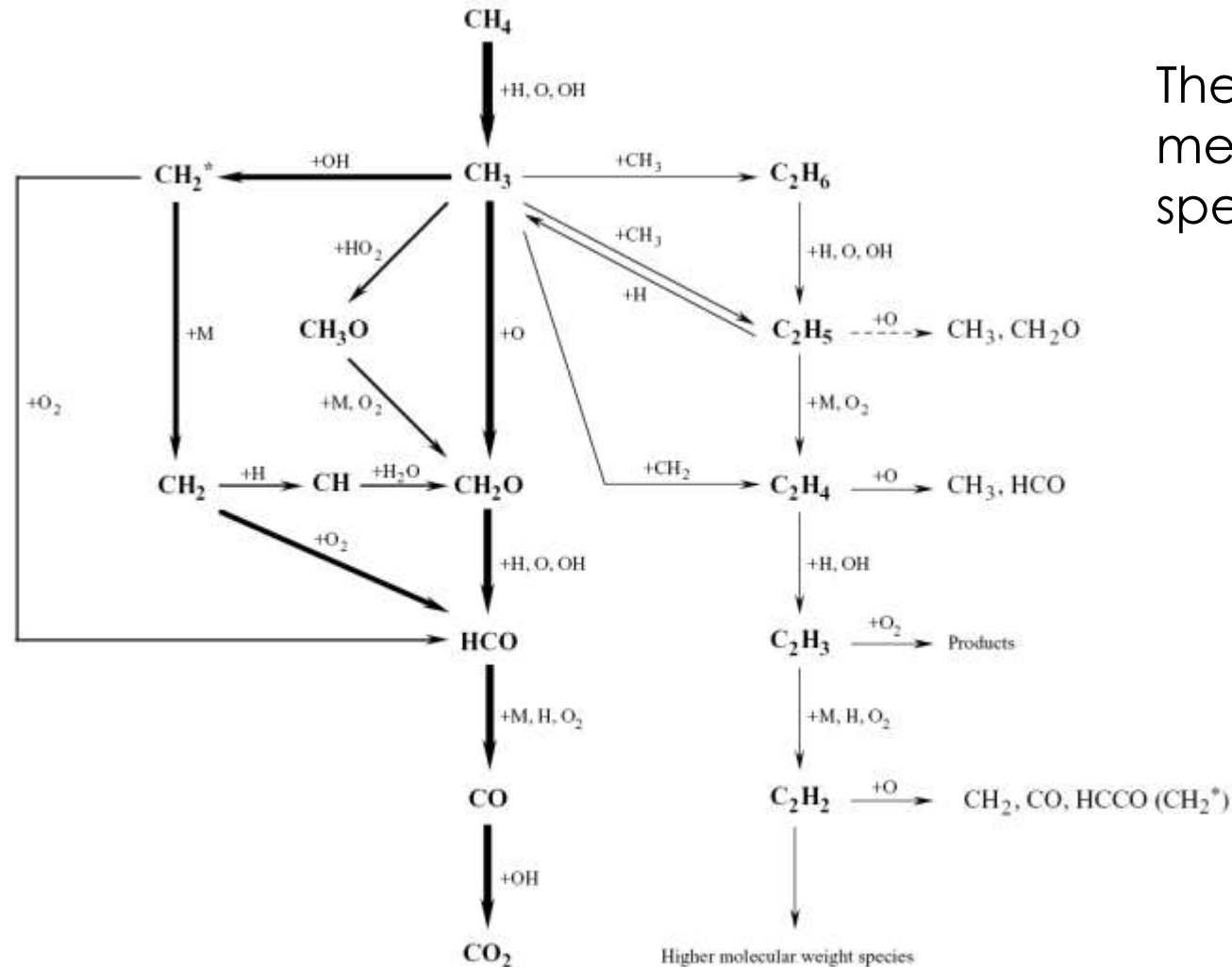


Figure 3.3.1. Reaction pathways in methane flames (adapted from Warnatz 1981).

Heat of reaction

Fuel	Chemical Formula	Molecular Weight (m_w) (integrals)	Enthalpy of Formation, Δh_f (kJ/kmol)	Higher Enthalpy of Reaction, Δh_R (MJ/kmol)	Lower Enthalpy of Reaction, Δh_R		Stoichiometric Air–Fuel Ratio, ϵ_{st}
					(kJ/kg)	(MJ/kmol)	
<i>Alkanes</i>	C_nH_{2n+2}						
Methane	CH ₄	16	−74,898	−891	−50,047	−802.9	17.16
Ethane	C ₂ H ₆	30	−84,724	−1561	−47,519	−1428.8	16.02
Propane	C ₃ H ₈	44	−103,916	−2221	−46,387	−2045.4	15.60
Butane	C ₄ H ₁₀	58	−124,817	−2880	−45,771	−2660.2	15.38
Pentane	C ₅ H ₁₂	72	−146,538	−3538	−45,384	−3274.3	15.25
Hexane	C ₆ H ₁₄	86	−167,305	−4197	−45,134	−3889.3	15.16
Heptane	C ₇ H ₁₆	100	−187,946	−4856	−44,955	−4504.4	15.10
Octane	C ₈ H ₁₈	114	−208,586	−5515	−44,820	−5119.5	15.05
Cetane	C ₁₆ H ₃₄	226	−454.3	−10,711	−44,000	−9963.4	14.88
<i>Aromatics</i>							
Benzene	C ₆ H ₆	78	+82,982	−3304	−40,605	−3172	13.20
Toluene	C ₇ H ₈	92	+50,032	−3950	−40,967	−3774	13.43
Xylene	C ₈ H ₁₀	106	+18,059	−4598	−41,276	−4378	13.60
Naphthalene	C ₁₀ H ₈	128	+150,934	−5233	−39,455	−5057	12.87
<i>Alcohols</i>	$C_nH_{2n+1}OH$	$C_nH_{2n+2}O$					
Methanol	CH ₃ OH	32	−201,301	−764	−21,114	−676	6.44
Ethanol	C ₂ H ₅ OH	46	−235,466	−1410	−27,742	−1278	8.85
Propanol	C ₃ H ₇ OH	60	−235,107	−2069	−31,504	−1893	10.30

Some observations

- The enthalpies of reaction for many hydrocarbon fuels on mass basis is similar, between 40-50 MJ/kg
- The stoichiometric air-fuel ratio for many hydrocarbons lie between 13:1 to 17:1
- Some fuels have positive formation of enthalpies
- All fuels have negative enthalpies of reaction
- The enthalpies of reaction for alcohols is less than non-oxygenated fuels (as oxygen does not provide any energy of reaction)
- Commonly used fossil fuels are a mixture of several hydrocarbon compounds (discussed later)

AE4-238 Aero Engine Technology
Combustion-IV: Types of combustion

Prof. Dr. Arvind Gangoli Rao

Contents of this lecture

- Adiabatic flame temperature
- Effect of pressure and temperature
- Types of combustion

Adiabatic flame temperature

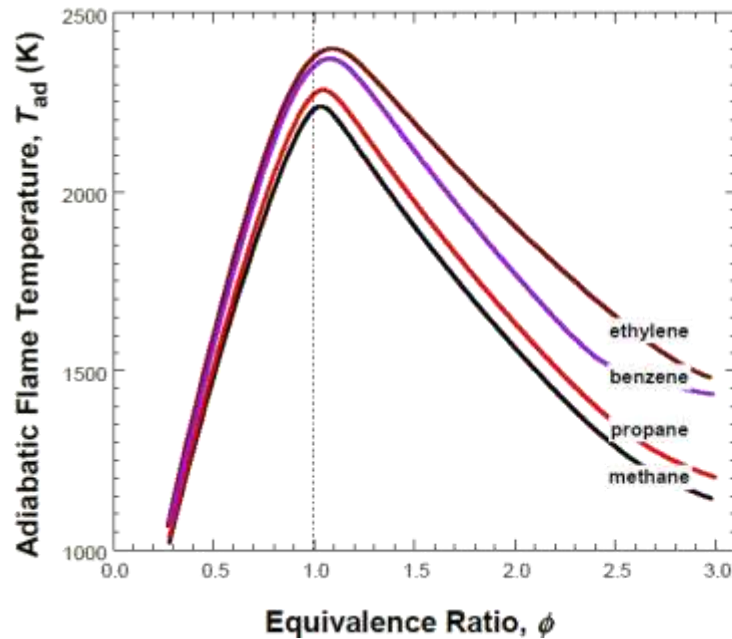


Figure 1.2. Adiabatic flame temperature of methane, propane, ethylene and benzene at 1 atm pressure.

Hai Wang, "combustion chemistry", 2012

$$H_{prod}(T_{ad}) - H_{reac}(T_0) = 0$$

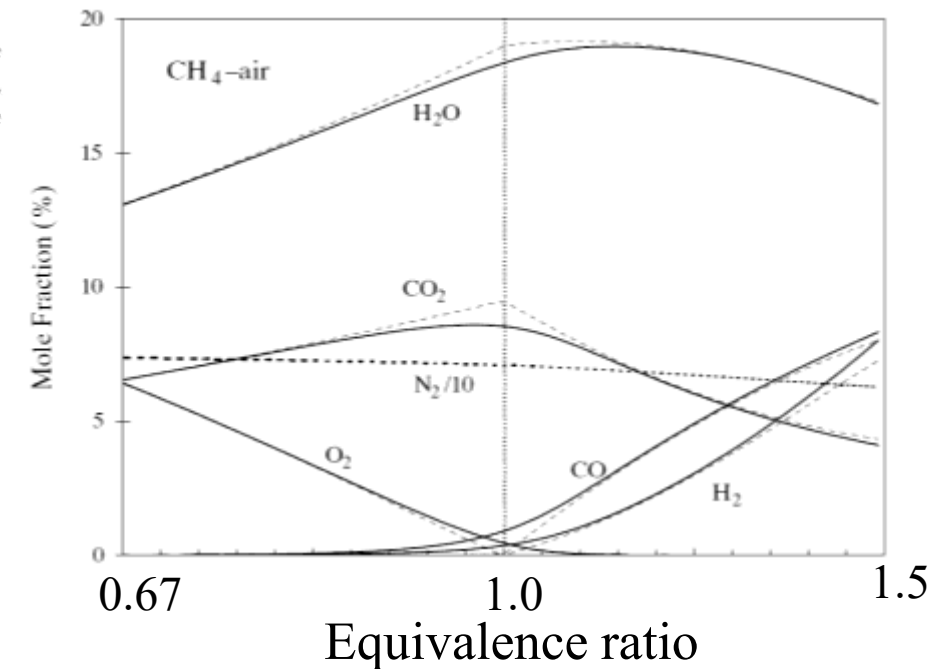
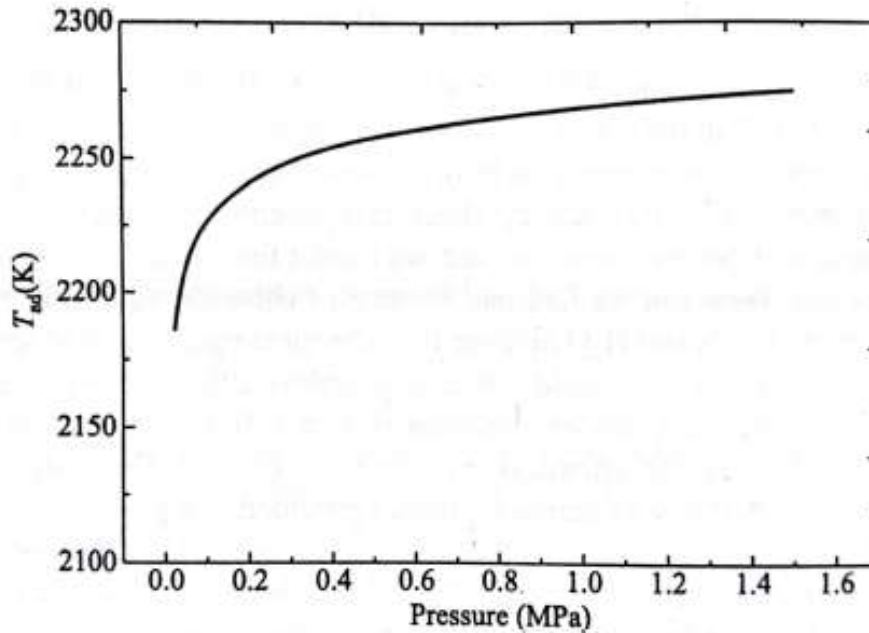


Figure 1.4.7. Concentrations of major product species as functions of Φ for methane–air mixtures at STP (– – Frozen equilibrium; — Dissociation equilibrium).

Effect of Pressure



Effect of pressure on adiabatic flame temperature

Gas turbine combustion: Lefebvre & Ballal

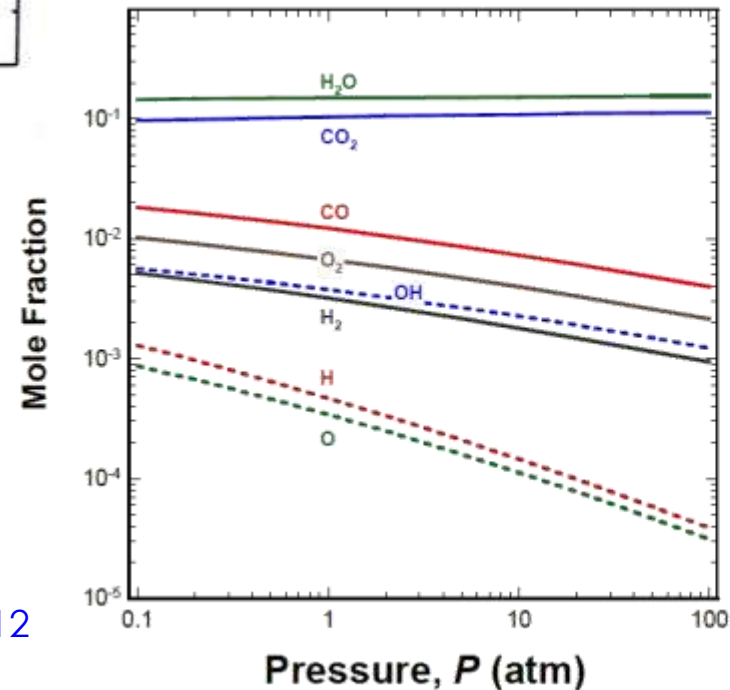


Figure 1.4. Mole fractions of equilibrium products computed for propane combustion in air at the unity equivalence ratio.

Effect of Pressure

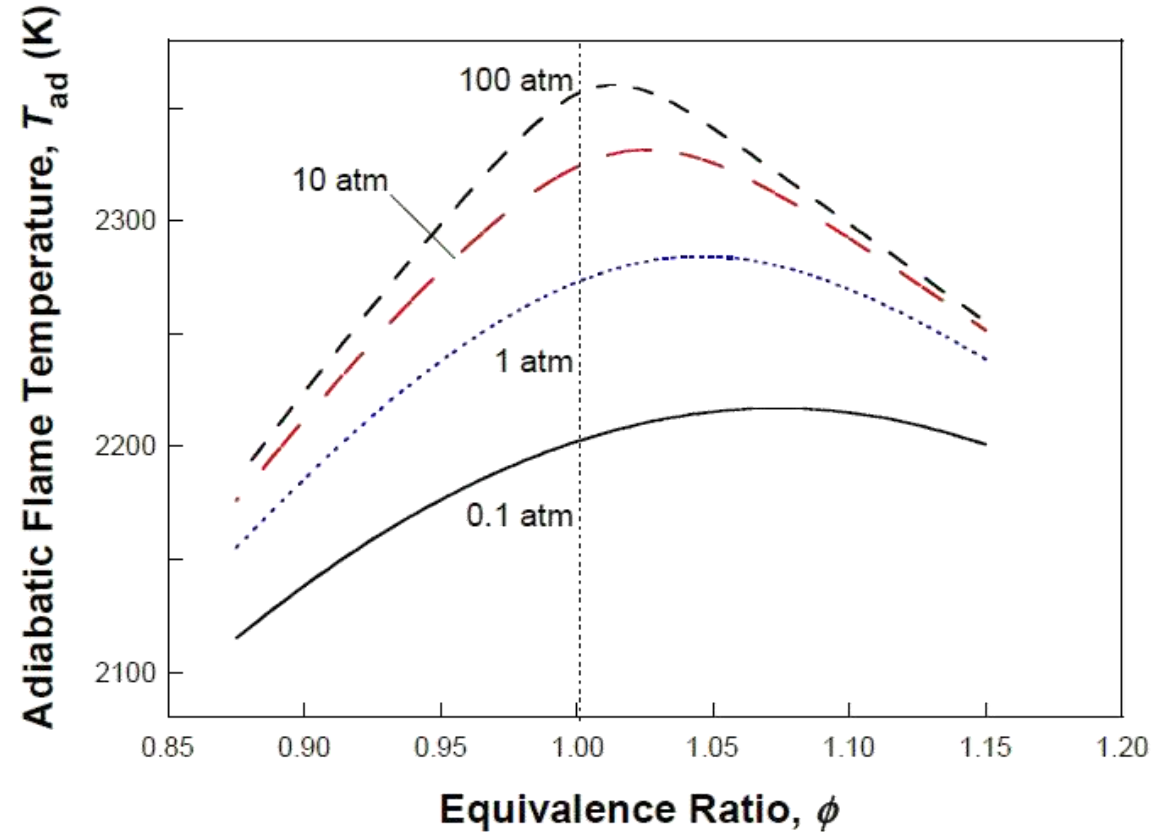
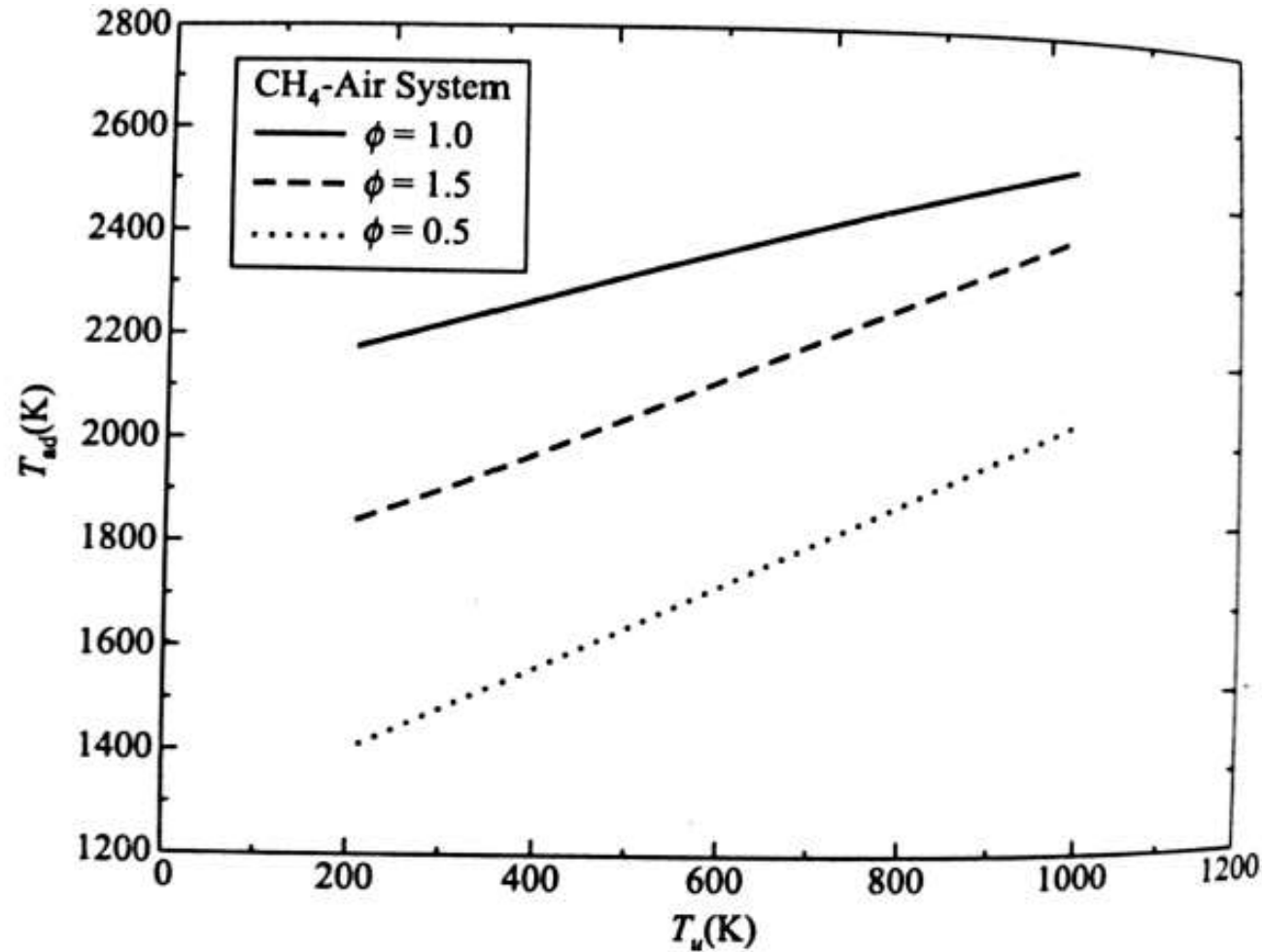


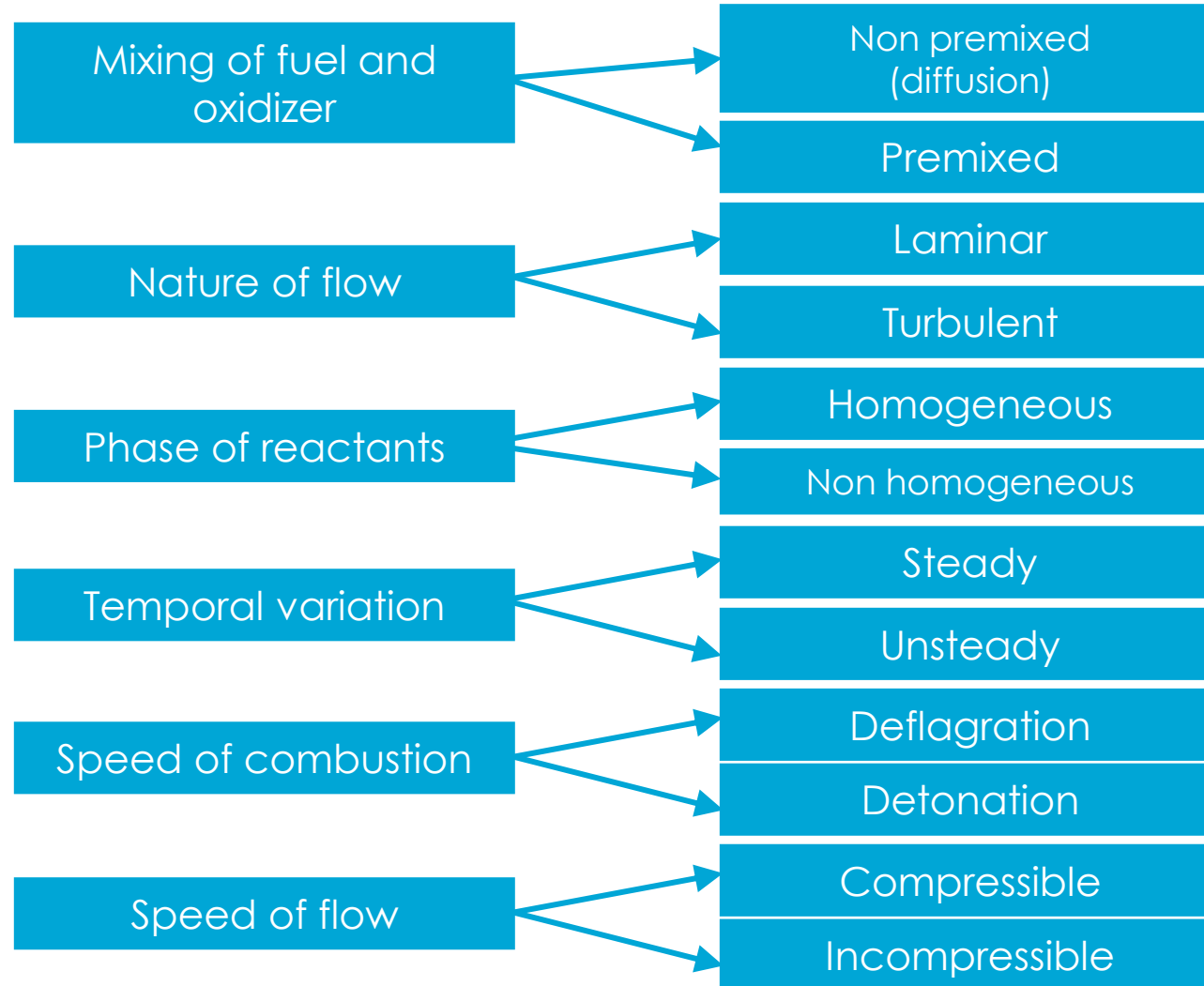
Figure 1.3. Adiabatic flame temperature of propane combustion in air at several pressures.

Effect of Inlet Temperature

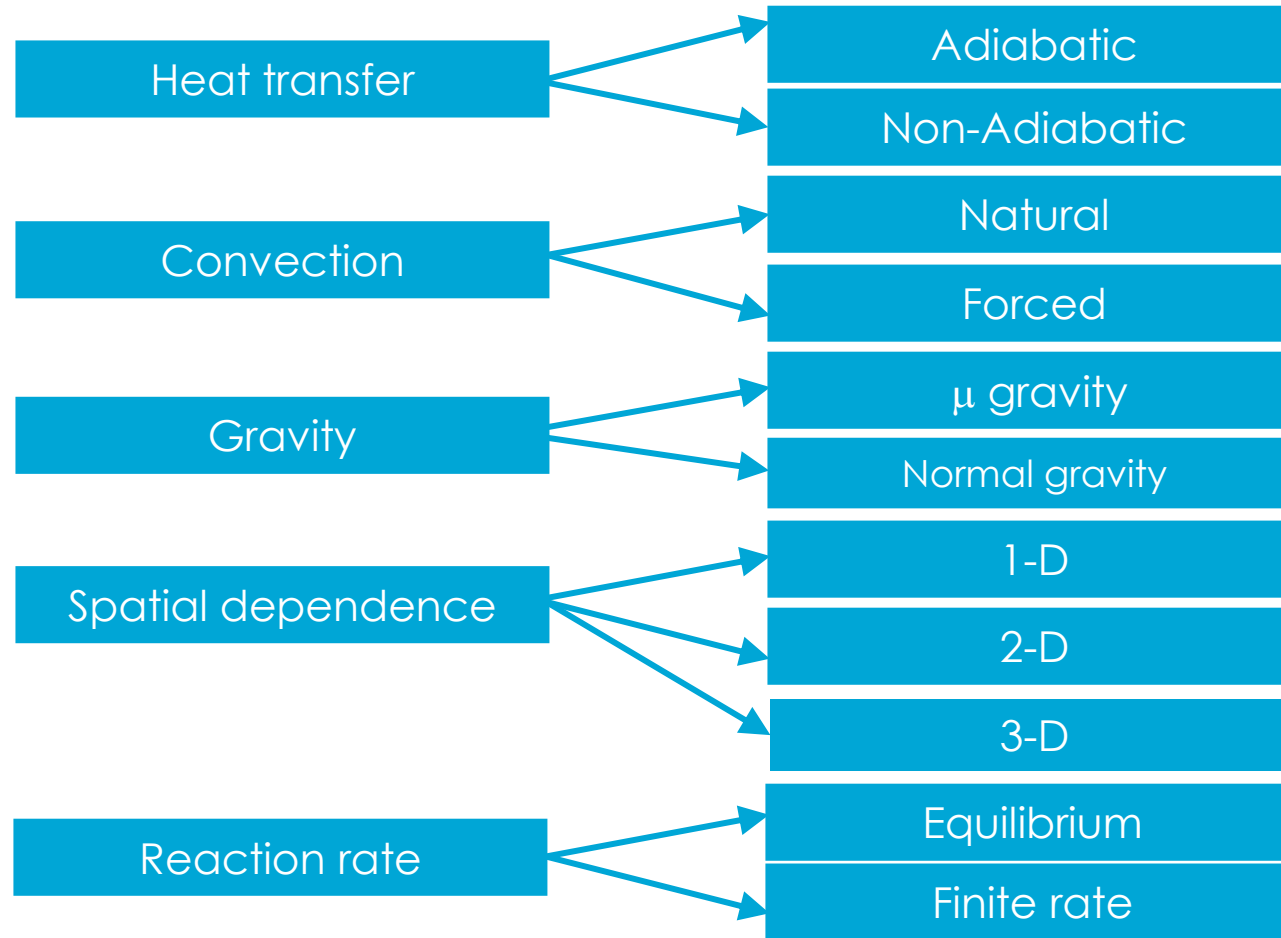


Effect of inlet air temperature and equivalence ratio on adiabatic flame temperature

Types of Combustion



Types of Combustion



AE4-238 Aero Engine Technology

Aero Engine Combustor-V: Combustor fundamentals

Prof. Dr. Arvind Gangoli Rao

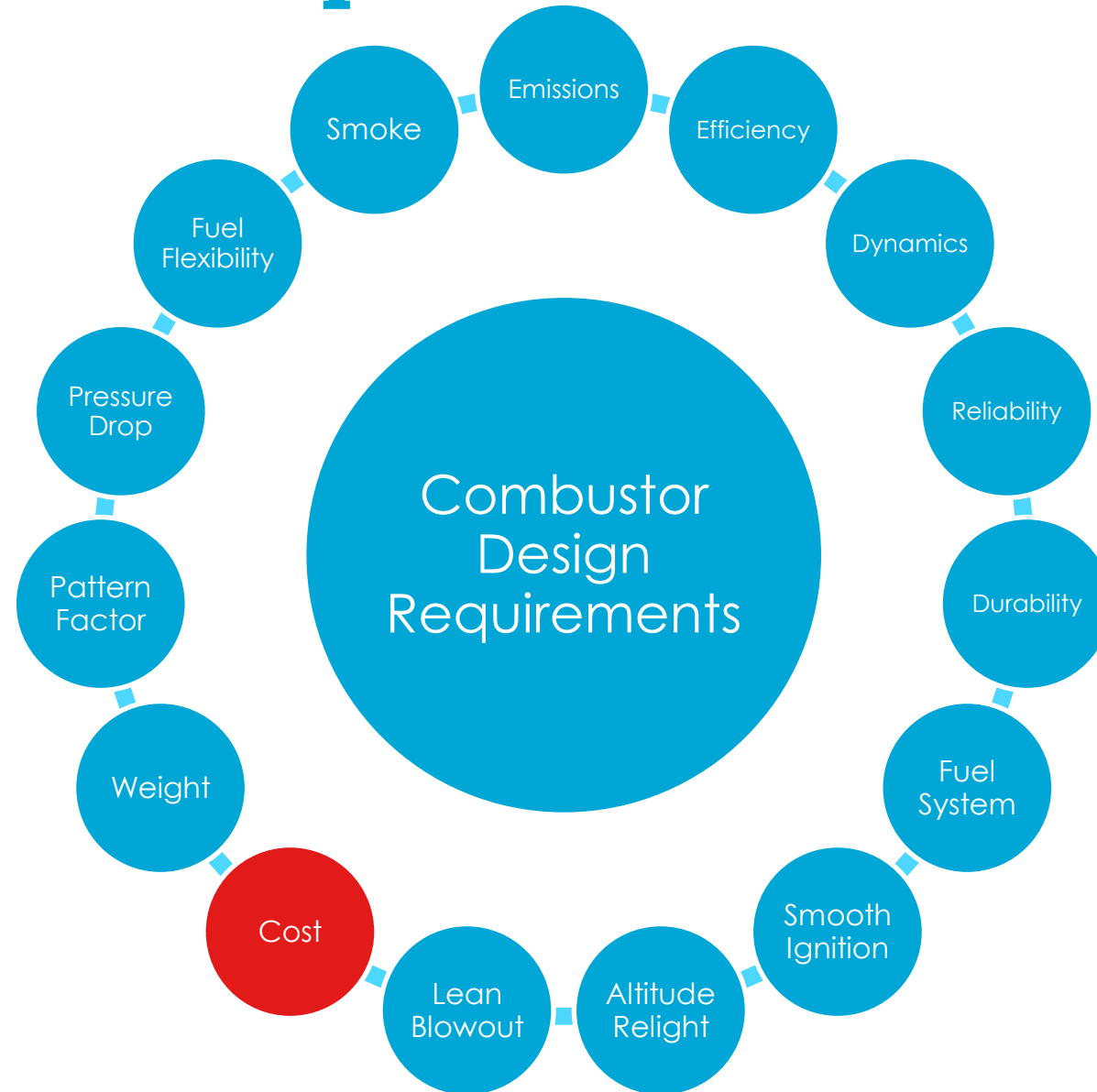
Contents of this lecture

- Basics of gas turbine combustor
- Zones within a combustor



What are the important design criteria for gas turbine combustor

Combustor requirements



Gas Turbine Combustor

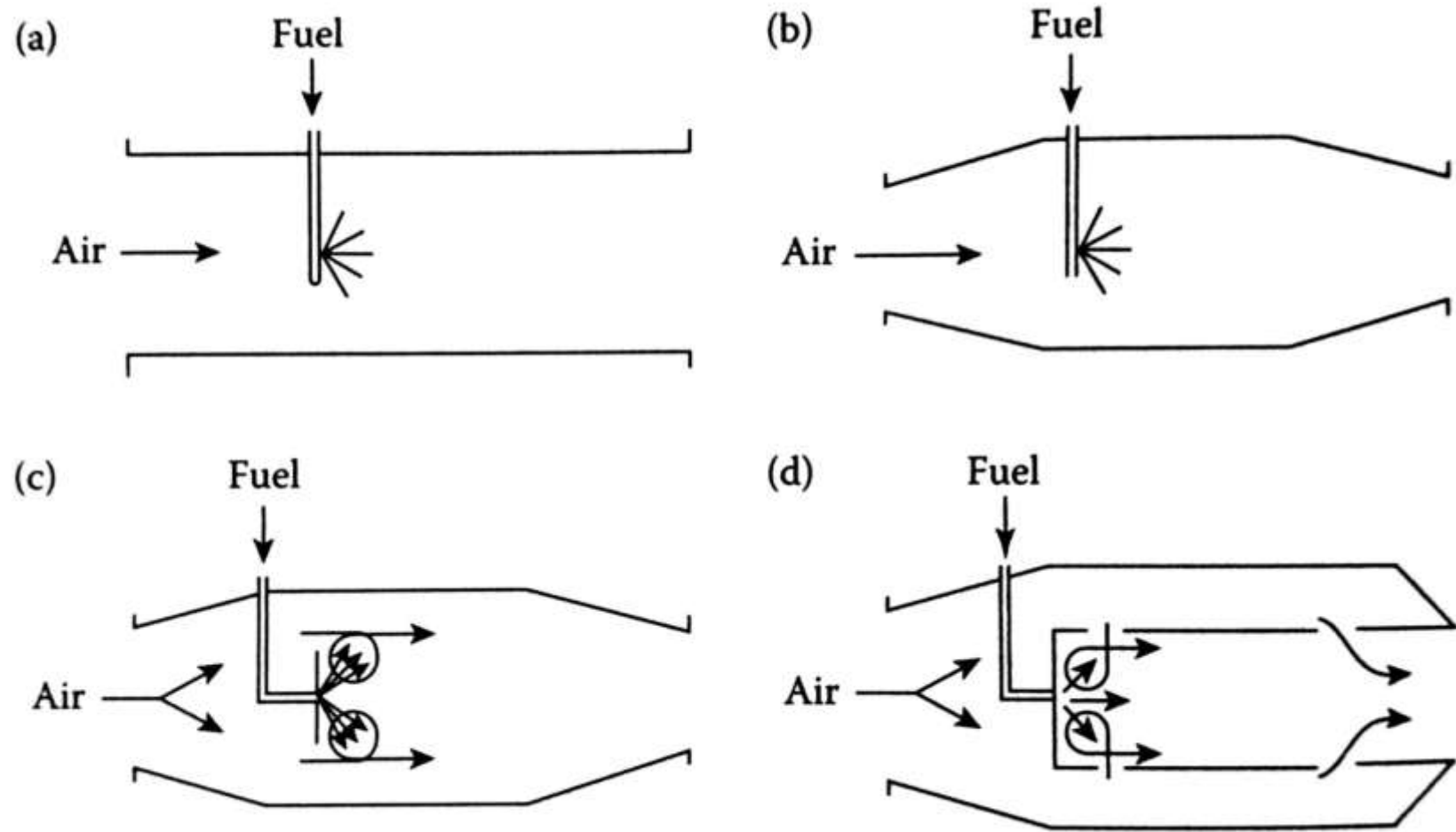
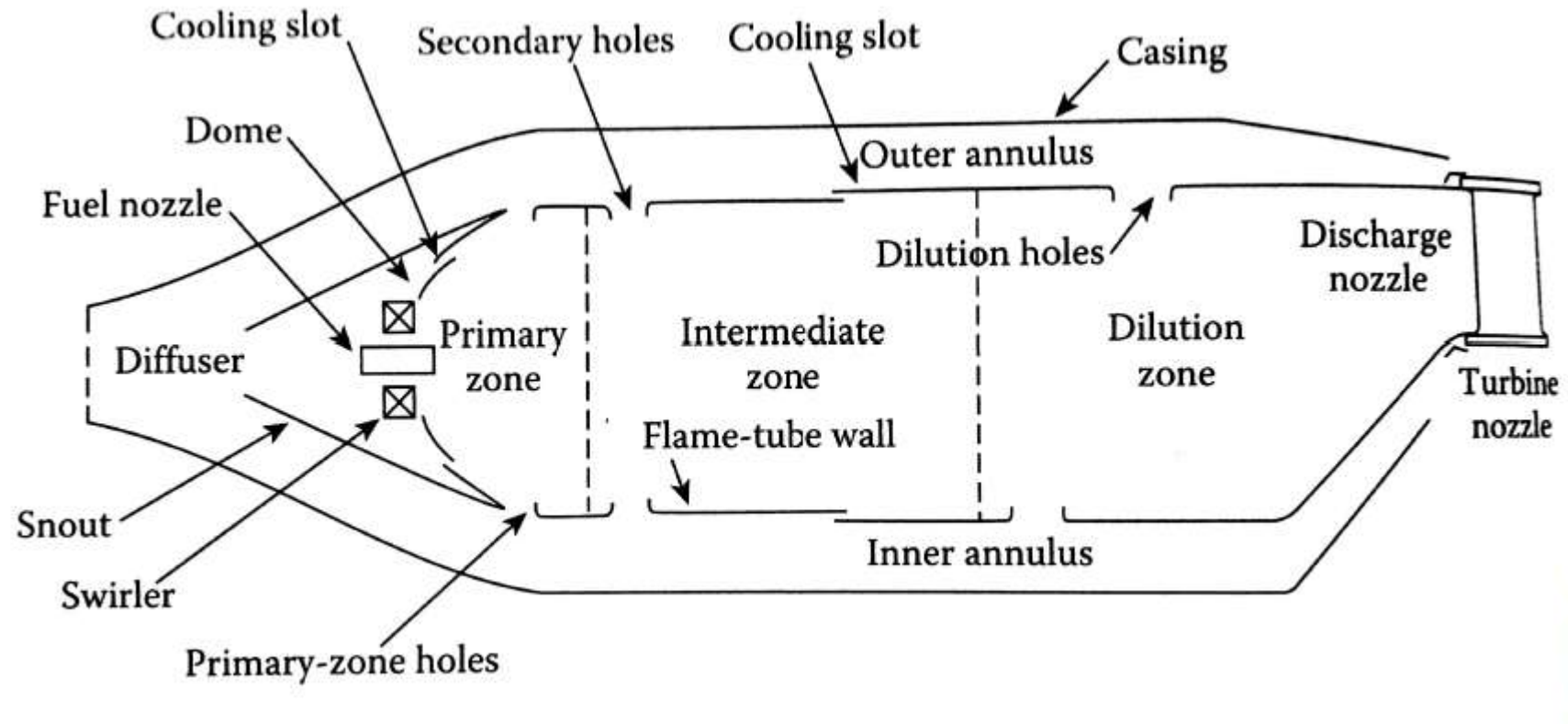


FIGURE 1.1

Source: Gas Turbine Combustion, Lefebvre & Ballal

A conventional combustor



Can type combustion chamber

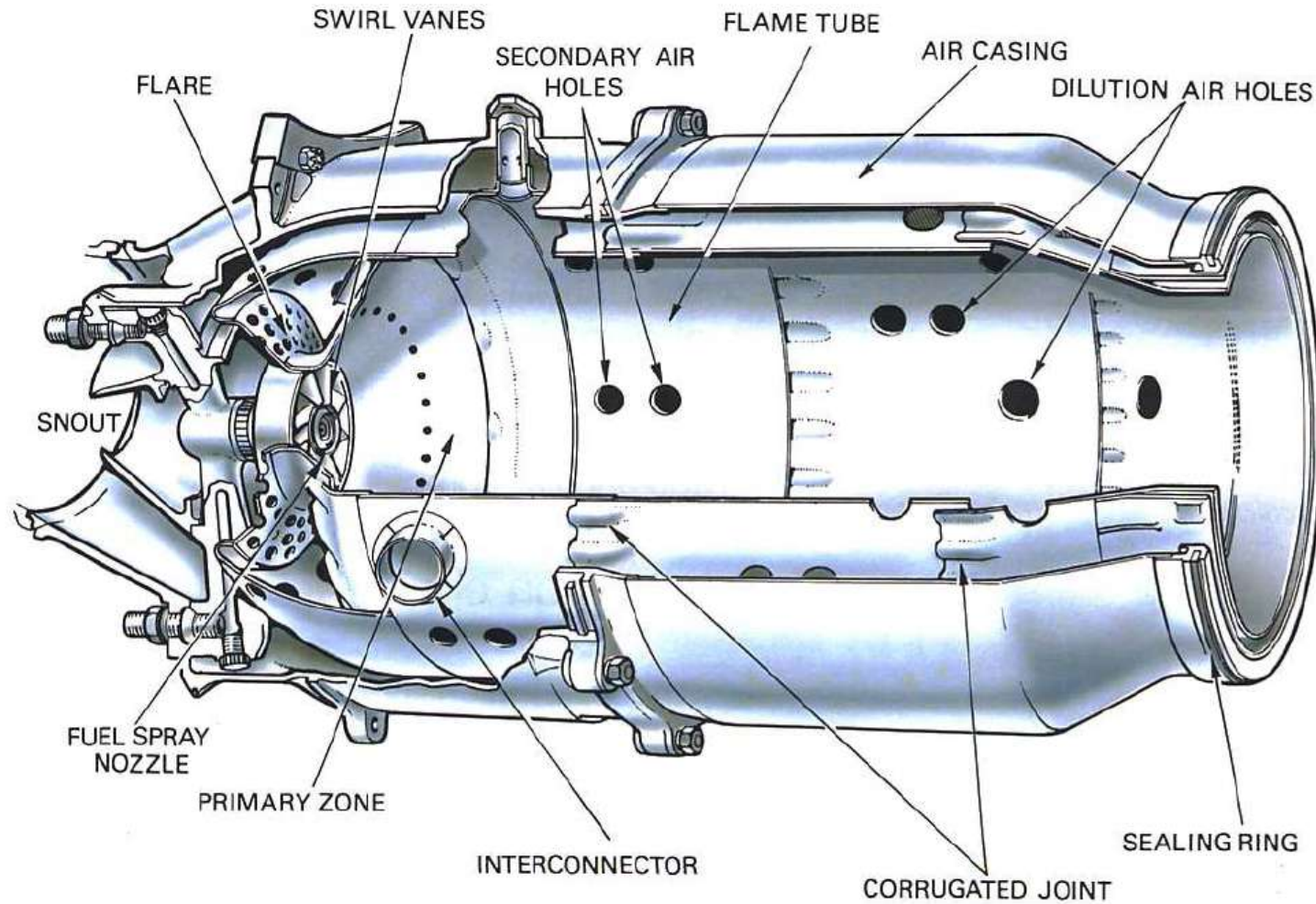
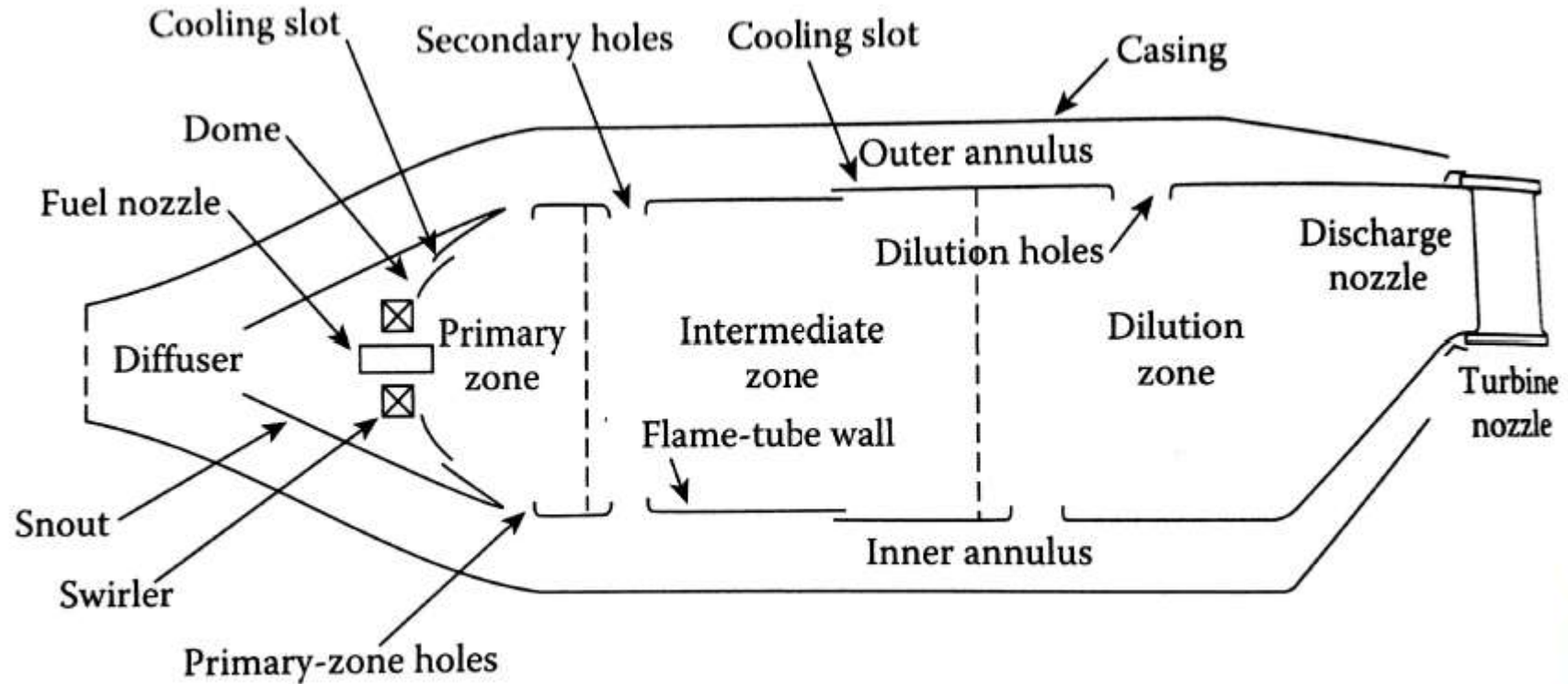


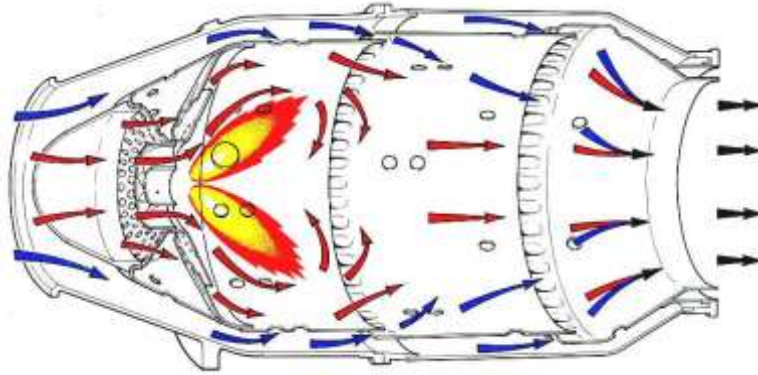
Fig. 4-1 An early combustion chamber.

Courtesy: Rolls Royce

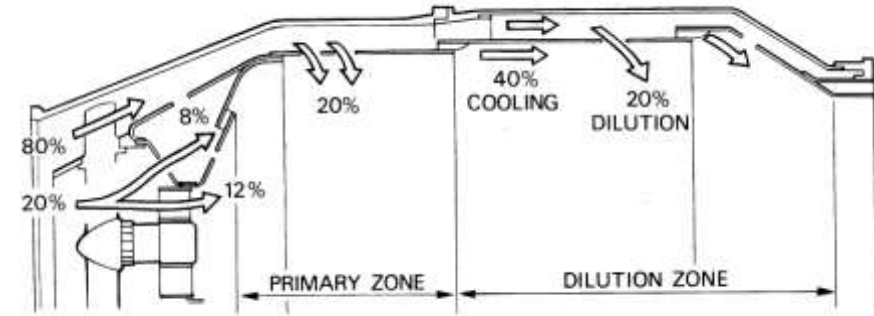
Aero Engine Combustor (conventional)



Conventional Combustor



General air flow pattern



Typical repartition of the air flow

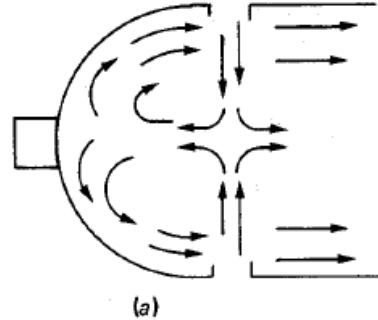
Primary Zone:-

- Anchors the flame
- Provides sufficient time, temperature and turbulence

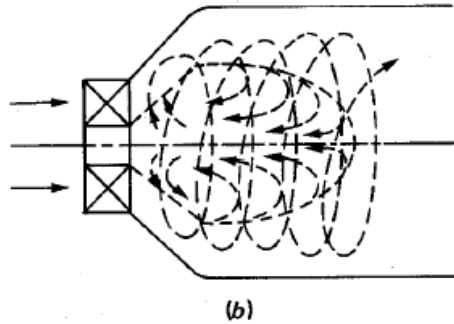
Three Step Process

- Endothermic dissociation of the fuel
- Exothermic formation of CO & H₂O → more than 80% of energy
- Exothermic conversion of CO to CO₂ → 75 % of the combustor length

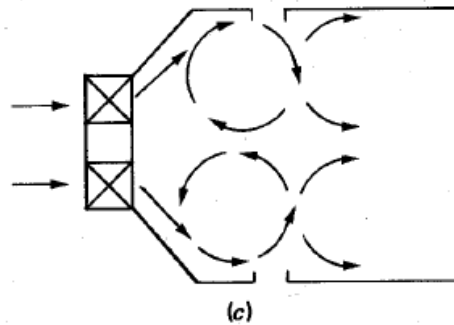
Primary zone flame stabilization



Stabilization by opposing jets



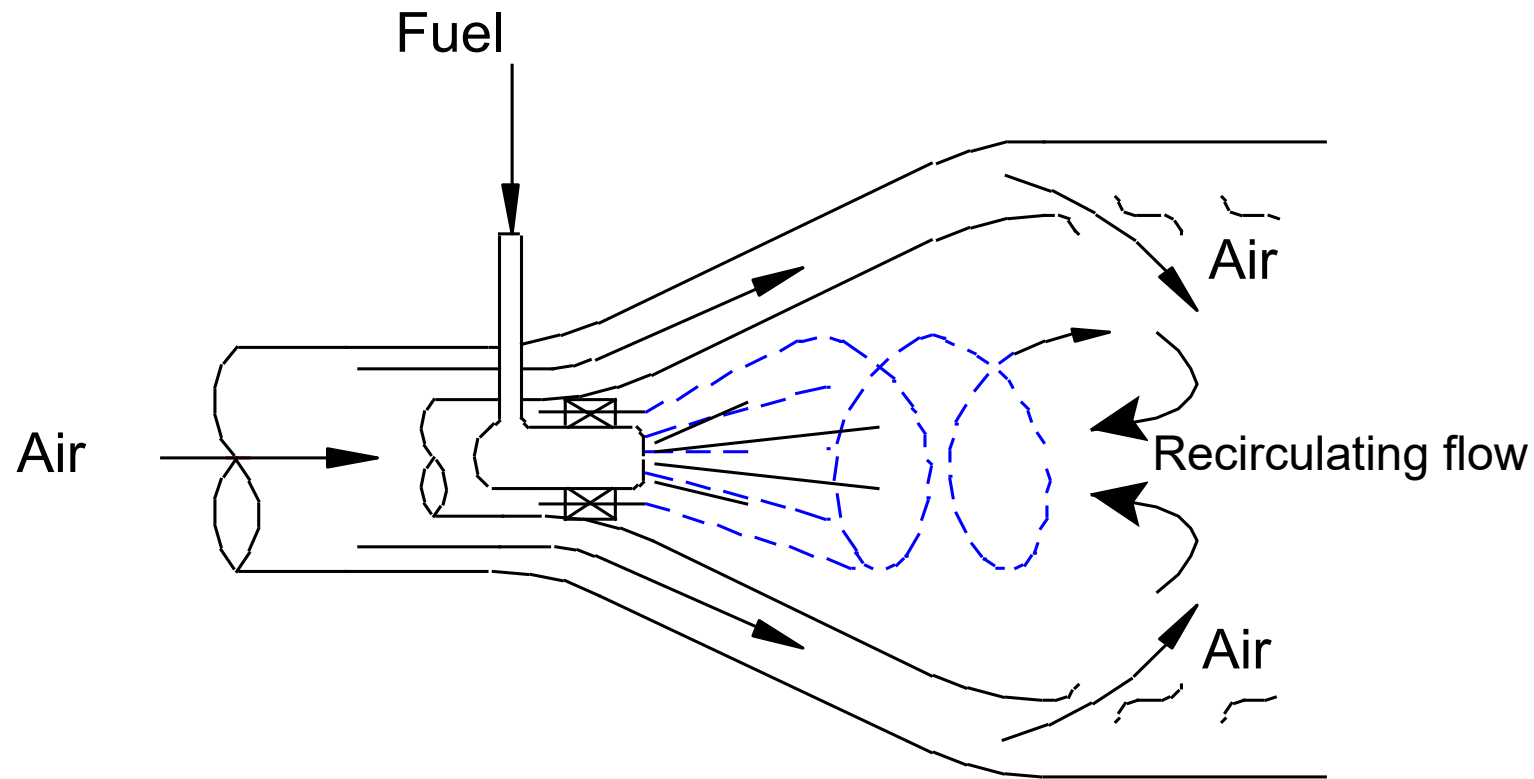
Stabilization by swirling jets



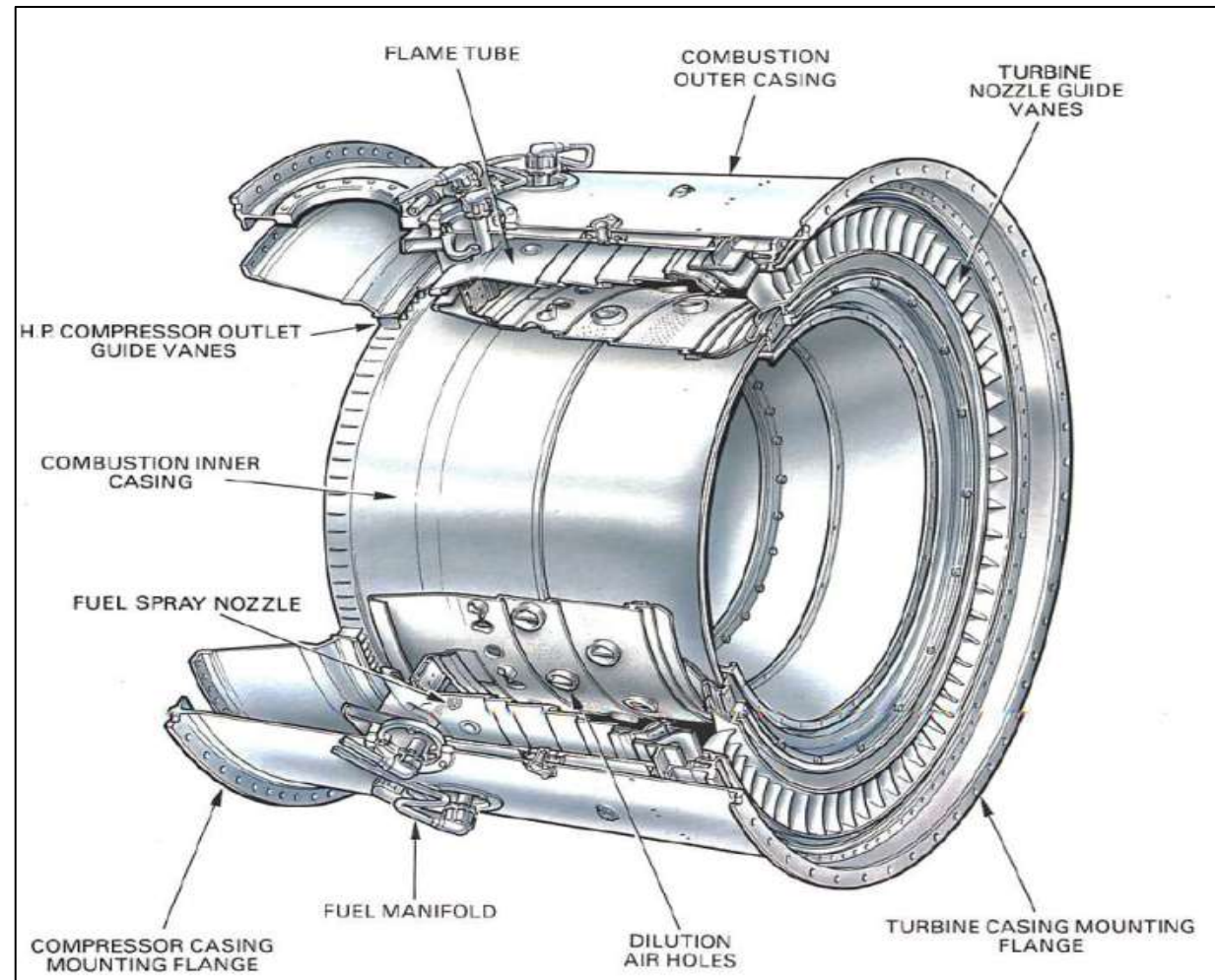
Stabilization by swirling flow assisted by secondary injection

Primary Zone

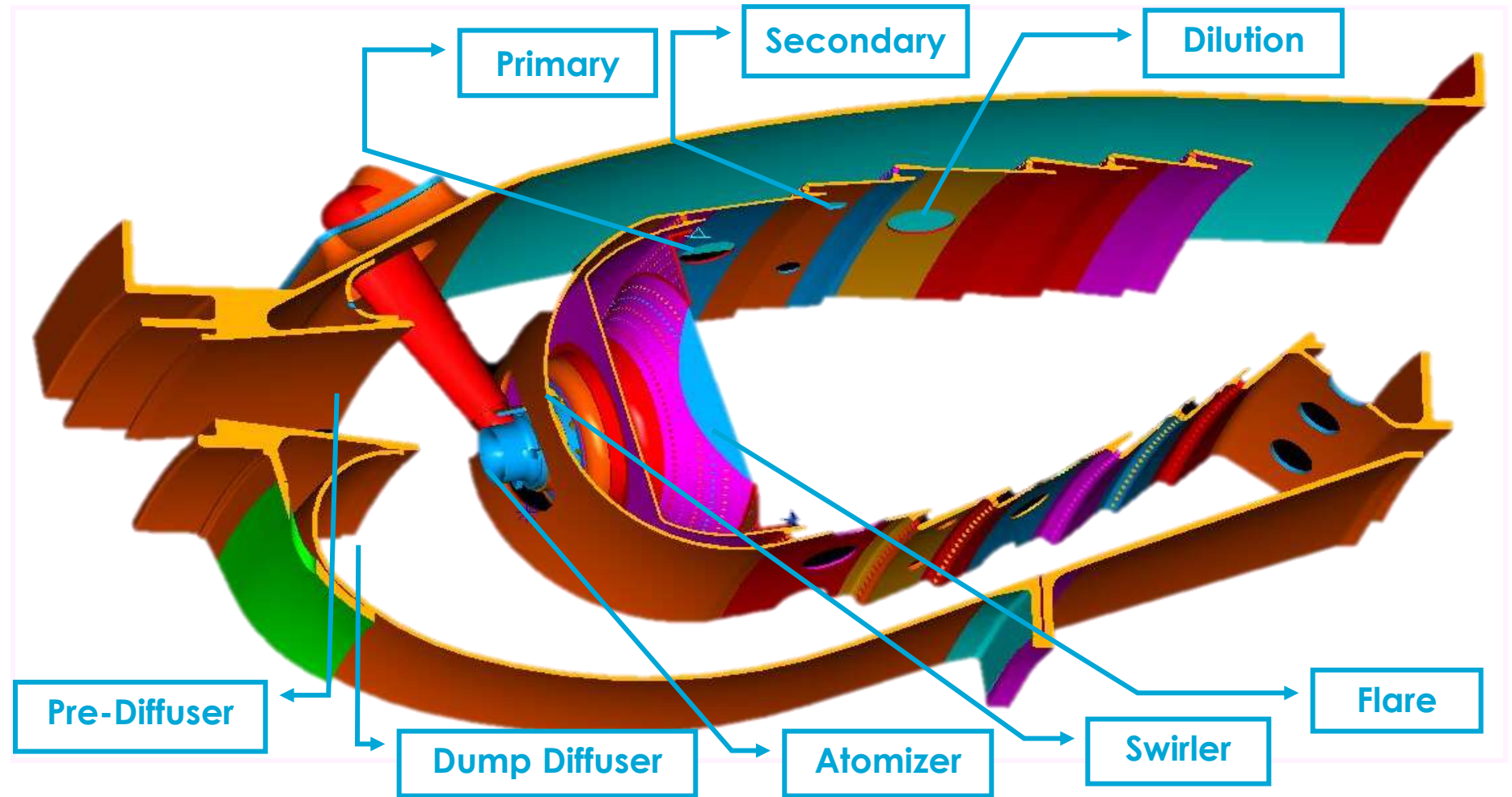
Flame stabilization in a conventional combustor



Courtesy: Rolls Royce



Annular combustor



Conventional combustor

1. Primary Zone

- Anchors Flame
- Provides sufficient time, mixing, temperature for “complete” oxidation of fuel
- Equivalence ratio near $\phi=0.8-1.0$

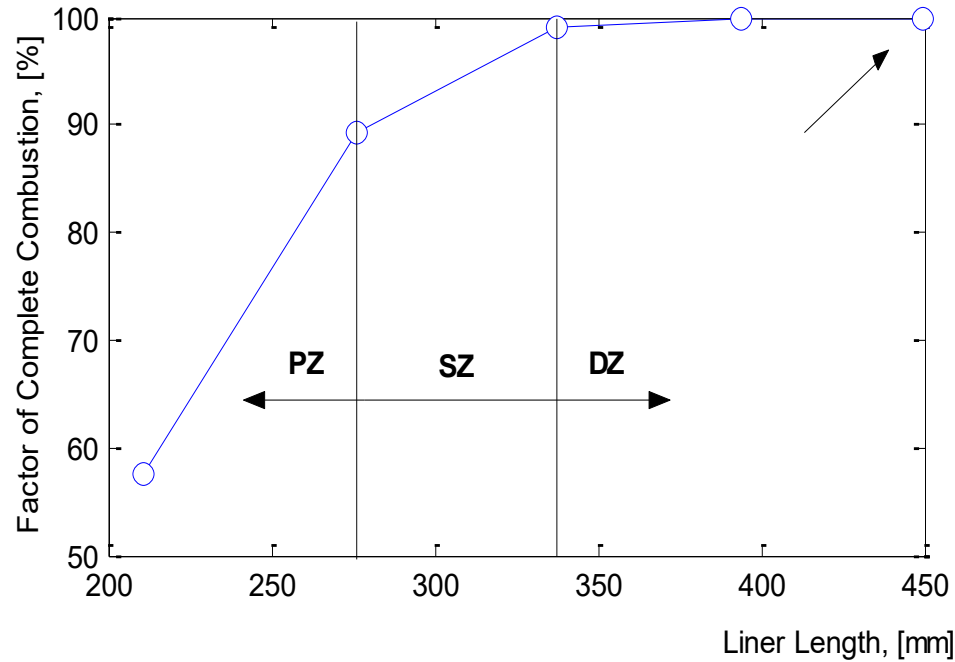
2. Intermediate (Secondary Zone)

- **Low altitude** operation (higher pressures in combustor)
 - Recover dissociation losses (primarily $\text{CO} \rightarrow \text{CO}_2$) and Soot Oxidation
 - Complete burning of anything left over from primary due to poor mixing
- **High altitude** operation (lower pressures in combustor)
 - Low pressure implies slower rate of reaction in primary zone
 - Serves basically as an extension of primary zone (increased t_{res})
- $L/D \sim 0.7$

3. Dilution Zone

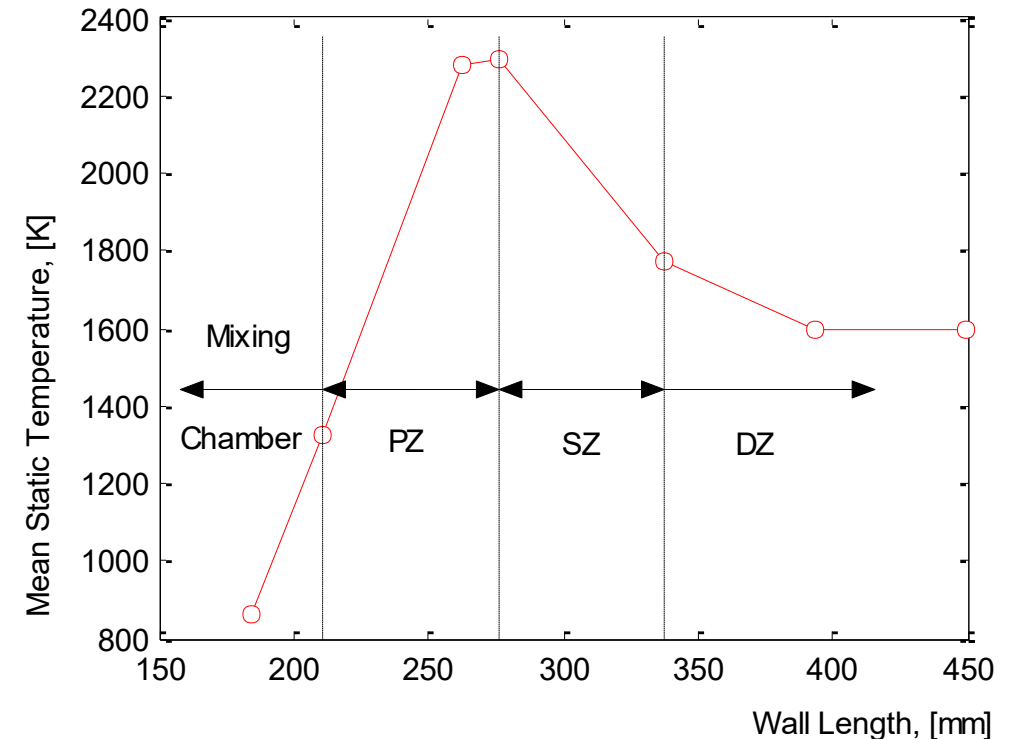
- Mixing air to lower temperature to acceptable value for turbine
- Tailor temperature profile (low at root and tip, high in middle)
- Uses about 20-40% of total ingested core mass flow
- $L/D \sim 1.5-1.8$

Temperature variation



Factor of complete combustion in a generic combustor at full power

Temperature distribution within a conventional combustor



S. A. Shakariyants, "Generic Methods for Aero-Engine Exhaust Emission Prediction", Delft University of Technology, 2008, ISBN 978-90-9023346-8.

AE4-238 Aero Engine Technology

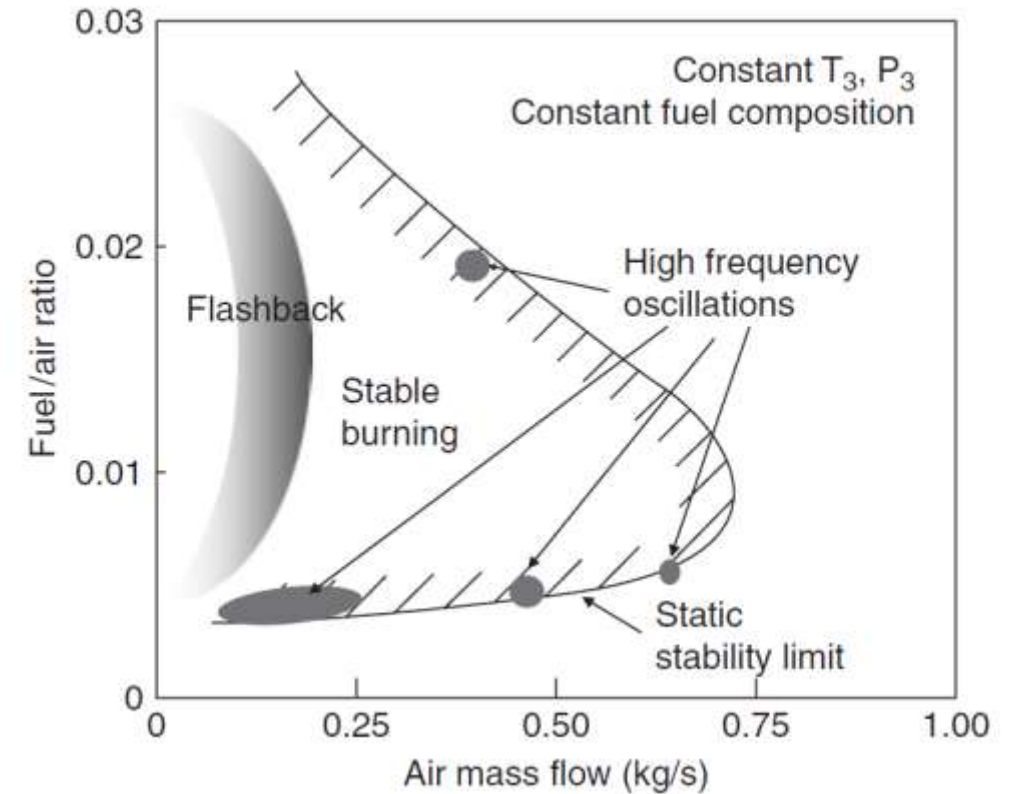
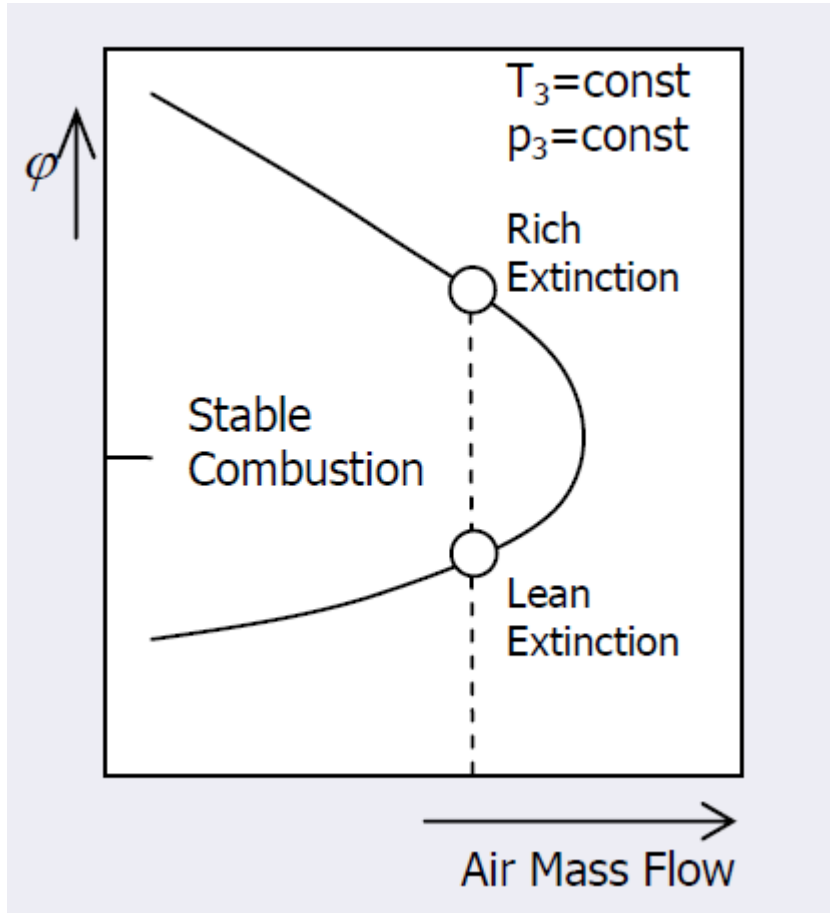
Aero Engine Combustor-VI: Flammability Limits

Prof. Dr. Arvind Gangoli Rao

Contents of this lecture

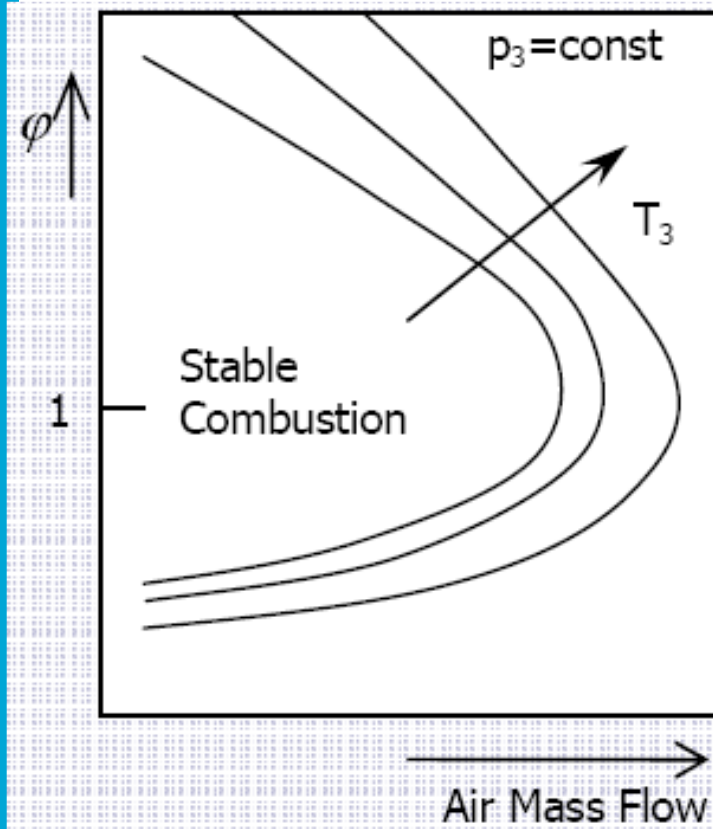
- Flammability limits
- Ignition and propagation

Flame stability

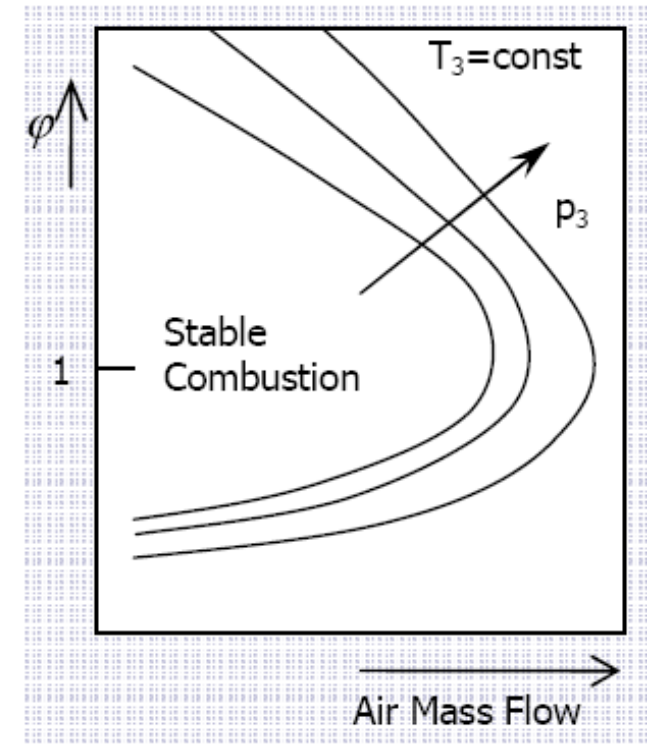


The gas turbine combustor must always operate well within this stability limit

The flammability limit



Effect of temperature on the flammability



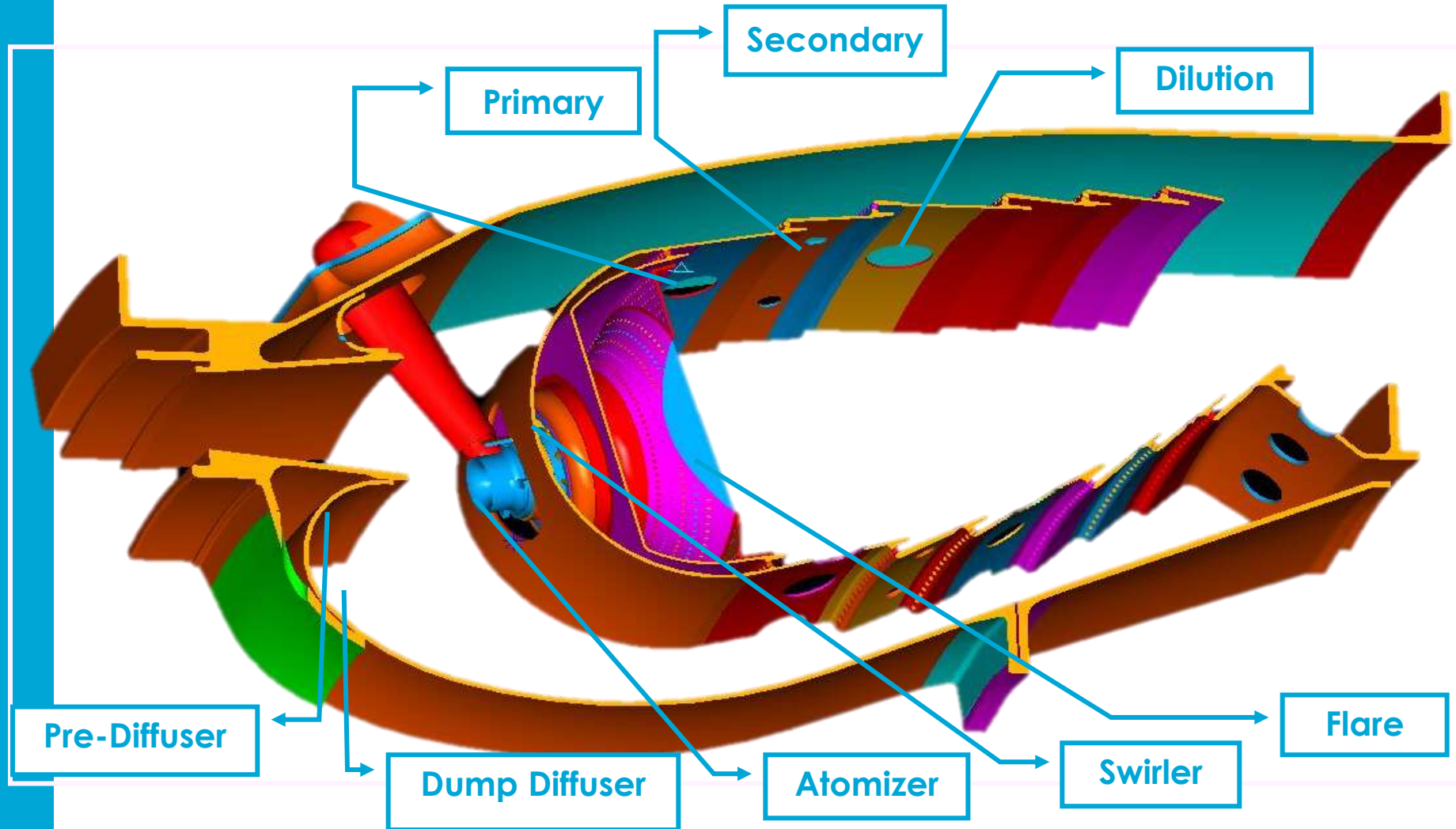
Effect of pressure on the flammability



Operating condition

Engine Family	Aircraft	Take-Off Thrust, [kN]	Overall <i>AFR</i> at TO	Overall φ at TO
CFM56-7	B737 NG	91.6	54.0	0.27
RB211-535	B757	163.3	52.3	0.28
CF6-80E1	A330	297.4	49.3	0.30
PW4000-112"	B777	396.6	43.1	0.34

Annular combustor



Conventional combustor

1. Primary Zone

- Anchors Flame
- Provides sufficient time, mixing, temperature for “complete” oxidation of fuel
- Equivalence ratio near $\phi=0.8-1.0$

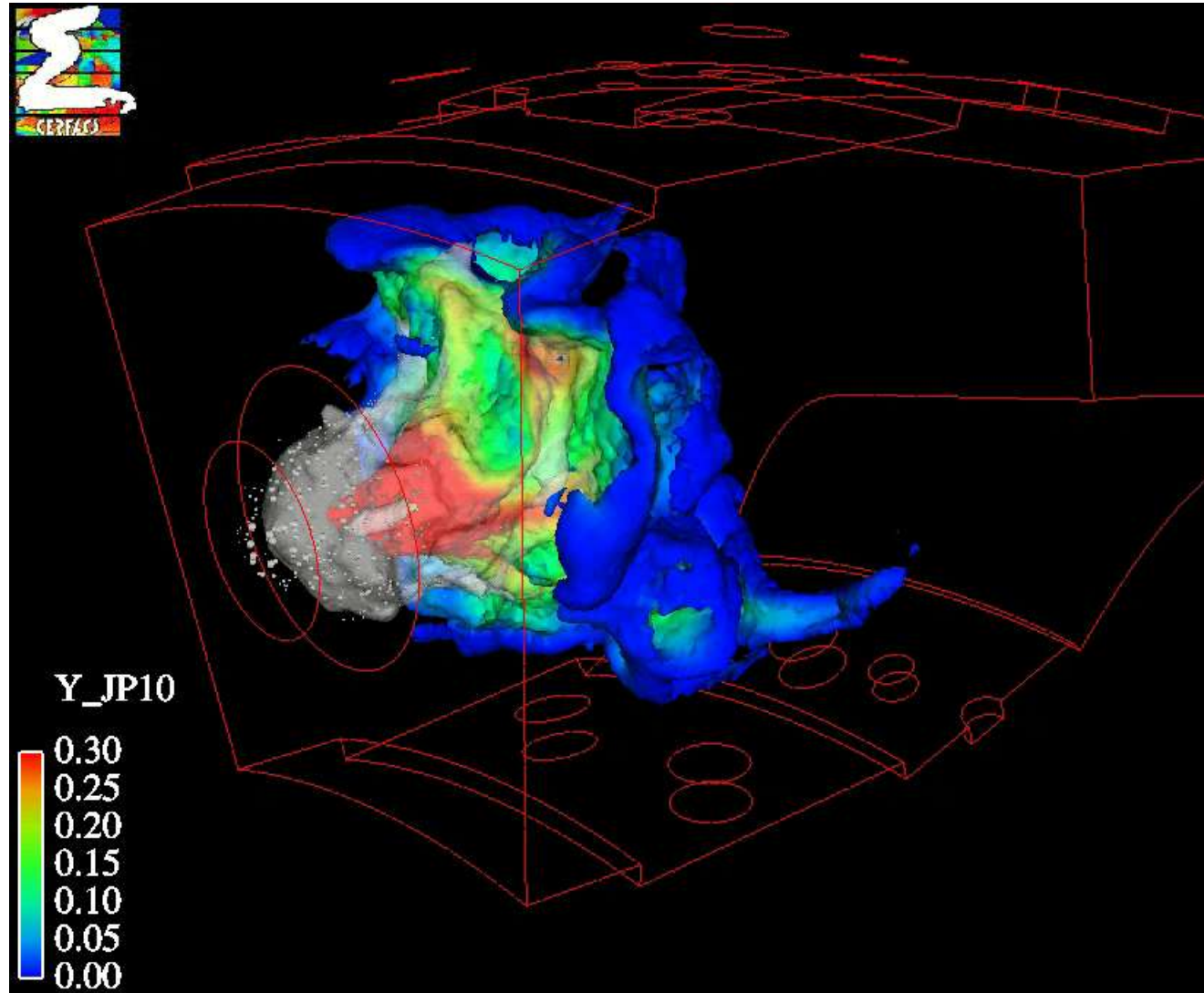
2. Intermediate (Secondary Zone)

- **Low altitude** operation (higher pressures in combustor)
 - Recover dissociation losses (primarily $\text{CO} \rightarrow \text{CO}_2$) and Soot Oxidation
 - Complete burning of anything left over from primary due to poor mixing
- **High altitude** operation (lower pressures in combustor)
 - Low pressure implies slower rate of reaction in primary zone
 - Serves basically as an extension of primary zone (increased t_{res})
- $L/D \sim 0.7$

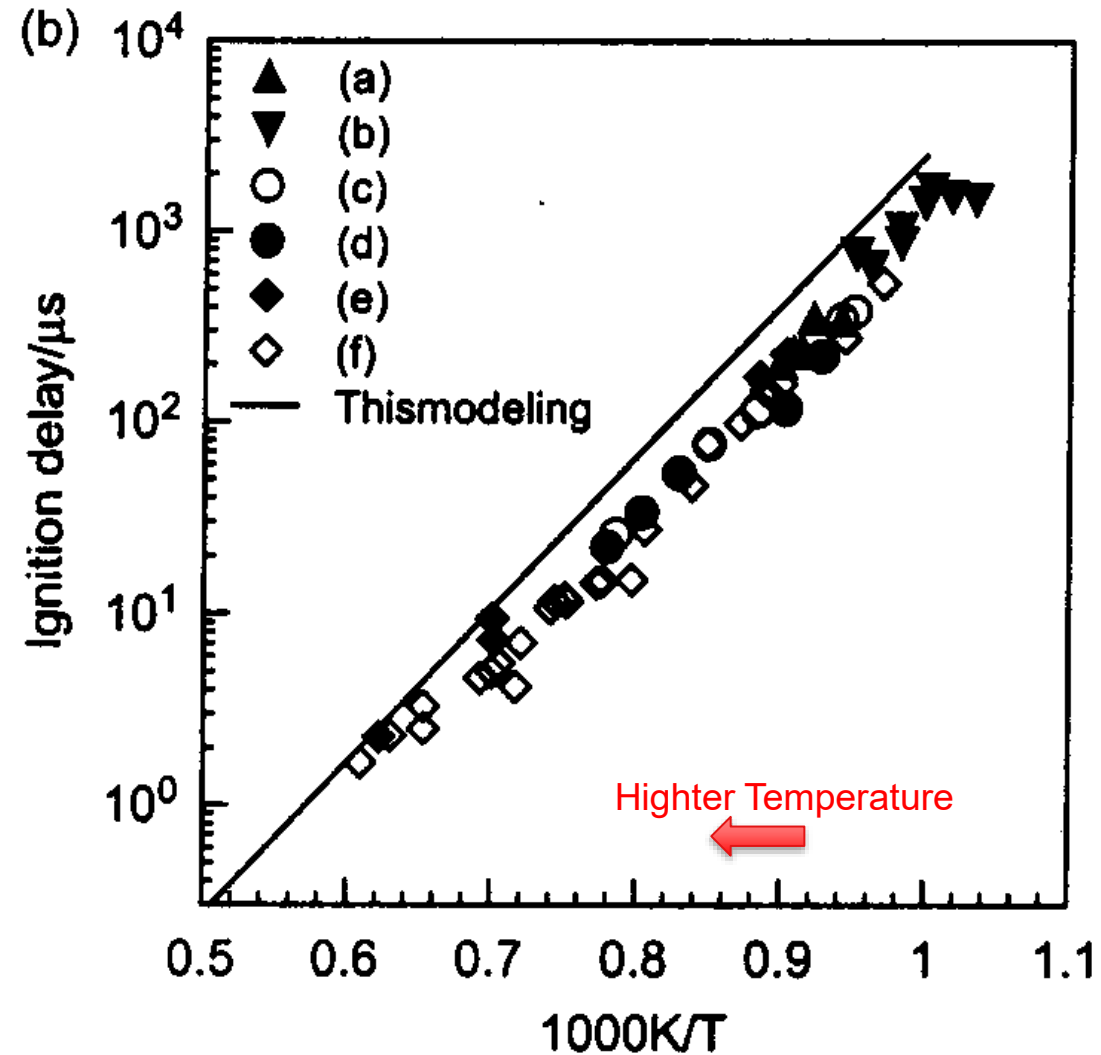
3. Dilution Zone

- Mixing air to lower temperature to acceptable value for turbine
- Tailor temperature profile (low at root and tip, high in middle)
- Uses about 20-40% of total ingested core mass flow
- $L/D \sim 1.5-1.8$

Combustor Simulation



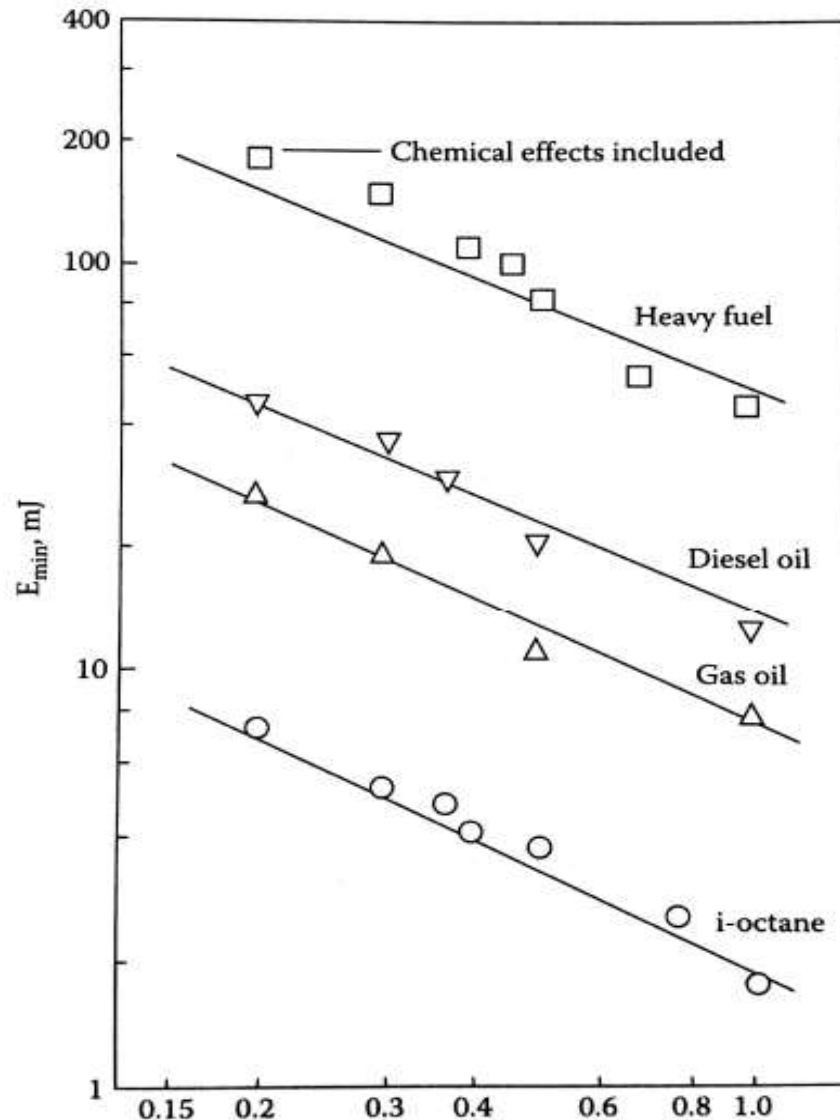
Ignition delay



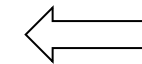
Ignition Delay of Kerosene/Air Mixture at 20 atm. (Dagaut and Cathonnet, 2006)

Activation Energy

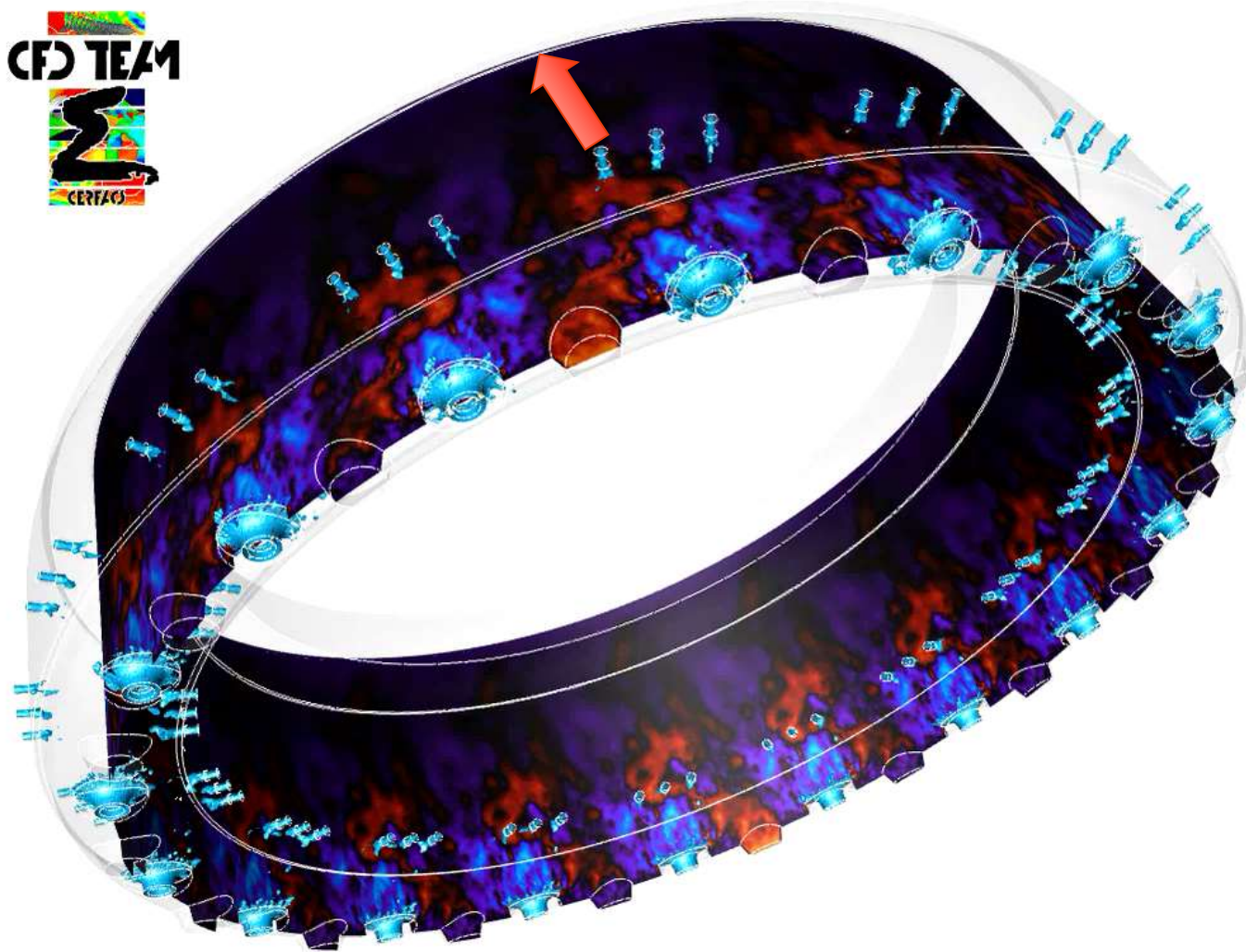
Activation energy (mJ)



Variation of activation energy with fuel type and equivalence ratio.



Ignition in an annular combustor



Courtesy:
CERFACS, France

AE4-238 Aero Engine Technology

Aero Engine Combustor-VII: Operating Characteristics

Prof. Dr. Arvind Gangoli Rao

Combustor efficiency

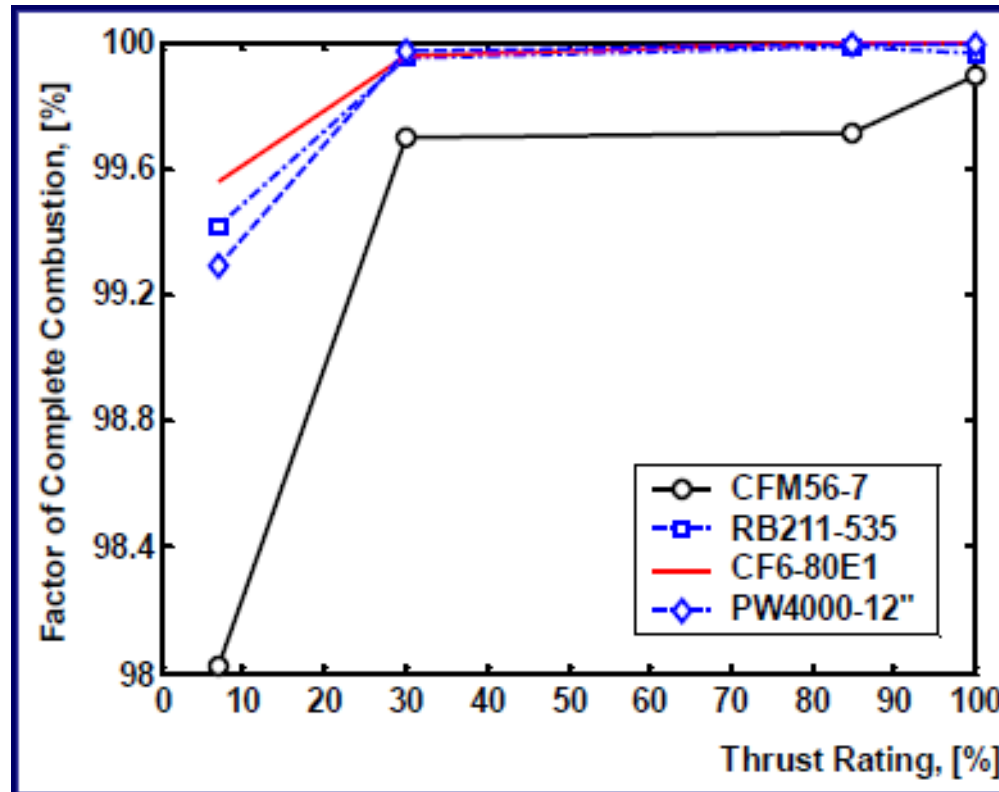
- Combustion efficiency, $\eta_b = \text{Actual Enthalpy Rise} / \text{Ideal Enthalpy Rise}$

$h = \text{heat of reaction (sometimes designated as } Q_R) = 42,800 \text{ KJ/Kg}$

$$\eta_b = \frac{\bar{c}_P \left[(\dot{m}_a + \dot{m}_f) T_{t4} - \dot{m}_a T_{t3} \right]}{\dot{m}_f \cdot h}$$

- General Observations:
 1. $\eta_b \uparrow$ as $p \uparrow$ and $T \uparrow$ (because of dependency of reaction rate)
 2. $\eta_b \downarrow$ as Mach number $\uparrow$ (decrease in residence time)
 3. $\eta_b \downarrow$ as fuel/air ratio $\downarrow$ (only in a certain range)

Estimated combustor efficiency



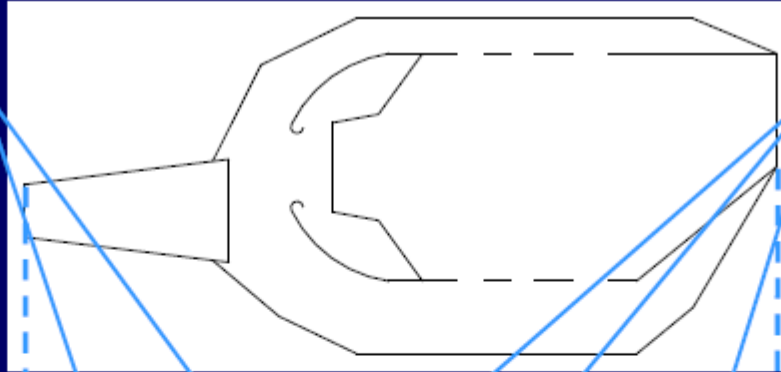
Variation of typical combustion efficiency with engine thrust rating
Note: These values are generic estimations and not from experiments.

Pressure losses

Friction-induced loss:

$$\Delta P_{fr} = 0.5 \rho V^2 \frac{L}{d_e} \xi$$

$$\xi = \frac{0.3168}{Re^{0.25}}$$



Macro-flow losses:

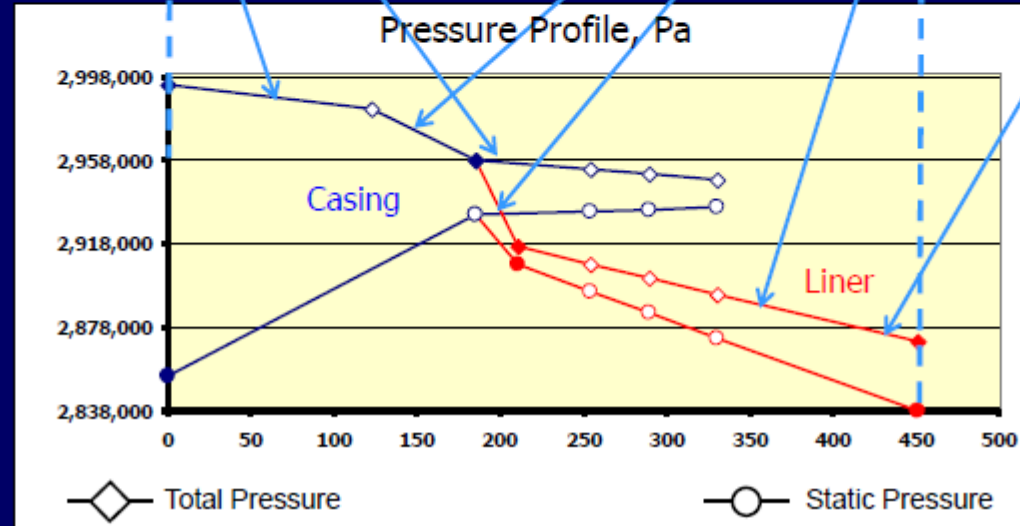
$$\Delta P_{dif} \approx 30\%$$

$$\Delta P_{sw} \approx 35\%$$

$$\Delta P_{liner} \approx 35\%$$

Hot loss:

$$\approx q_{ref} \left(\frac{T_4}{T_3} - 1 \right)$$



Characteristics of Aero Engine Combustor

- Primarily turbulent combustion
- External ignition (e.g. spark) is required to light off only, then combustion is self-sustained
- Constant (nearly) pressure combustion
- Pressure losses penalize cycle performance and have to be limited ($\approx 4\%$, during cruise & 7% during T-O)
- Typically, low relative volume (intensity up to 150 MW/atm-m^3) and structural weight:
- Fast volume and heat dynamics, low surface-to-volume ratios
- Low residence time for low NO_x ($\sim 5 \text{ ms}$)
- No heat rejection, no work extraction from/in combustor:
- Combustor is insulated, heat losses penalize cycle performance

Characteristics of Aero Engine Combustor

Fuels:

- Gaseous & liquid
- Natural gas & products of gasification (power generation)
- Kerosene & synthesized liquid (aviation)

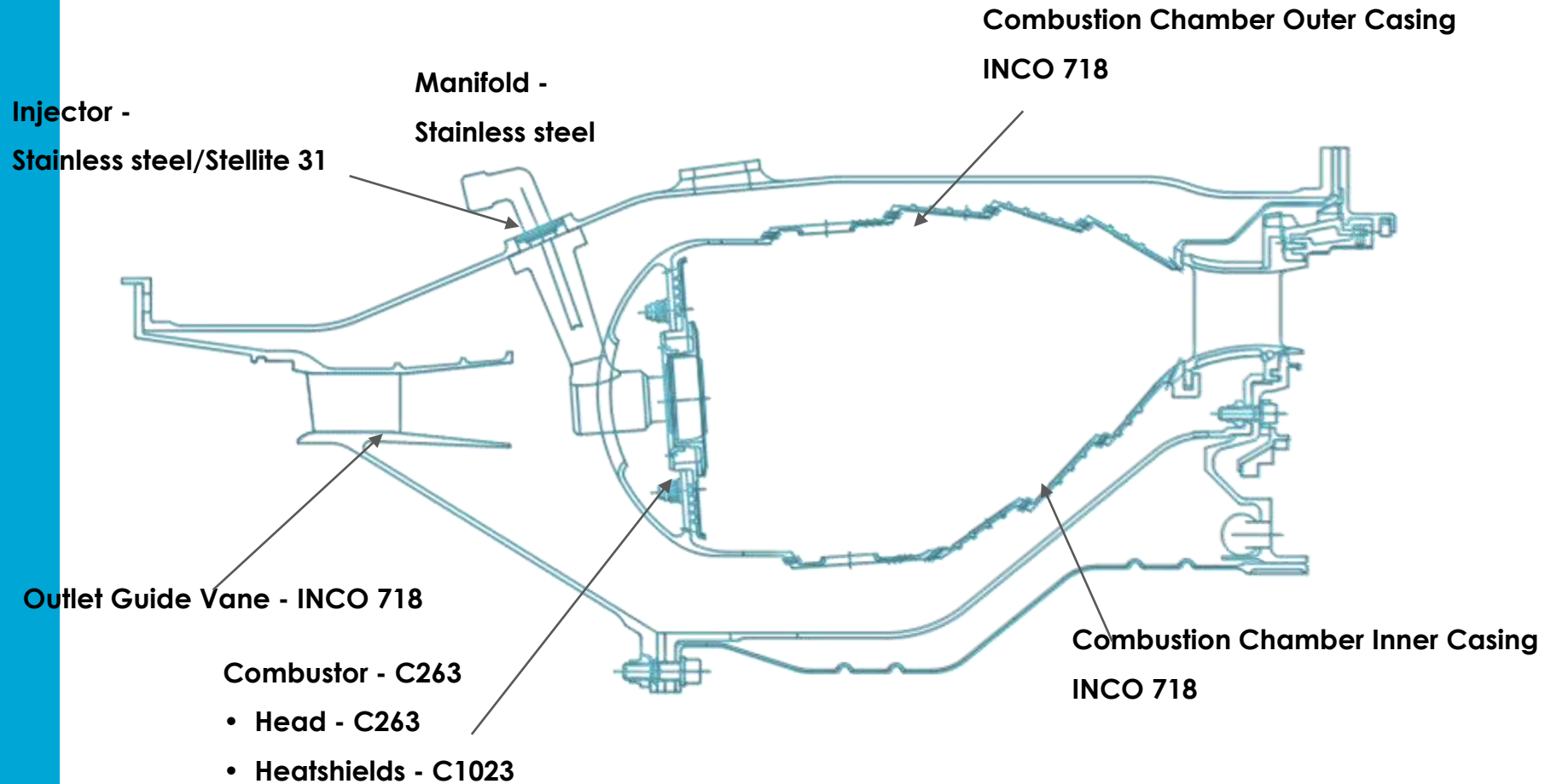
Elevated/high pressure and air inlet temperature:

- Aircraft propulsion: up to 40-60 bar & >900 K
- Power generation: <30 bar & <750 K
- Micro turbines: 3.0-4.5 bar & up to 1050 K (due to recuperation)

Combustor “types”:

- Conventional
- RQL, Lean Burn, staged combustors, etc
- Alternative technologies: Lean Direct Injection, Flameless Combustion, etc
- Use of catalysts & inhibitors

Combustor Material



AE 4238

AE4-238 Aero Engine Technology

Aero Engine Combustor-VIII

Prof. Dr. Arvind G. Rao

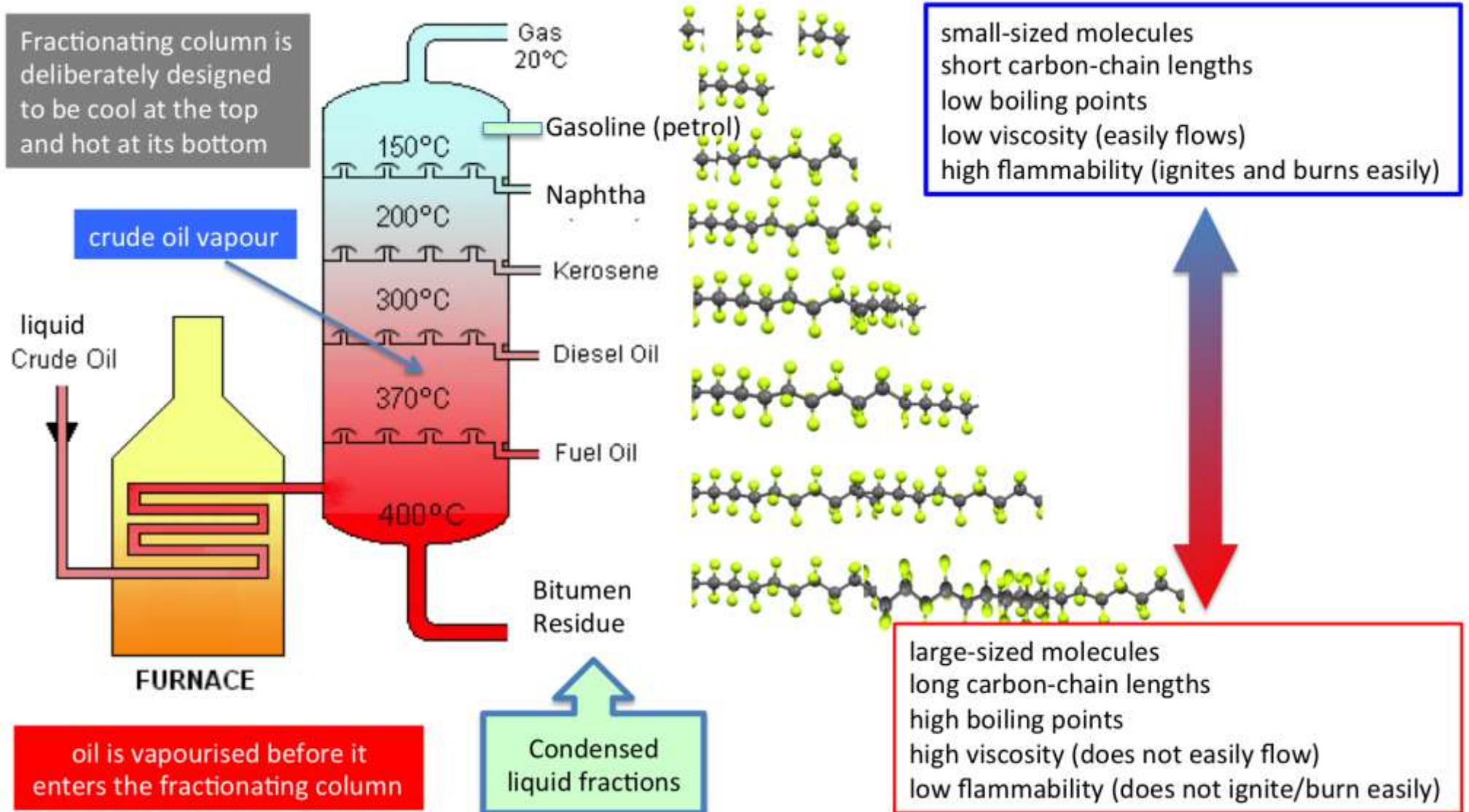
Topics covered so far

- Enthalpy Balance
- Gas Turbine Combustor types
- Adiabatic flame temperature
- Types of combustion
- Flammability limits
- Working of Conventional Gas turbine Combustor
- Characteristics of gas turbine combustors
- Combustor materials

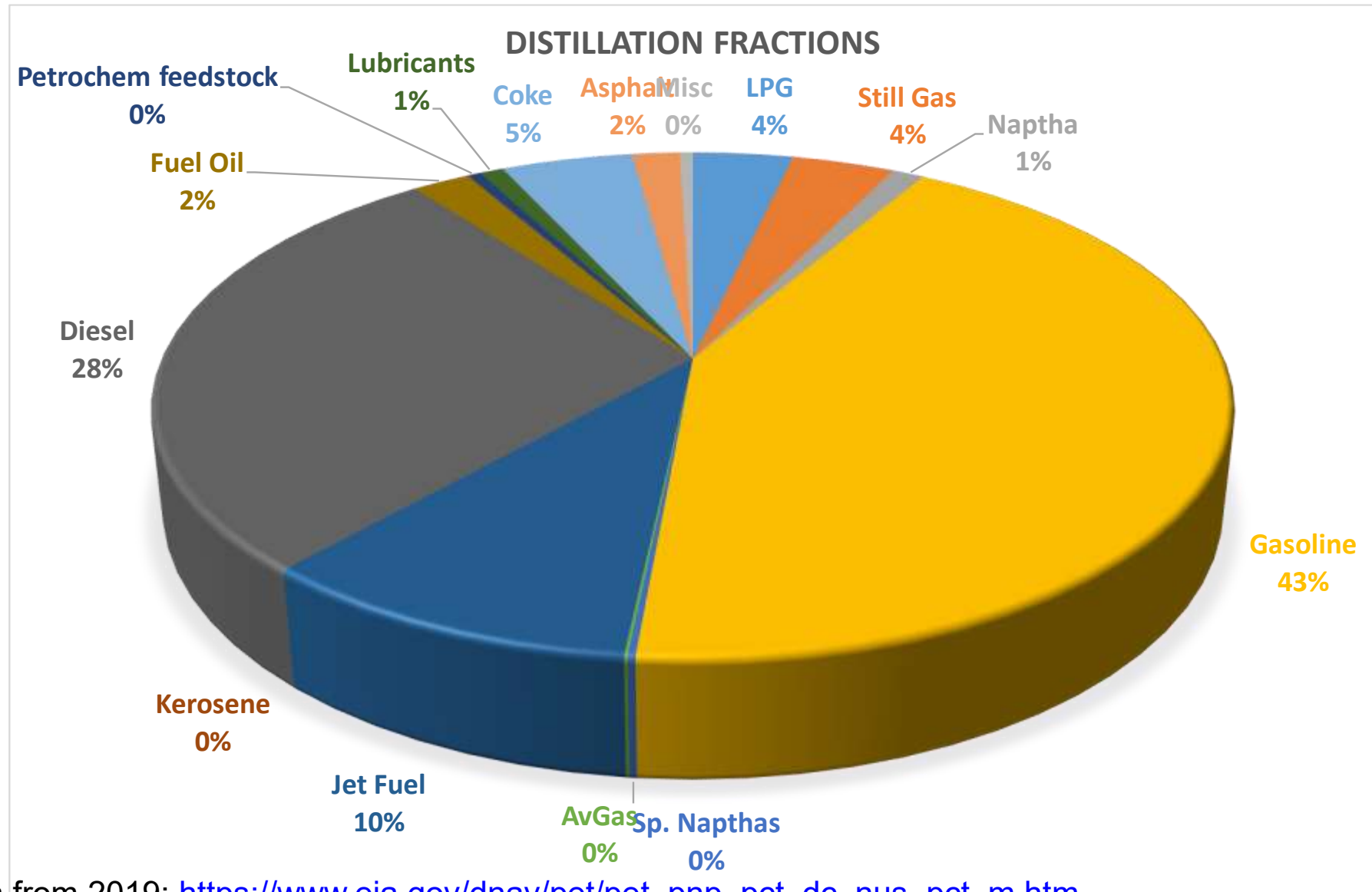
Topics to be covered in this lecture

- Fuel
- Emission Regulations
- Emission Characteristics
- RQL Combustors
- Lean Combustion

Distillation of Crude oil

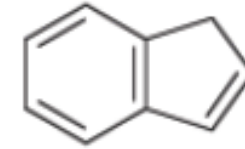
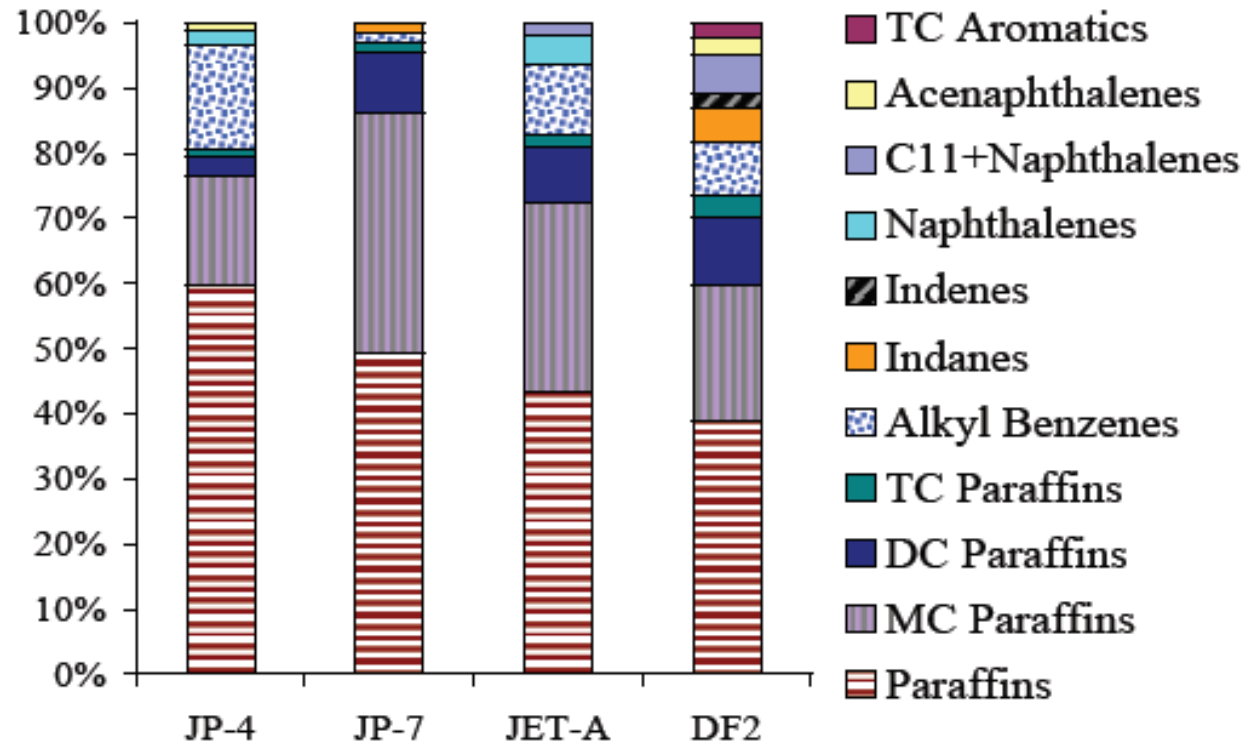


Fractions from distillation of Crude Oil (USA)

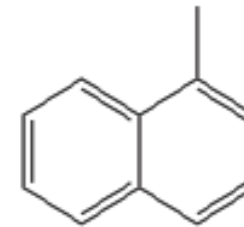


Data from 2019: https://www.eia.gov/dnav/pet/pet_pnp_pct_dc_nus_pct_m.htm

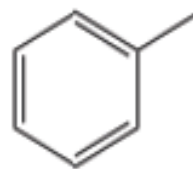
Fuel composition



indene

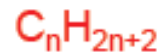


1-methyl naphthalene

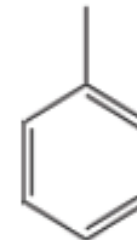


n-propyl benzene

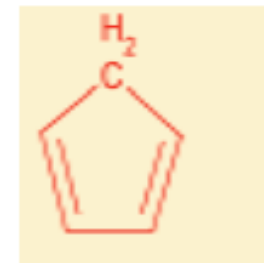
Developed
for SR71



alkanes



toluene



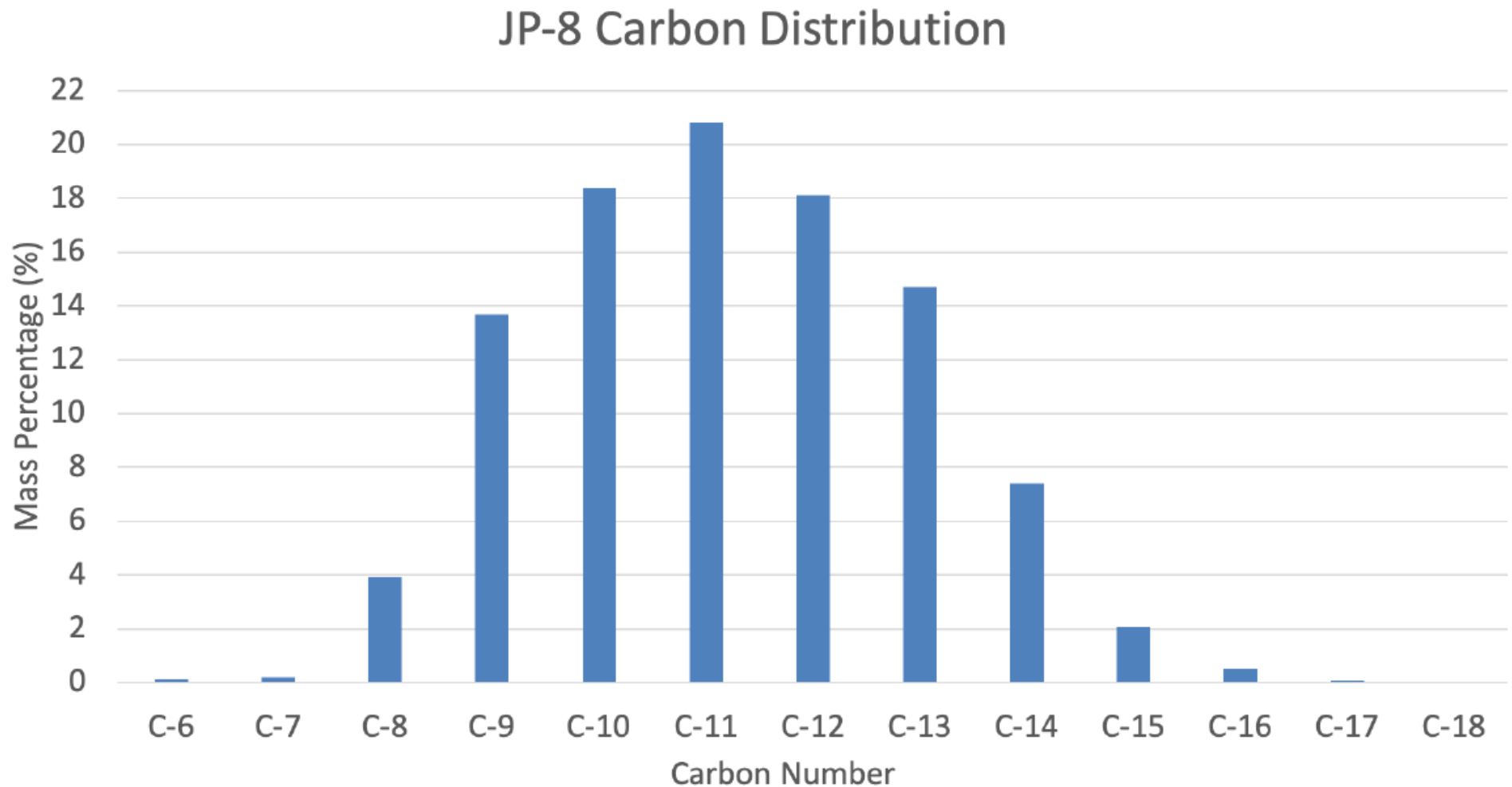
cyclopentadiene

MC Paraffins: Mono chain paraffins; DC paraffins: Di-chain Paraffins; TC: Tri-chain Paraffins

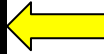
Ref. Lindstedt P., Markaki V., Robinson R. Detailed and Simplified Chemical Kinetics of Aviation Fuels and Surrogates.

MACCCR/AFOSR Fuel Research Review, Princeton, 2010

Typical fuel composition



Combustion type



Diffusion burning:

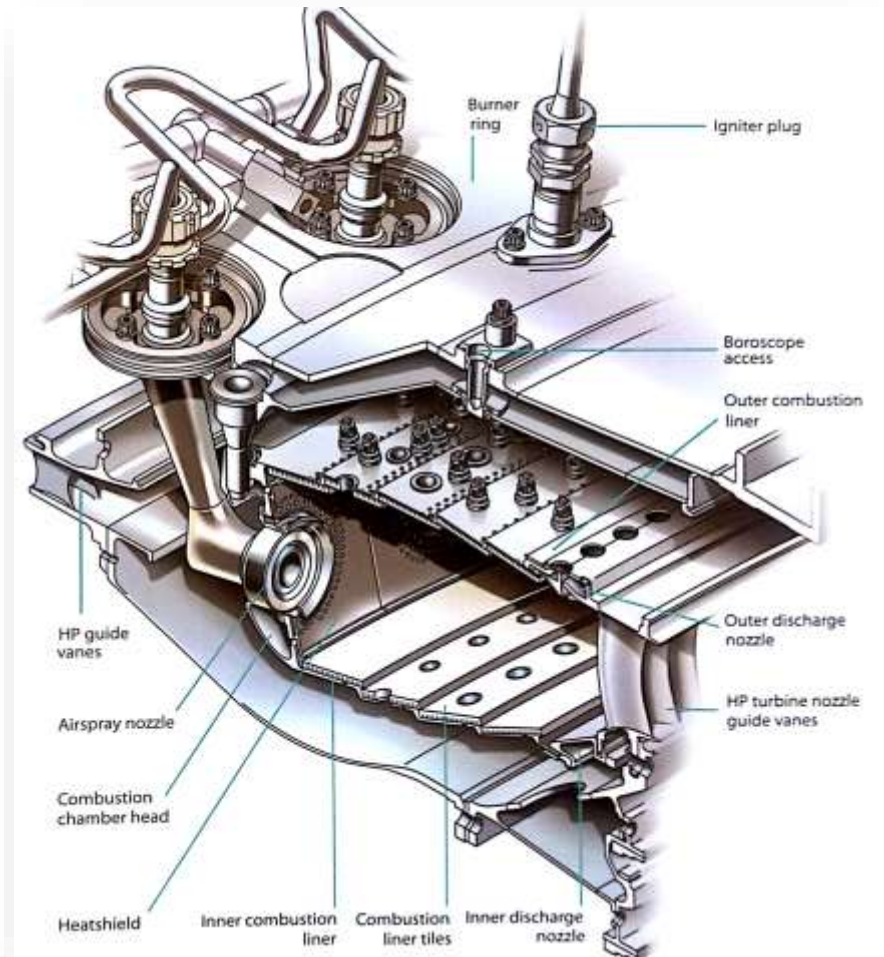
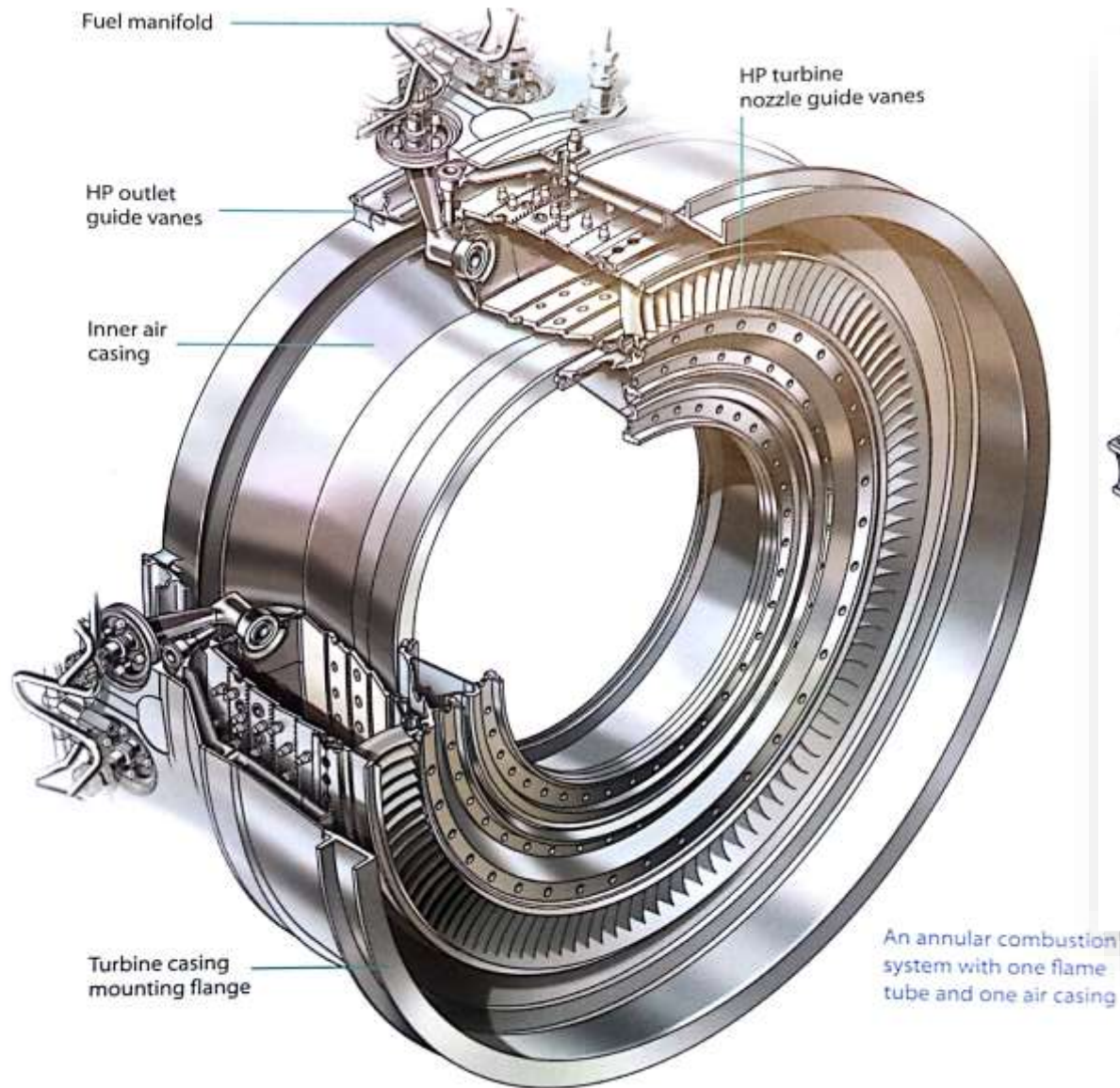
- burning at $ER \sim 1$
- flashback free
- very stable
- reliable ignition
- broad operat. envelope
- high NO_x
- easy to control CO
- high soot

Premix burning:

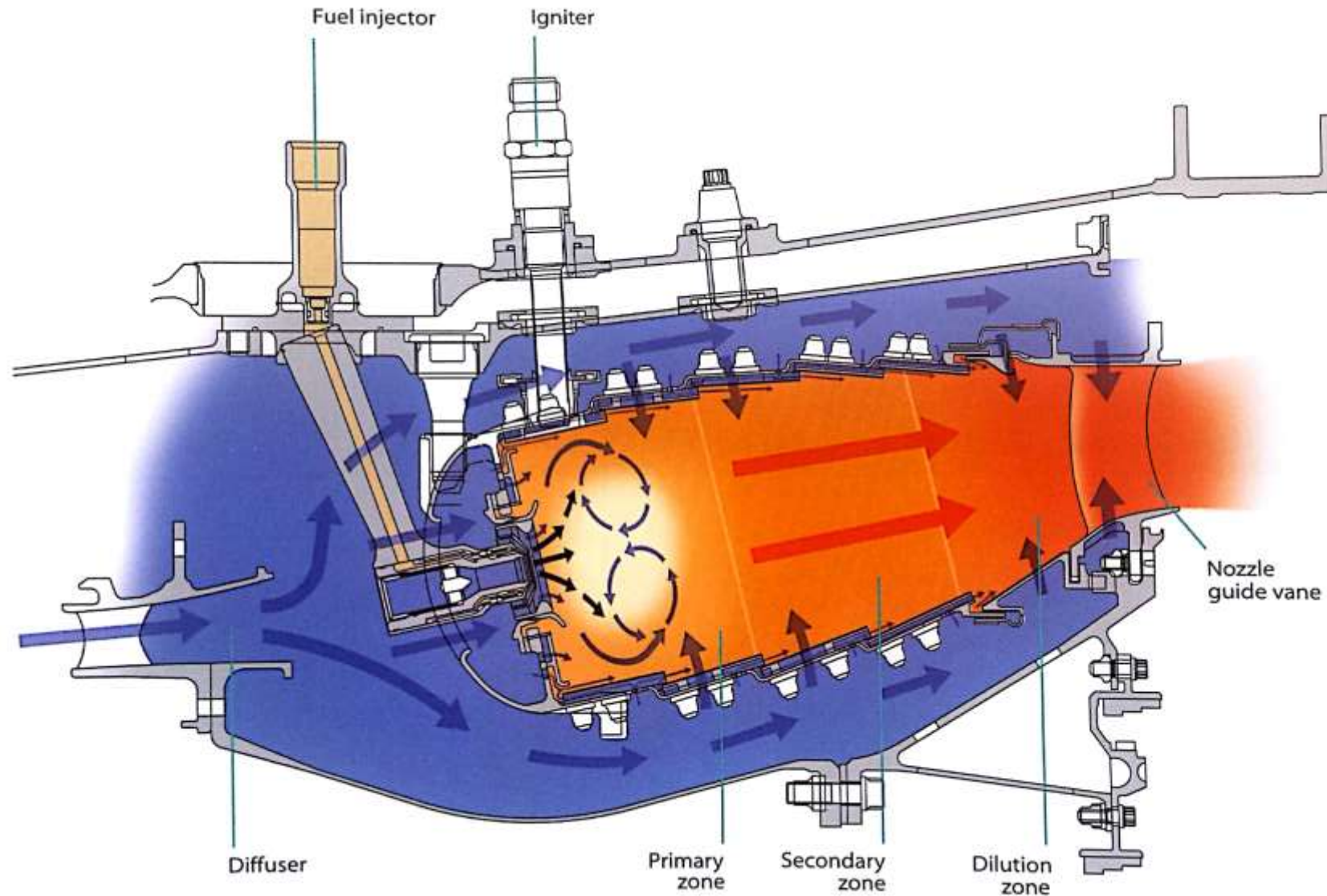
- burning at selected ER
- prone to flashback
- prone to instabilities
- unreliable ignition
- narrow operat. envelope
- low NO_x
- difficult to control CO
- low soot



Annular combustor



The gas turbine combustor



AE 4238

Engine Emissions

The LTO cycle

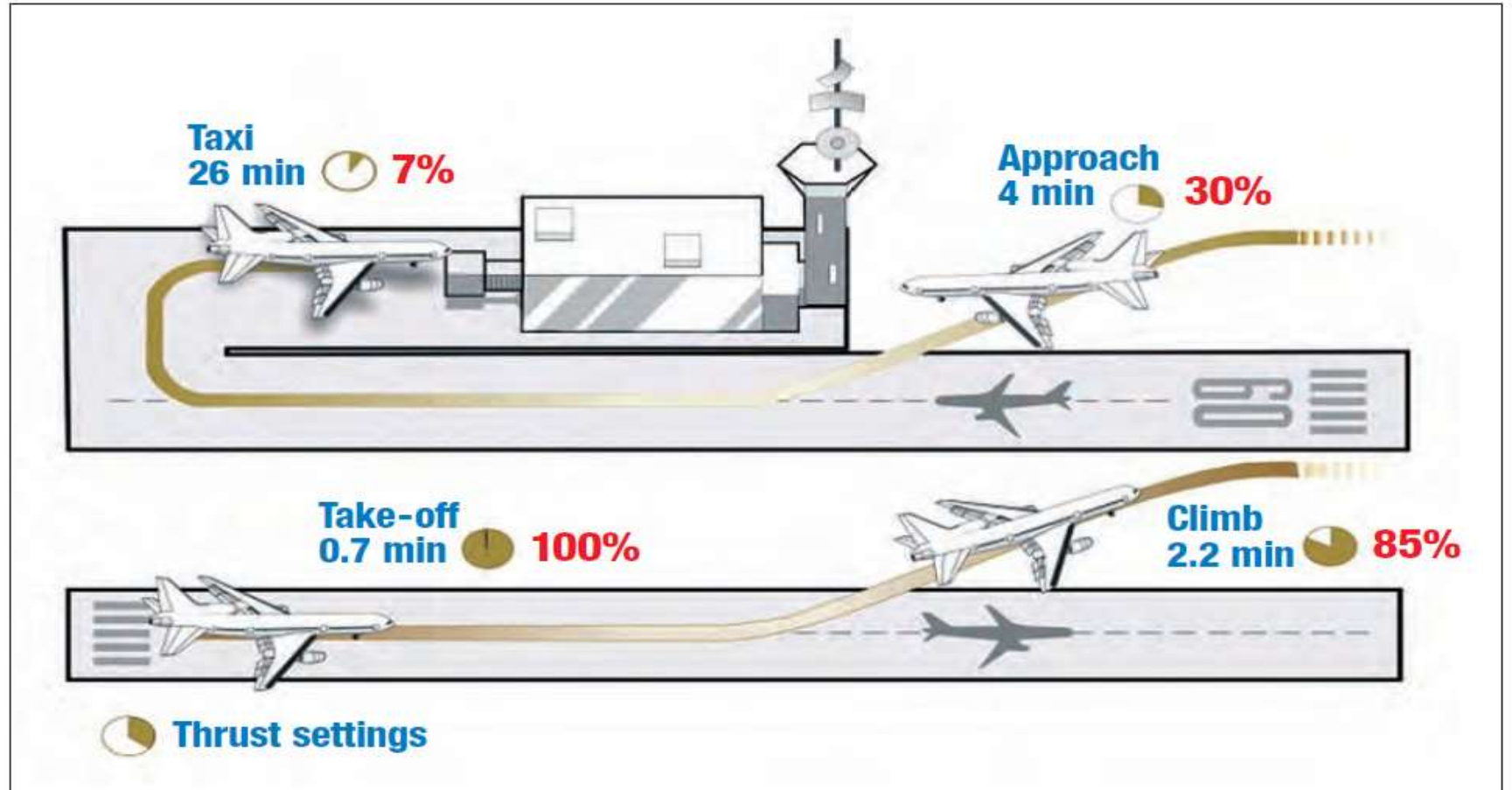


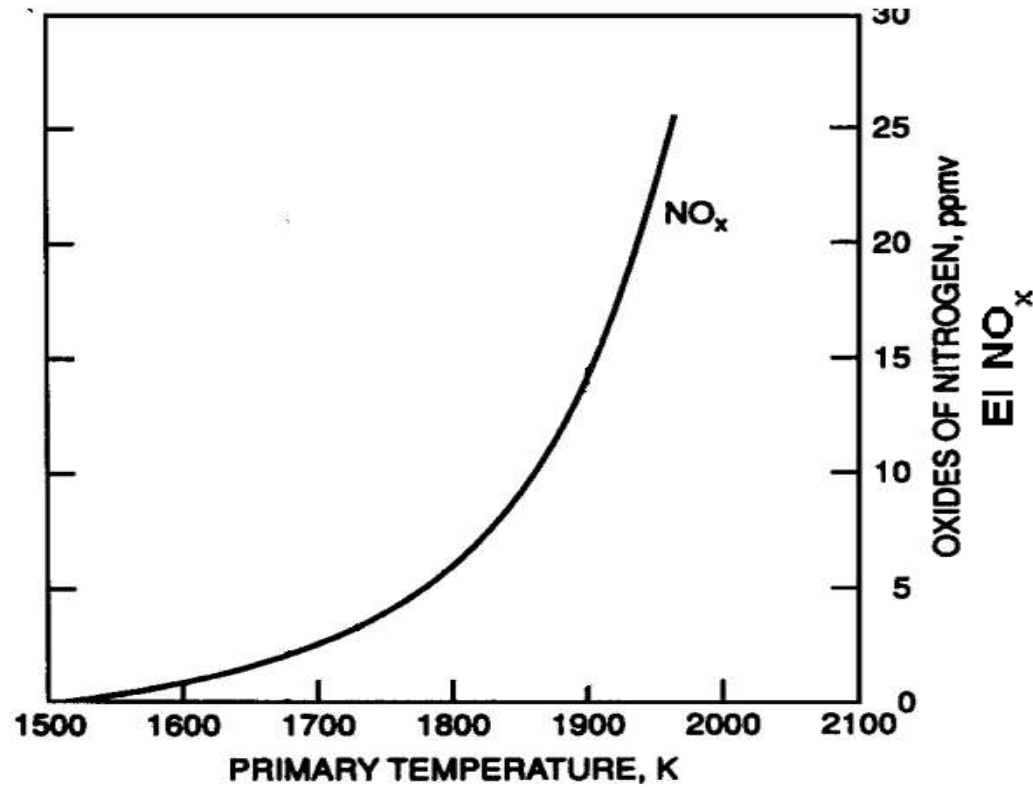
Figure 1: Illustration of ICAO emissions certification procedure in the LTO cycle.

Local Air Pollution

Pollutant	Health Effect
CO – Carbon Monoxide	● Cardiovascular effects, especially in those persons with heart conditions
HC – Unburned Hydrocarbons (a primary component of Volatile Organic Compounds, or VOC)	● Eye and respiratory tract infection ● Headaches ● Dizziness ● Visual disorders ● Memory impairment
NO_x – Nitrogen Oxides	● Lung irritation ● Lower resistance to respiratory infections

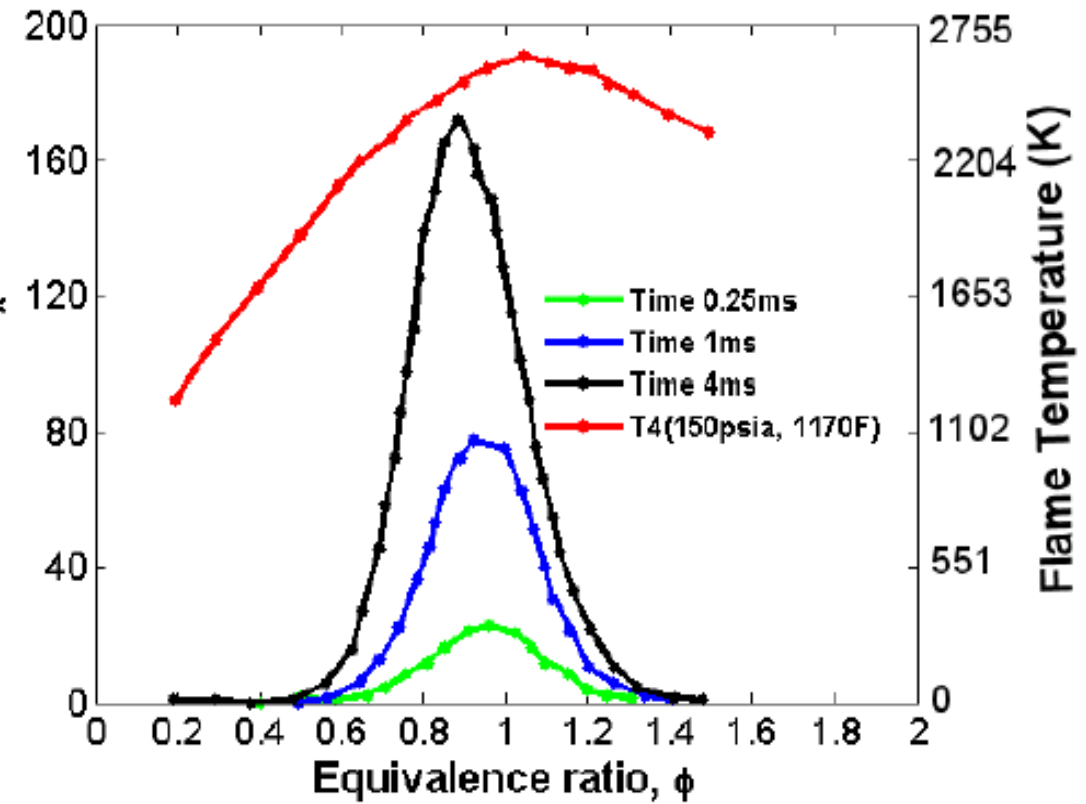
Pollutant	Health Effect
O₃ – Ozone (HC is a precursor for ground-level O ₃ formation)	● Lung function impairment ● Effects on exercise performance ● Increased airway responsiveness ● Increased susceptibility to respiratory infection ● Increased hospital admissions and emergency room visits ● Pulmonary inflammation, lung structure damage
PM – Particulate Matter (smoke is a primary component of PM.)	● Premature mortality ● Aggravation of respiratory and cardiovascular disease ● Changes in lung function ● Increased respiratory symptoms ● Changes to lung tissues and structure ● Altered respiratory defence mechanisms

NO_x Emission



Variation of NO_x with temperature

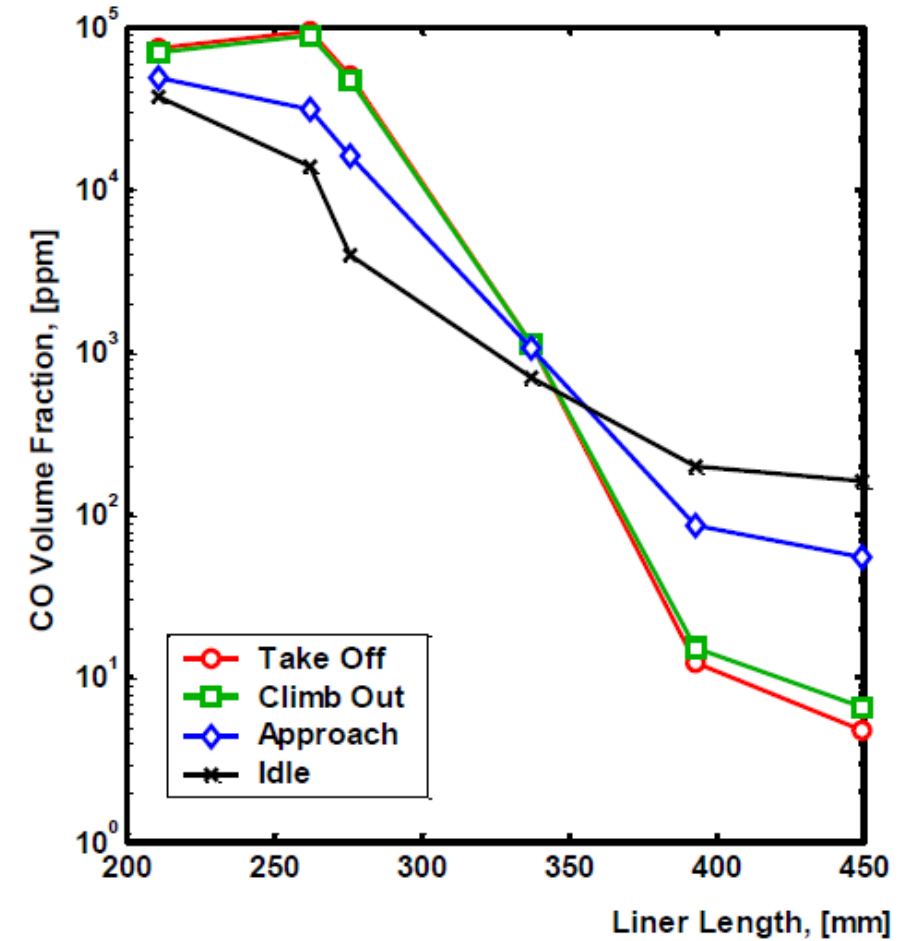
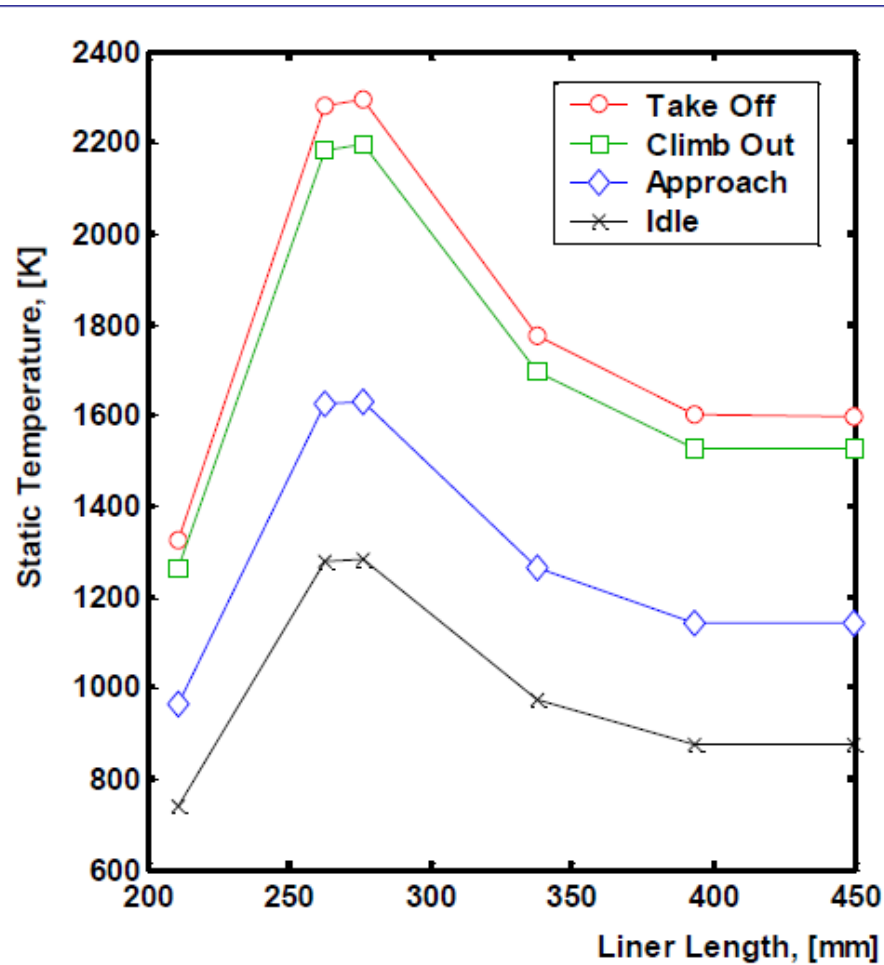
Source: Gas Turbine combustion, Lefebvre, Arthur H.; Ballal, Dilip R., 2010, Taylor & Francis



Variation of NO_x with equivalence ratio and residence time

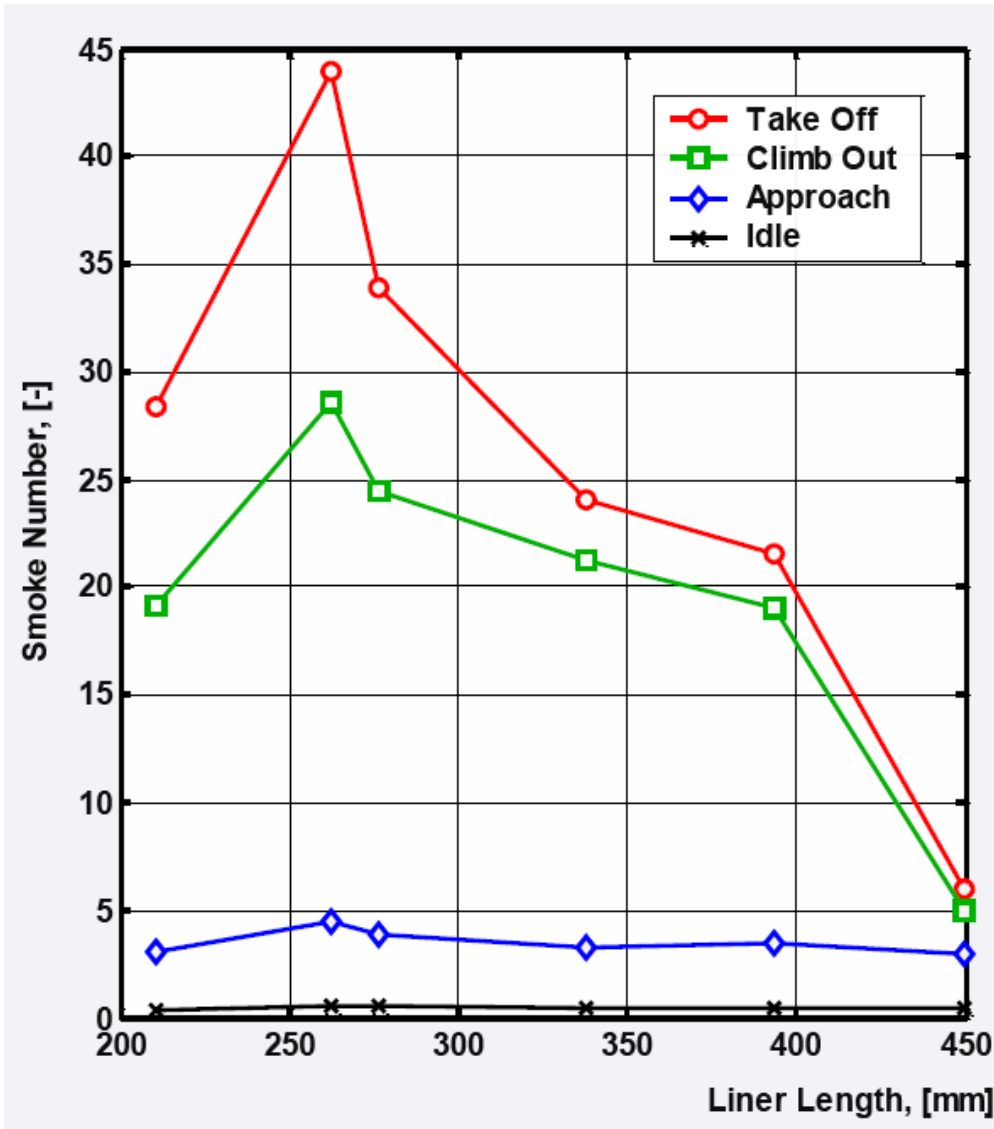
D.Dewanji "Flow Characteristics In Lean Direct Injection Combustors", TU Delft, 2012.

Combustor Temperature and CO Emission



Source: Shakariyants, et. Al. 2004, "Generic Geometry Definition of the Aircraft Engine Combustion Chamber", GT2004-53522, Proceedings ASME Turbo Expo 2004.

Soot Emission

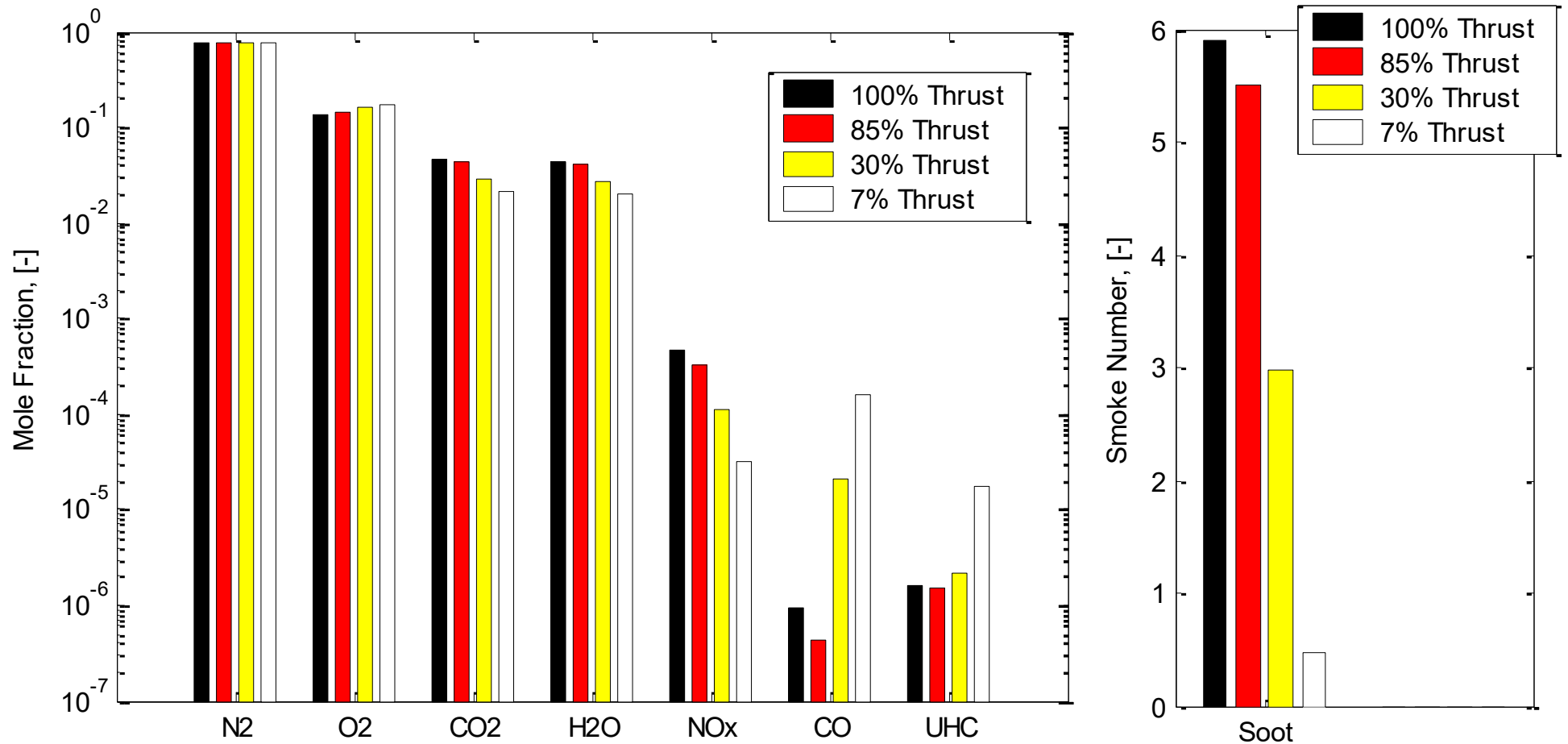


Controlling Factors:

- Fuel-to-air ratio
- Pressure
- Temperature
- Quality of fuel injection
- Mode of fuel injection

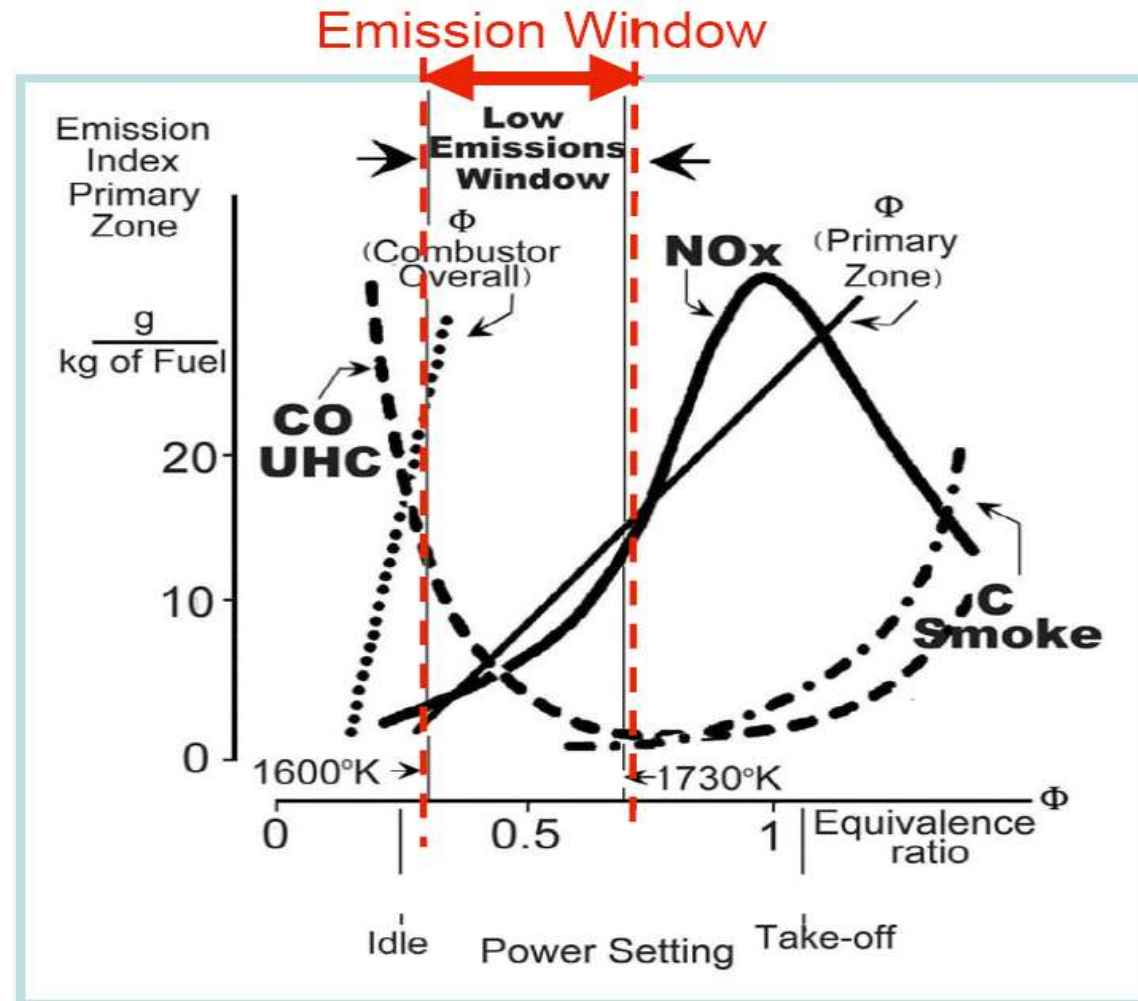
Source: Shakariyants, et. Al. 2004, "Generic Geometry Definition of the Aircraft Engine Combustion Chamber", GT2004-53522, Proceedings ASME Turbo Expo 2004.

Variation of Emission with Thrust



Source: Shakariyants, et. Al. 2004, "Generic Geometry Definition of the Aircraft Engine Combustion Chamber", GT2004-53522, Proceedings ASME Turbo Expo 2004.

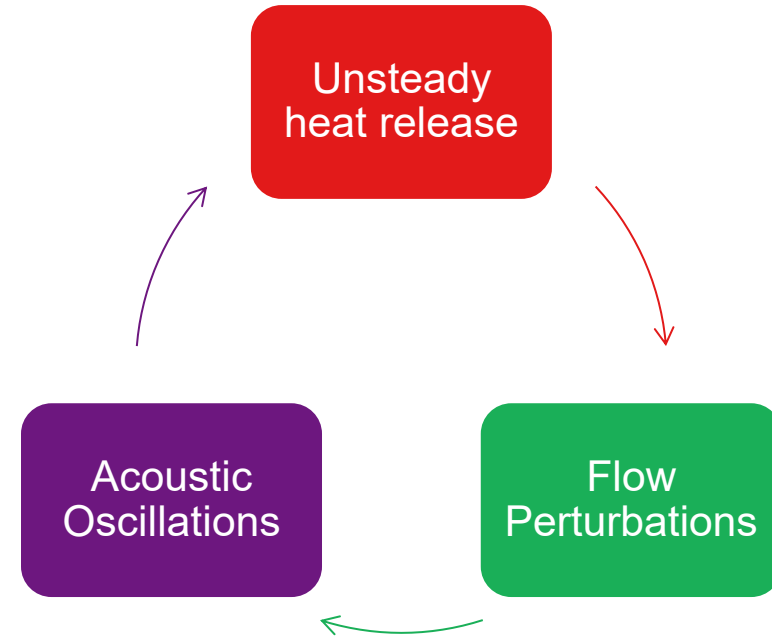
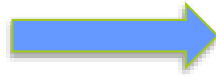
Emission Window



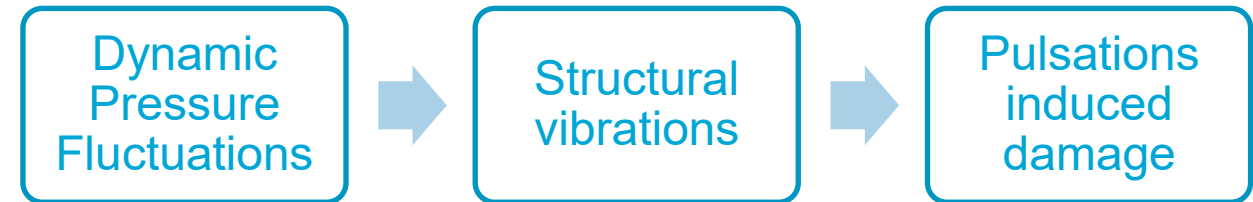
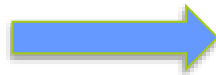
Thus there is only a narrow window in which the emission is low

Combustor instability

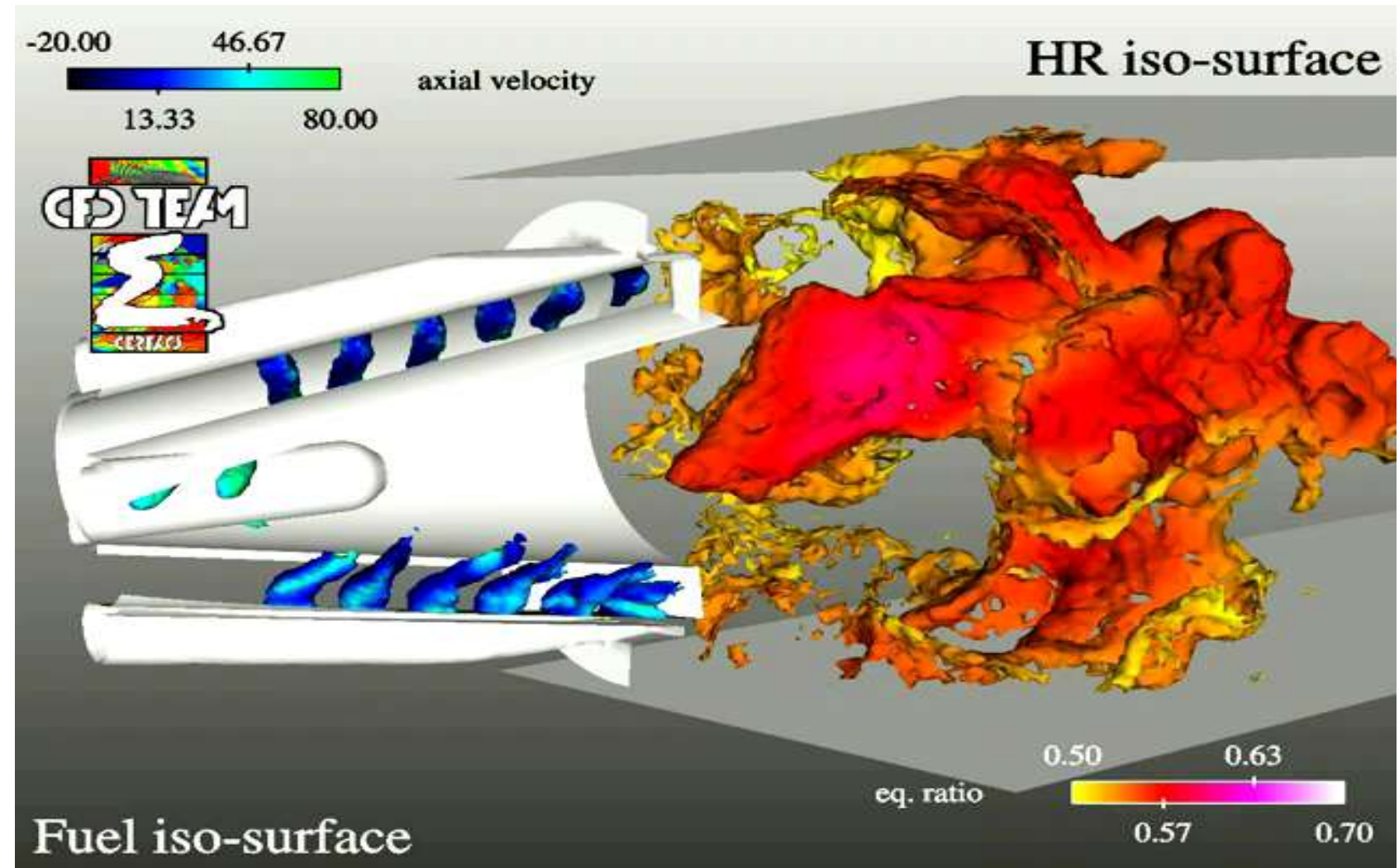
Cause



Effect



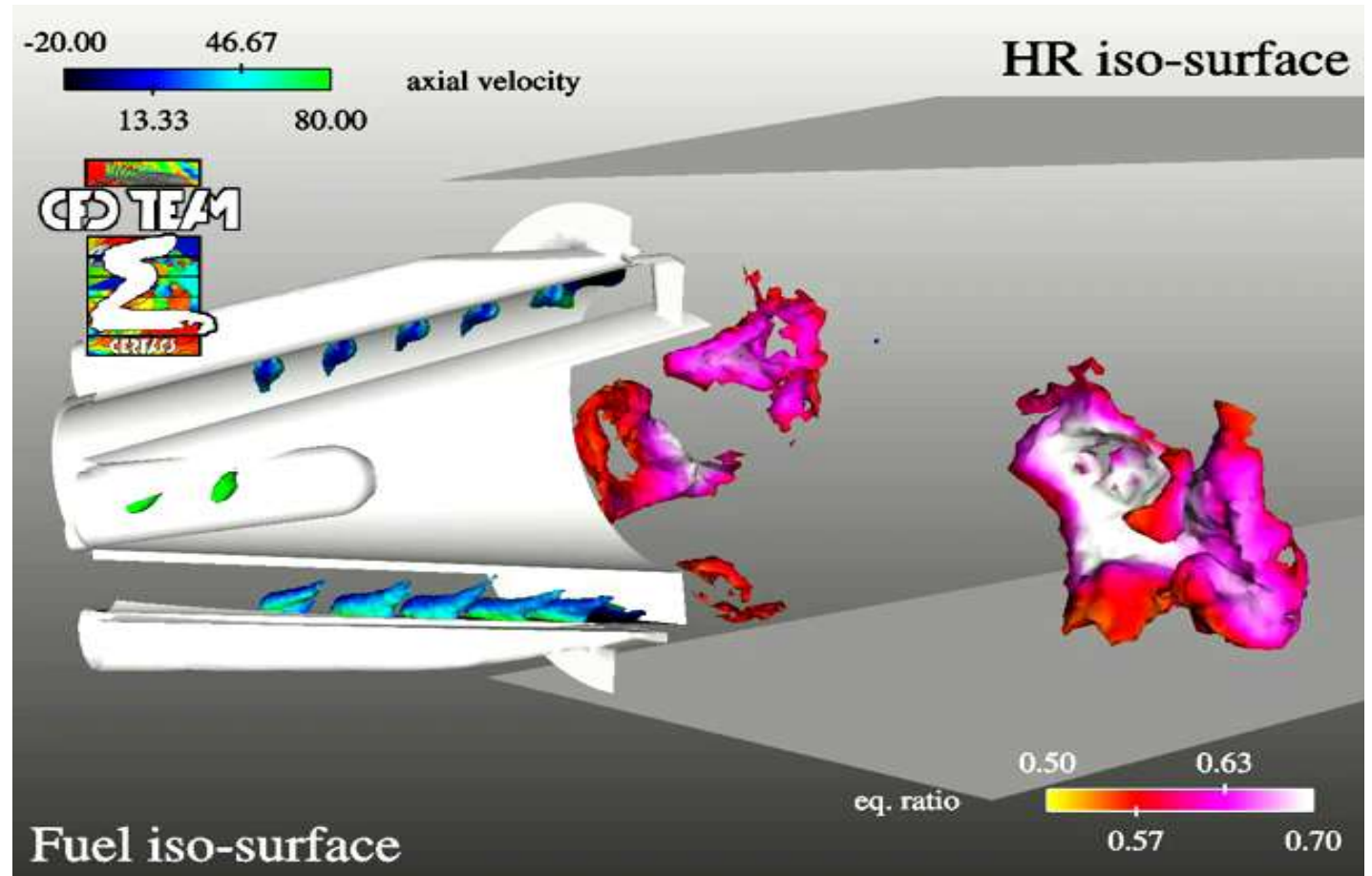
Stable combustion



Courtesy: CERFACS, France

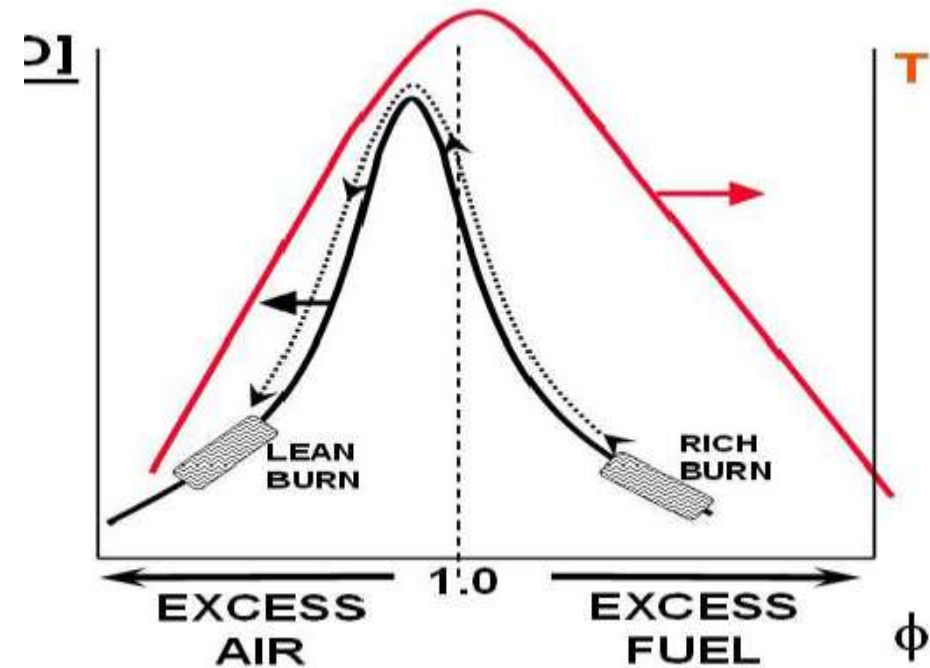
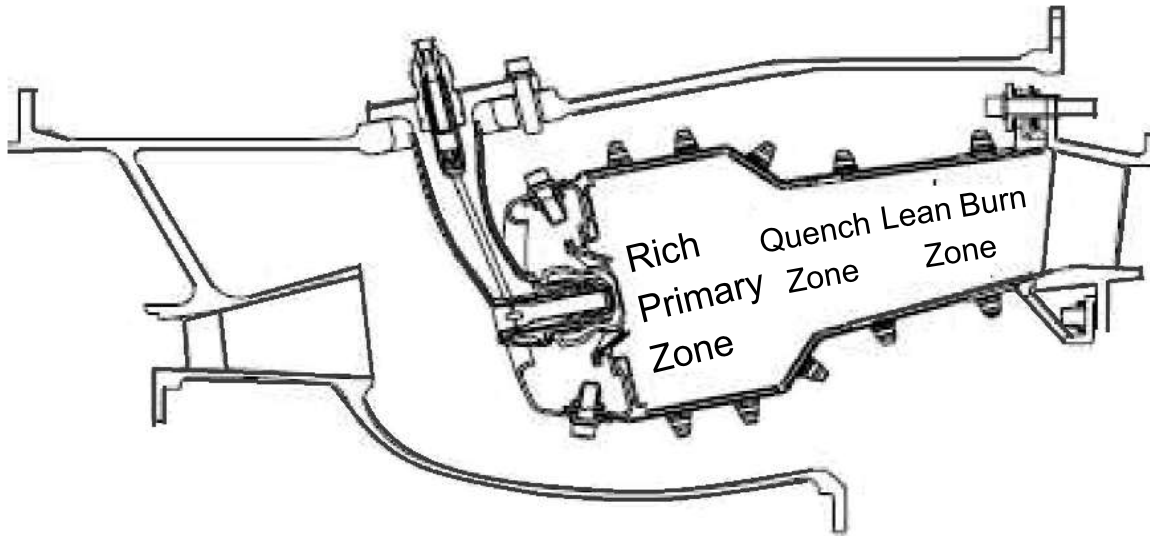
Combustion Instability (when operating close to the lean blow out limit)

Courtesy:
CERFACS, France



When operating close to the lean blow out limit, the variation in heat release gets coupled with the fluid mechanics and pressure fluctuations

Rich burn-Quick quench-Lean burn



RQL Combustor: Rolls Royce

Phase 5 Combustion Technology – Trent 900

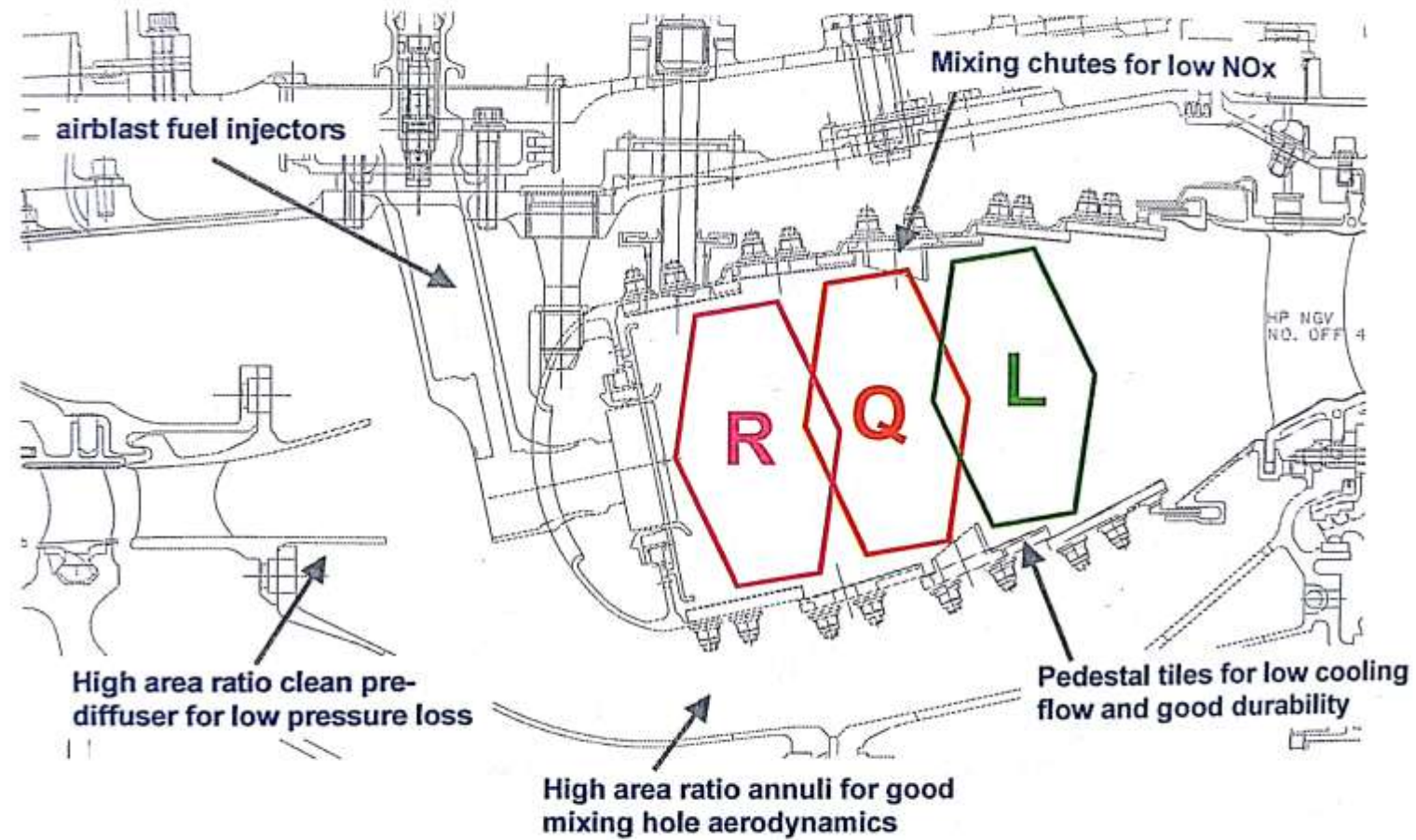
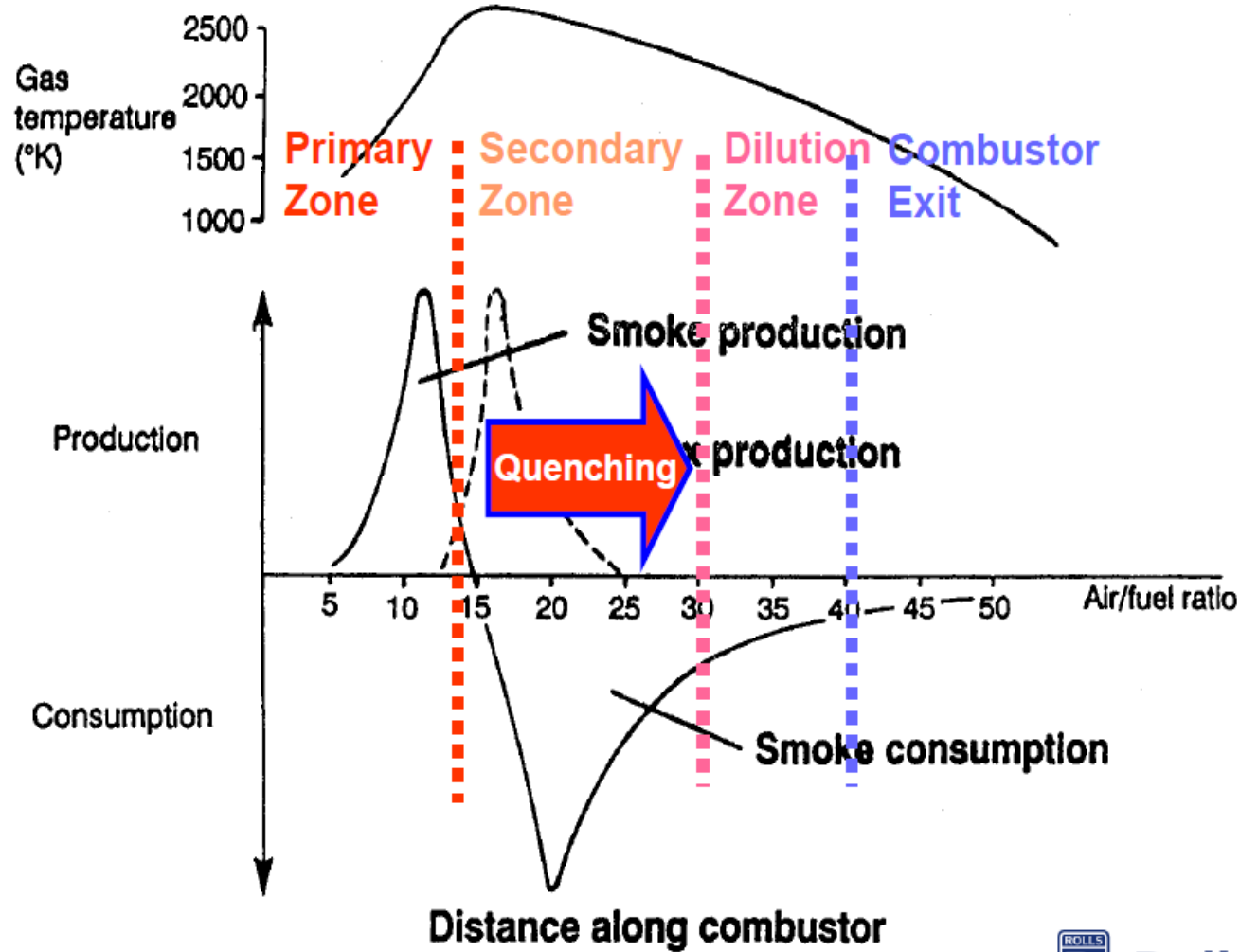


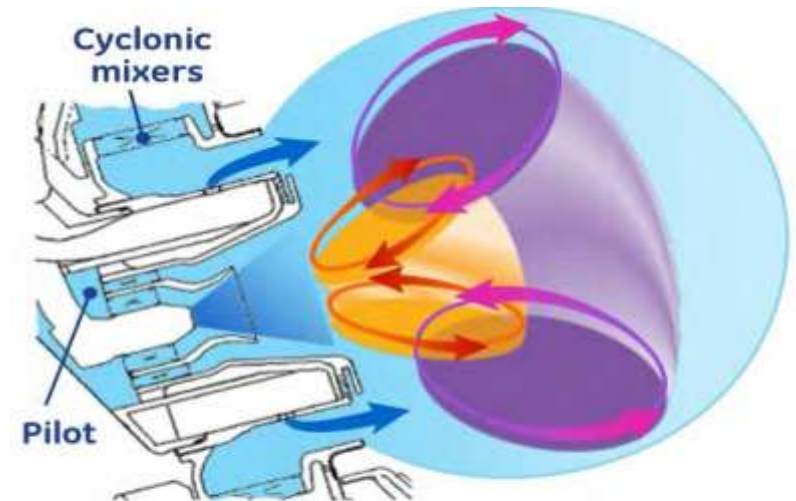
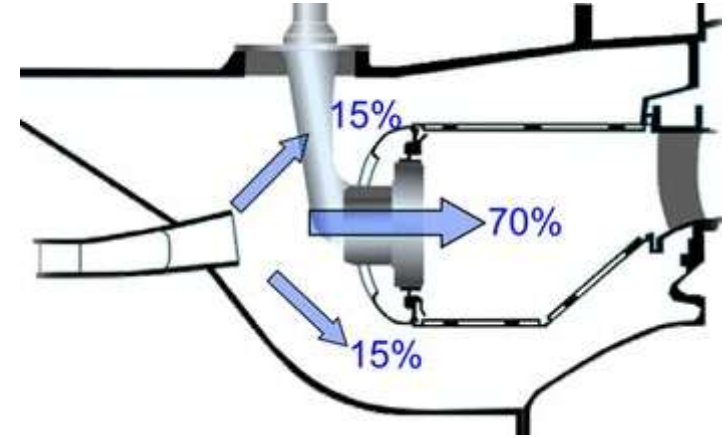
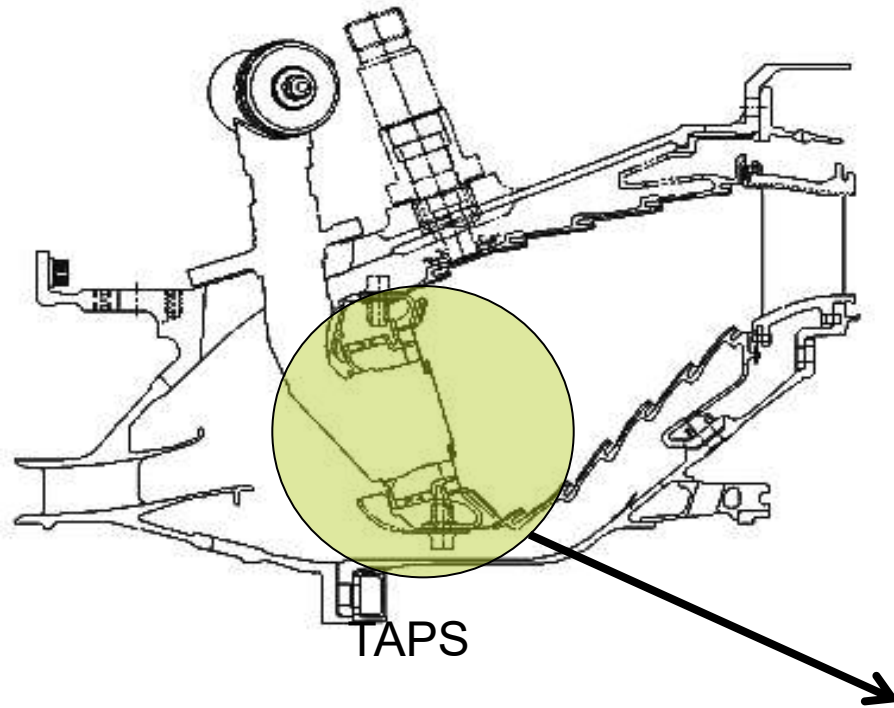
Fig. 9.4 RR Trent 900 combustor – applied RQL principle

RQL Combustor characteristics



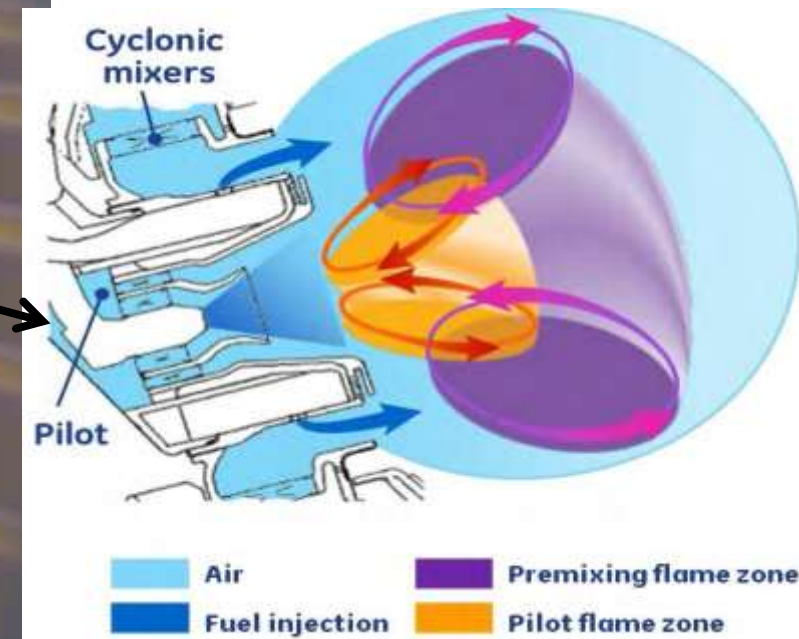
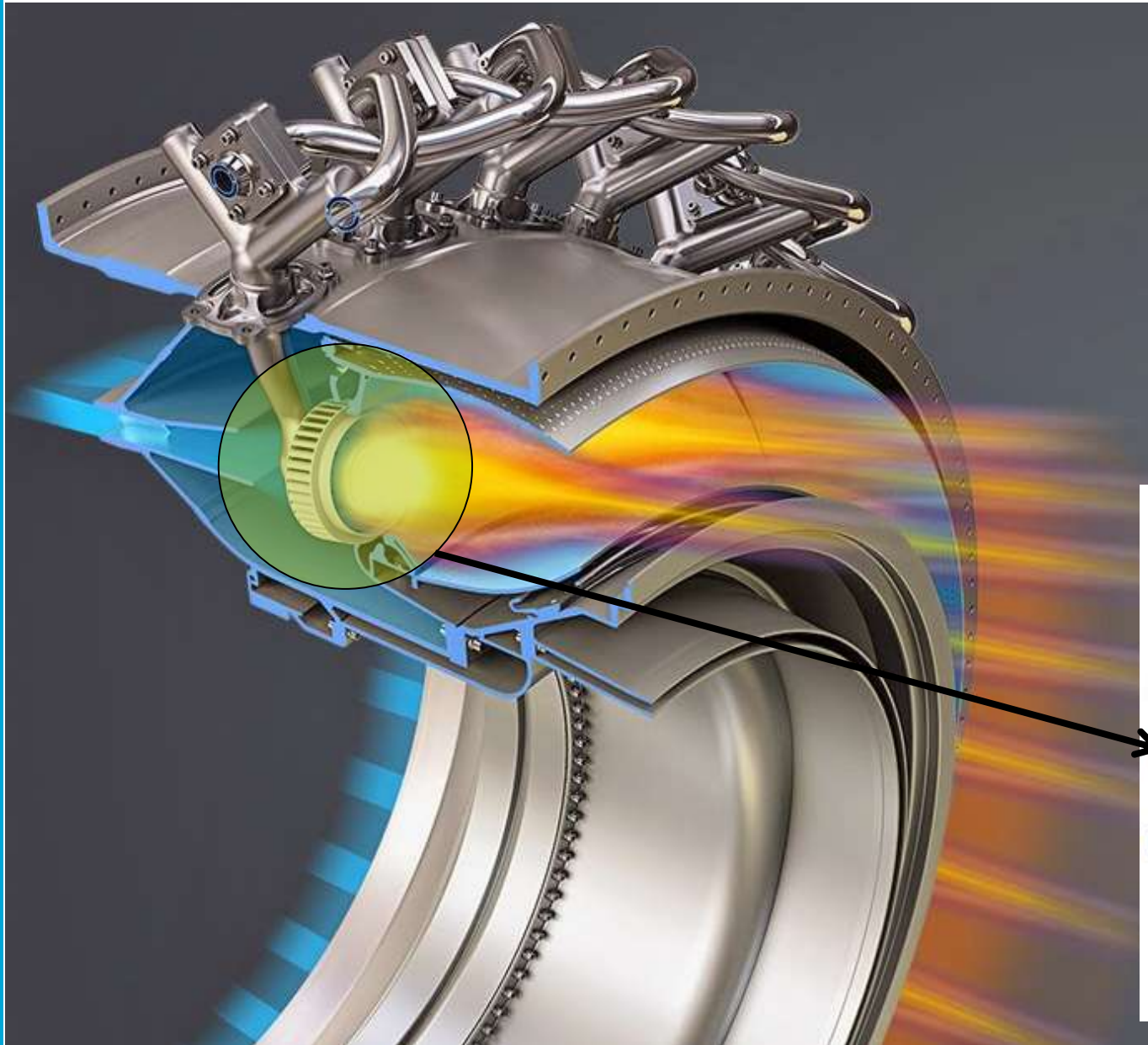
Rolls-Royce

TAPS combustor

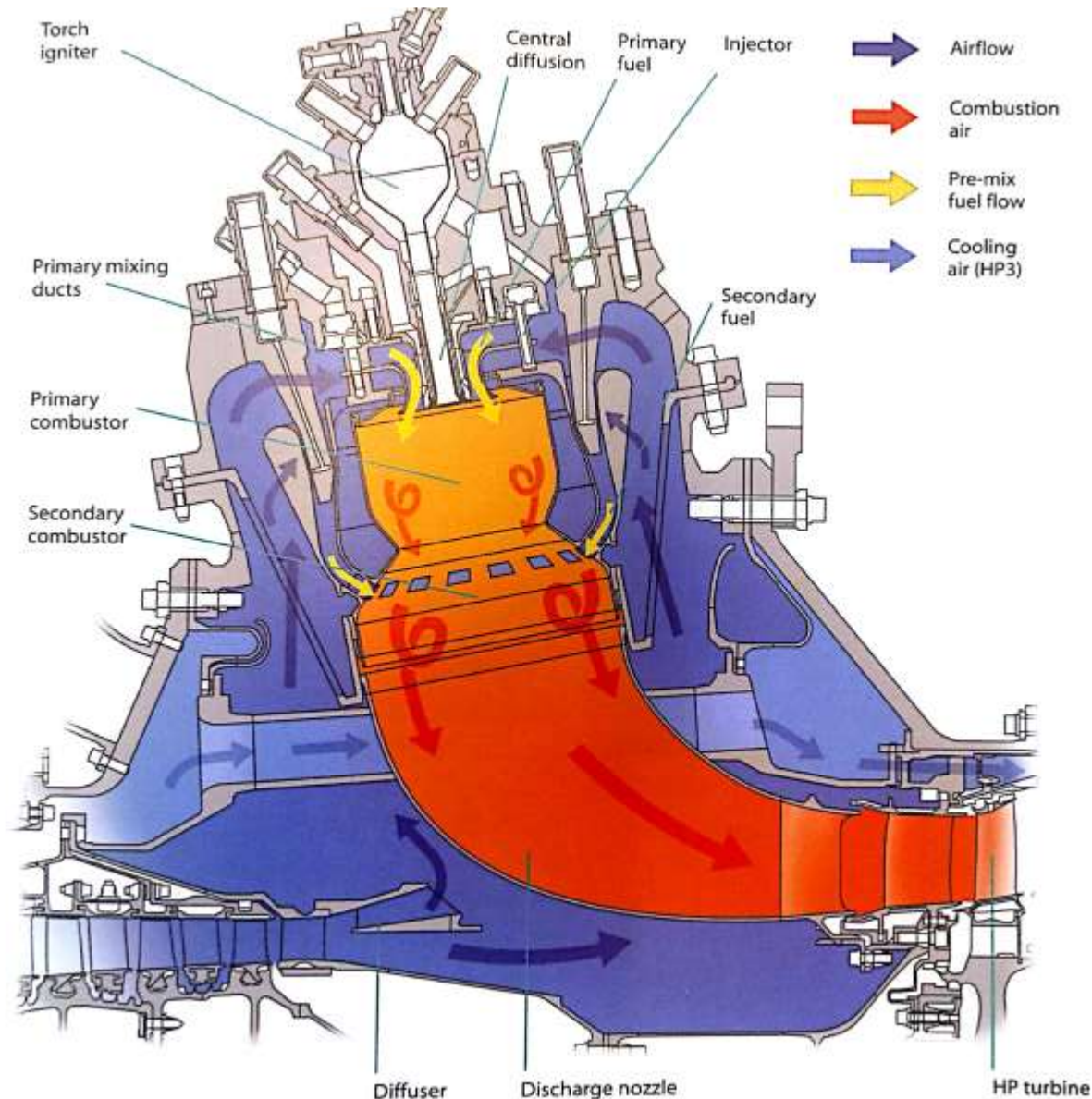


- | | |
|--|--|
|  Air |  Premixing flame zone |
|  Fuel injection |  Pilot flame zone |

GENx Combustor



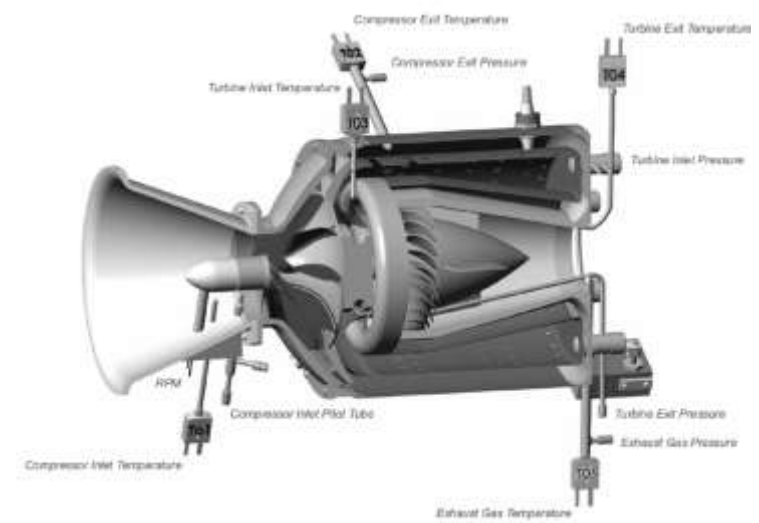
Industrial Gas Turbine Combustor



Comparison: Aero & Industrial

Criteria	Industrial	Aero
Weight & Size	Not critical	Critical
Operating conditions	Sea level	Wider than the flight envelope
No. Of cycles	30,000-50,000	6000-8000
Equivalent Operating Hours (EOH)	100,000-150,000	30,000-40,000
Emission regulations	Very stringent	Stringent
Critical point in design	Lean blowout	Relight
Fuel flexibility	Important	Not critical
Fuel type	Mostly Natural gas	Kerosene
Active air distribution	Possible	Not possible
Residence time	$\cong 10\text{-}30\text{ ms}$	$\cong 5\text{ ms}$
Combustor Type	Mostly Can Type	Mostly Annular type
Combustion philosophy	Mostly Lean, reheat	RQL/TAPS/ Staged

AE 4238



Arvind G. Rao A.Gangolirao@tudelft.nl

